# Capital Regulation as a Barrier to Bank Consolidation*

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**Abstract**

While thousands of mergers have reshaped the United States banking sector since the 1990s, the U.S. still has more banks per capita than other large countries. We show that equity capital regulation has meaningfully depressed bank M&A activity by making leverage-increasing financing expensive. Using a 2001 policy change in accounting standards as an instrument for the post-merger regulatory capital of target banks, we find that a standard deviation increase in capital raises the probability of a bank being purchased within a year by approximately 50%. Estimating a structural model of banks choosing merger partners subject to capital constraints, we find that weakening capital requirements around M&A since 1992 would have led to more consolidation and 19% fewer operating banks in 2022. Fixed cost synergies realized from additional mergers are of similar magnitude to consumer welfare losses from lower competition in deposit and loan markets. However, pairing relief with additional antitrust screening could have retained efficiencies while protecting consumer surplus.

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# 1 Introduction

The United States has an extraordinary number of banks. Until 1994, the number of banks was rationalized by interstate branching regulation: state and federal laws that made it "easier for a New York bank to expand in Kuala Lumpur than Jersey City" (William Jefferson Clinton, 1994).[1] However, today, despite full interstate branching deregulation and thirty years of consolidation, the United States still has 4,000 banks: nearly as many banks as the entire European Union and more banks per capita than any country with a population exceeding ten million people (Figure A.1).

The continued abundance of banks is puzzling in the face of apparent private gains that consolidation could provide. The industry has features that would suggest large economies of scale that would make additional mergers efficient (Berger, 1998).[2] Moreover, mergers give the remaining, larger firms market power to raise markups on their consumers.

In this paper, we show that the regulation of bank capital structure constitutes an important barrier to bank mergers, then quantify the downstream consequences of that barrier on bank customers: depositors and borrowers. In order to limit losses from potential bank failures, regulators require banks to hold a minimum amount of equity capital against their assets. However, by capping the debt potential merger combinations can take on, capital regulation *also* caps the ability of banks to finance acquisitions in cash. We call the shadow cost that this friction imposes on potential mergers the regulatory capital wedge (RCW). We show causal evidence of the RCW using time variation in the accounting of how the market value of merger targets' assets is incorporated into regulatory capital ratios. Using this instrument, we find that a standard deviation increase in capitalization raises the one-year probability of a bank being purchased by 44%. The effect of RCW on bank consumers depends on how synergistic mergers are, and thus whether the prevented mergers help or harm consumers (Williamson, 1968). Embedding this friction in a structural model of endogenous mergers, we find that the allocative efficiency gains prevented by RCW exceed the negative effects of market power in the loan market that would result from greater consolidation.

We begin by providing causal evidence that consolidation since 1990 has indeed been profitable for banks. We build a population of five thousand mergers between commercial banks using Call Reports and NIC Transformations and Relationships data. 80% of the mergers since 1990 were done by community banks, a designation for banks holding fewer than $10B in assets that applies to 90% of currently operating U.S. commercial banks. In order to measure the causal effect that mergers have on the merging parties, we use difference-in-differences to compare the outcomes of the combined bank (the *pro forma* entity) around the merger date to untreated controls. We demonstrate that realized mergers raised the average profits of the combined bank by 14% (15bps of combined assets) two years after the merger, both by reducing costs (non-interest expense) and raising revenue (net interest income). Prices rise slightly for consumers: the average interest rate

[1] Paraphrasing the remarks of Thomas Labrecque, then CEO of Chase Manhattan Bank.

[2] Rationales for economies of scale vary but include declining marginal costs of servicing additional deposits and loans or quality benefits of larger ATM and branch networks (Saloner and Shepard, 1995; Gowrisankaran and Stavins, 2004; Dick, 2007; Ho and Ishii, 2011).

paid on loans increases 8bp one quarter after the merger date but reverts within two years, and the estimate of deposit rates relative to prior to the merger are insignificant at the 95% confidence level but negative. The effects survive the relabeling of merger timing to bank holding company consolidation dates and a stacked difference-in-differences design using every unmerged bank as a control. Regardless of aggregate effects for consumers, the fact that in equilibrium realized mergers are profitable leaves open the possibility that bank M&A activity is constrained by costs unaccounted for in measures of future profits.

Armed with this evidence of profitable mergers, we present and empirically test a novel explanation – that minimum capital regulation constrains the financing of cash mergers. The mechanism is as follows. While mergers can be paid for with either stock or in cash, private firms struggle to pay targets with stock because their stock is illiquid. As a consequence, firms choose to use cash for acquisitions. However, unlike other firms, banks face strict minimum requirements for the amount of equity capital they must hold to support their assets. Bank regulators – Federal Reserve, OCC, and FDIC – impose the minimum on banks for reasons unrelated to M&A: reducing the likelihood of bank failure and limiting the losses to the FDIC's Deposit Insurance Fund in the event of a failure. Funding cash mergers with debt risks breaching minimum capital requirements, and funding with newly issued equity is costly as in other industries, and so potentially profitable cash mergers are untenable. We refer to the combination of these forces as the regulatory capital wedge (RCW). 80% of banks are privately held, making their stock illiquid. Accordingly, just 24% of acquisitions by private buyers since 1990 were conducted exclusively in stock.

To test empirically for RCW constraining bank mergers, we use the 2002 phase-out of Pooling of Interests accounting to create variation in regulatory capital ratios. In our main specification, we leverage the policy change to show that the book, regulatory value of equity shifts merger propensity holding market values fixed. Banks are regulated on the book value of their assets, rather than the true market value of assets if they were to be sold. Pre-2001, acquirers could use Pooling of Interests accounting to account for their acquisitions and avoid the marking of a merger target's assets to market. However, the 2001 FASB phase-out of Pooling of Interests meant that 2002-onward, mergers had to revalue the target at its fair market value, rather than its accounting book value. The "unrealized gains" on a target's assets, namely the difference in their market and book value, consequently shift bank merger propensity without shifting the profits that could be realized from acquiring the bank. The magnitude of unrealized gains is predictive of when a bank becomes a merger target, and the relationship is positive and significant only after 2001. Constructing an approximation of the post-merger Tier 1 Leverage Ratio in the two regimes using available data on the market value of bank held-to-maturity (HTM) and available-for-sale (AFS) securities, we use unrealized gains to instrument for the post-merger capital ratio. The second stage suggests that a standard deviation increase in a bank's Tier 1 Leverage Ratio causes a 94bp (44%) increase in the chance that the bank is a merger target in the next 12 months.

To study the consequences of RCW for efficiency and consumer welfare, we develop a model of heterogeneous banks considering mergers to reduce costs or increase market power of the com-

bined bank before competing for deposits (depositors) and loans (borrowers). The construction allows the pair-specific differences in merger synergies and in the constraints imposed by capital regulation for each pair to be disciplined transparently by parameter values so that the effect of changing capital regulations can be evaluated. Banks are simultaneously auctioned off to one another in a merger market. Each bank places bids on all banks with fewer assets than it has and sets its reservation value of remaining independent. With a realistic simplification of banks to consider each potential merger in isolation and once-and-for-all, the process allocates selling banks to their highest merger surplus buyer.[3] Surplus has two components: profit synergies and one-time costs. Mergers can increase efficiency – reduce the fixed costs of the combined entity – and can let the newly combined firm and its competitors raise markups. But, mergers also incur the RCW: a one-time cost of increasing bank leverage that reduces any merger's potential surplus. After the merger market clears, banks earn profits from lending at a spread above the risk-free rate and offering deposits at a spread below the risk-free rate.[4]

Estimation of the structural model proceeds in two parts. We estimate depositor and borrower demand with Berry (1994) and use an assumption of Bertrand Nash conduct to identify bank-time-level quality, marginal costs, and fixed cost parameters. We then use these parameters as data to estimate merger market parameters via simulated maximum likelihood. Maximum likelihood maximizes the likelihood of seeing the realized merger pairs and level of fixed costs of merged and unmerged banks in the data, using data from the year prior and the year after mergers to identify parameters. In other words, merger pairs are the consequence of selection on gains. In maximizing the likelihood of merger pairs and fixed costs with the model specified, we accommodate unobserved heterogeneity in shocks to each firm's standalone cost efficiency absent mergers that would otherwise bias the estimation of merger synergies.[5]

Our model estimates suggest that many efficiency-enhancing mergers are constrained by capital regulation. Those forces appear in two aspects of the model estimates: parameters governing the profit synergies realized from mergers and parameters of RCW. The baseline model specification allows for two synergies: reductions in fixed cost that come from the merging firms operating as one and the increase in profits coming from pricing as a single entity. While the intercept of fixed cost synergies implies that any particular pair could become *less* efficient, mergers with larger

[3]When the highest surplus buyer of a seller is a seller themselves, the auctions clear such that the final buyer absorbs both sellers. The ex post resolution of chains presents a partial identification problem, where bank A purchasing banks B and C could have either been the highest surplus buyer for B and C, or B could have been the highest surplus buyer for C, paid the one-time merger cost, then A absorbed both. We report parameters with the assumption that allocations were created with no chains needed to be resolved. Re-simulations of the merger market at the estimated parameters assign only 3% of pairs as chains, consistent with this being a realistic assumption for estimation.

[4]In the model, banks are not concerned with any increase or decrease in the risks of their portfolios created by mergers. That assumption is consistent with the empirical existence of binding capital regulation. Conventional wisdom of bank regulators and practitioners is that regulation is binding due to the risk shifting incentive created by the government-provision of deposit insurance so that banks themselves do not internalize the risk created by taking deposits to extend loans.

[5]Because banks choose mergers based on increases in future profits, by specifying a model of competition and hence specifying the model of future profits, the model setup also allows for a variant where selection on gains can be done on pair-specific unobservables. Namely, the pair-specific utility of a given merger pair can be estimated from real profit outcomes.

branch overlap drive fixed cost reductions. The parameter estimate suggests that the fixed costs could decline by an additional 2pp of seller assets annually when the seller's branches were entirely geographically overlapping with buyer's versus when they were entirely distinct.

Concurrently, for privately held firms, being constrained by regulatory capital minimums imposes a sizable one-time cost on potential M&A. The one-time cost of implementing a merger in the lowest quantile of post-merger capital corresponds to 11% of seller assets. That figure is 60% of the present value of future synergies from branch overlap or 1.2x the standard deviation of the pair-specific merger surplus shock, declining in the pair's Tier 1 Leverage Ratio post-merger.

Providing capital regulation relief around mergers both would have created efficiency while harming consumer welfare. Removing the one-time capital cost of performing cash mergers as a privately held firm increases the merger rate by 42% since 1992. Since the vast majority of banks are privately held, the estimate reflects not only that the cost of capital regulation is large but also that such a policy would target many firms who would find mergers otherwise profitable. The counterfactual world would have reduced the number of community banks over the past thirty years and result in 19% fewer operating banks in 2022. Along the way, the intervening years would have increased total welfare to the detriment of consumers. Fixed cost synergies realized in each period from mergers would translate to $11B in additional producer surplus. At the same time, the increased consolidation would have reduced the welfare of depositors and borrowers by nearly $6B.[6]

In designing optimal policy to trade off cost efficiencies with consumer welfare, pairing temporary capital regulation relief with traditional antitrust merger screening tools can retain some efficiency benefits of bank consolidation while limiting consumer welfare harm. In demonstrating this fact, we consider the counterfactual where RCW disappears while antitrust review also adopts a lower, national deposit cap. Currently, each state sets their own deposit cap: a maximum amount of deposits that a single bank can hold within the state, which, by default, is 30% nationwide but states have discretion to raise and lower. While banks can grow organically to exceed this cap, the cap is in place specifically to deny mergers that would cause the combined entity to exceed that market share. Imposing a 25% cap alongside removing RCW since 1992 would have both realized fixed cost synergies between 1992 and 2022 and raised consumer surplus.

In sum, the results suggest that regulating an industry's capital structure exerts *ex ante* merger deterrence. In the frictionless world considered by Modigliani and Miller (1958), firm capital structures are independent of firm value and so changes in capital structure will not affect a firm's optimal choices. However, given the empirical reality where cash mergers are more feasible for privately held firms, regulating capital structure becomes a potent policy tool that improves upon antitrust alone in two ways. The policy tool stops mergers before they are proposed to the an-

[6]Including a probability of bank default or financial stability in the calculation of social surplus would require two assumptions left to future work. First, given the diversification benefits of holding larger portfolios of risky loans, how much safer would a single merged bank be compared to a set of smaller community banks performing the equivalent lending and deposit taking? Second, given temporary capital relief tautologically implies lower capital ratios of the combined bank in the near-term, how should short-term increases in bank default probabilities before banks recapitalize from increases in profits be weighed against the diversification?

titrust authority. In industries where transaction-by-transaction review is costly or administratively constrained, a rule-based policy that automatically deters some anti-competitive transactions could lessen the load of antitrust merger review. But additionally, as only a model of endogenously proposed M&A can highlight, while antitrust acts on mergers to approve, approve with remedies, or deny, tuning or targeting relief from financing constraints can allow for a merger subsidy that can raise efficiency.

*Related literature:*

**Merger waves**. Policymakers and antitrust regulators have raised concerns about consolidation or "roll up" of several industries, especially industries that receive government funding (Cumming, 2026; Kessler, 2026). In healthcare markets, prior work examining hospitals (Dafny, 2009; Gaynor et al., 2013; Dafny et al., 2019), dialysis centers (Wollmann, 2024), and fertility clinics (La Forgia and Bodner, 2025) has shown that consolidation harms consumers. However, in other contexts, consolidation also has cost efficiency and productivity benefits (Jeziorski, 2014; Clark and Samano, 2022; Demirer and Karaduman, 2026). Theoretically, the cost efficiency of consolidation can come from economies of scale or from efficient firms buying inefficient firms (Jovanovic and Rousseau, 2002; Maksimovic and Phillips, 2001).[7]

We show that capital regulation, while not explicitly designed to limit consolidation, makes such a roll-up untenable for U.S. banks. A typical rationale for a lack of roll-up is that the Source of Strength Doctrine (noa, 2010) prohibits meaningful entry of private equity into the banking sector. While the doctrine could contribute, we establish that bank-to-bank mergers are also constrained by regulation. The requirement that a buyer of a bank must be well-capitalized means that the debt, the engine of private equity roll-ups in other industries (Kaplan and Stein, 1993; Axelson et al., 2009, 2013; Shivdasani and Wang, 2011), can only be applied sparingly (Johnston-Ross et al., 2025). In this way, such financing frictions are preventing many mergers from even being proposed by competitors to be brought before antitrust regulators.

**Welfare of bank mergers.** We contribute two-fold to the literature on the welfare effects of bank mergers. First, crucially compared to past calculations, we consider the welfare consequences of bank mergers as an endogenous, equilibrium outcome targeted by policy. In examining the mergers that *have* been realized, Pakes et al. (2024) finds that consumer welfare is lower than prior to mergers in a sample of markets that experienced a single bank merger relative to markets experiencing none, and Ho and Ishii (2011) evaluate aggregate changes in consumer welfare as the market evolved since 1990.[8] Second, we show reduced form evidence of cost efficiencies that

[7] As for the motivations for mergers, across industries, mergers can be considered potentially value enhancing investment, and so financing constraints (Harford, 2005; Almeida et al., 2011; Gorbenko and Malenko, 2018) and market timing (Shleifer and Vishny, 2003; Li et al., 2018) both play a role in inhibiting or encouraging mergers. Managerial entrenchment, implicit in the design of the structural model but not the paper's focus, can affect both buy-side (Roll, 1986; Morck et al., 1990; Masulis et al., 2007; Harford et al., 2012; Yim, 2013) and sell-side firms (Jenter and Lewellen, 2015; Kotia, 2025).

[8] Without explicitly calculating welfare, a large body of work has examined prices and quantities on the loan side Berger et al. (1998); Sapienza (2002); Kahn et al. (2005); Di Patti and Gobbi (2007); Erel (2011); Nguyen (2019); Brunetti et al. (2026) and the deposit side Prager and Hannan (1998); Bord (2018); Focarelli and Panetta (2003);

increase producer surplus.[9] Wheelock and Wilson (2000); Hannan and Pilloff (2009); Beccalli and Frantz (2013); Akkus et al. (2016) all suggest that these cost efficiencies drive mergers. We explicitly quantify producer surplus and trade it off with consumer welfare.

**Capital regulation**. In showing that bank capital regulation affects bank investment in M&A, the paper documents a novel channel by which bank capital regulation affects bank operations. Prior work studies its effect on entry and exit (Corbae and D'Erasmo, 2021; Corbae and Levine, 2025; Corbae, 2025), but M&A is far more common in the modern U.S. banking sector than entry and exit. Our empirical design isolates distance from the effect of "microprudential regulation" (Hanson et al., 2011) on M&A while putting aside its effect on day-to-day bank operations, which are plentiful but not the focus of the paper (Berger and Udell, 1994; Gorton and Winton, 1995; Peek and Rosengren, 1997; Gorton and Winton, 2017; Gropp et al., 2019; Fraisse et al., 2020; Carlson et al., 2022; Berrospide and Edge, 2024).[10]

Most related to our paper is Granja et al. (2017) that shows that bank capitalization is associated with which banks purchase banks from FDIC receivership. We isolate the mechanism pertaining not just to capitalization, but a bank's binding to its regulatory ratio, extend the study to the entire population of banks as potential targets using an instrumental variables strategy, and quantify implications of that phenomenon for the sector.[11]

**Endogenous bank industry structure**. Economically, we contribute a quantification of RCW, a novel, policy-relevant financing friction, within a model of endogenous M&A between banks. In estimating the model, we can document quantitative importance of microprudential regulation in mediating consolidation among U.S. banks. Computationally, we contribute a tractable model of endogenous M&A where 1) buying and selling firms need not be assigned *ex ante* and 2) product market competition in prices is considered during the first stage selection of merger partners. Past work has proposed one of three methods to accommodate endogenous M&A in firm decision-making. First, Gowrisankaran and Holmes (2004); Corbae and D'Erasmo (2025) place mergers in a dynamic model where a dominant incumbent is the only potential buyer of a competitive fringe of other incumbents. But, this method has limitations for studying consolidation between small community banks who perform the majority of M&A transactions.[12] Gowrisankaran (1999); Igami and Uetake (2020); Jeziorski (2023) allow for multiple imperfectly competitive firms to make take-it-or-leave-it merger offers to their competitors in a dynamic, sequential merger model. Given the

Granja and Paixao (2023); Liebersohn (2024).

[9]This is updated from Rhoades (1993); Berger (1998); Rhoades (1998); Berger et al. (1999); Focarelli et al. (2002); Houston et al. (2001); Focarelli and Panetta (2003) to include twenty more years of mergers.

[10]In using an instrument for capital to study the propensity to merge in the short-run to identify effects, we also do not model how capital structure itself affects bank operations, for example, whether it causally mediates risk-shifting, whether bank deposits discipline the use of funds, or whether deposit insurance creating moral hazard (Calomiris, 1990, 1999; Calomiris and Jaremski, 2019; Calomiris and Chen, 2022).

[11]Prior work has investigated the first half of the RCW mechanism, that illiquid stock incentivizes acquirers toward cash mergers (Faccio and Masulis, 2005; Brau and Fawcett, 2006; Bargeron et al., 2008; Massa and Xu, 2013; Huang et al., 2024).

[12]Unlike our model, these models and others allow for sophisticated entry, exit, and investment decisions by banks, including into new geographic markets, which will not be the focus of our work (Corbae and D'Erasmo, 2021, 2024; Corbae and Levine, 2025; Corbae and D'Erasmo, 2025; Oberfeld et al., 2025)

number of operating firms considered, the model is feasible but computationally intensive in our setting.

To make progress, we consider simultaneous allocation of banks to merging partners with an assumption that firms do not consider the prospect of potential M&A by other players. The model adapts the logic of papers considering and correcting for the selection problem created by endogenous firm entry (Bresnahan and Reiss, 1991; Seim, 2006; Ciliberto and Tamer, 2009). Related work by Perez-Saiz (2015); Aguirregabiria et al. (2016); Uetake and Watanabe (2020) construct models of "entry by merger," where revealed preference inequalities identify parameters because entrants and incumbents are *ex ante* assigned. We show that, by specifying an allocation rule with several micro-foundations, buyers can evaluate being both a buyer and a seller in the same market simultaneously. That nuance is paid for with a requirement that banks do not consider the externalities of a given merger decision on other players or that they may consider multiple simultaneous mergers within a single period.

# 2 Data

In order to quantify the reasons for a lack of larger banking sector consolidation, we use standard banking datasets to track bank outcomes on a quarterly basis and to construct the list of bank mergers.

## 2.1 Measuring bank outcomes

Our analysis combines information from several data sources, most of which are commonly used in the banking literature.

### 2.1.1 Bank-level balance sheets and income statements

We collect the universe of bank quarterly balance sheets and income statements ("Call Reports") from 1990 Q1-2024 Q4. We use these forms to track regulatory capital ratios, operating costs, and other important balance sheet characteristics we control for throughout our analyses (e.g., insured deposits-to-assets ratios and funding costs). When revenue and cost items are defined as year-to-date, we use the flow value of the item so that we can measure the revenue and cost of the filer quarter-by-quarter. Details of this procedure are provided in Appendix C.1 We also use these data to construct our primary source of exogenous variation in the capital-intensity of a potential acquisition: the fair value losses on the securities portfolio that would be booked in the event that a bank is acquired. This variable is constructed by summing the difference between fair value and amortized cost on each bank's held-to-maturity (HTM) and available-for-sale (AFS) securities portfolios, both of which are booked at cost for the purposes of calculating regulatory capital but are marked to fair value in the event the bank is acquired under the current accounting standards.

We obtain annual, branch-level information on deposit volumes spanning 1990 to 2024 from the FDIC's Summary of Deposits filings.

#### 2.1.2 Bank holding company level balance sheets, income statements, and ownership

We collect the analog to the Call Reports at the holding company level, the Y-9 series, over the same time period. We use these data primarily to track issuance activity of equity and long-term debt. We assign individual banks that file the Call Reports to their bank holding companies, if they have one, using the National Information Center relationships file. We do so by following the procedure outlined in the Federal Reserve's data dictionary.[13]

### 2.2 Constructing a bank merger database

We obtain information on the history of bank organizational relationships and merger transactions from the National Information Center's (NIC) "relationships" and "transformations" files. When studying financing methods, we use the population of bank mergers maintained by S&P Global Market Intelligence. The database of bank mergers filters out activities more aptly characterized as corporate reorganizations and recognizes mergers at the bank level.

#### 2.2.1 Merger transaction dates and merger deal considerations

We collect data from NIC on the history of ownership relationships between different banks and bank holding companies ("relationships" file) and on mergers and acquisitions ("transformations" file). The data are at the bank charter level, where each charter is assigned an RSSD. That RSSD-level is how we define a bank, but some of the largest "banks," namely all of the Global Systemically Important Banks (G-SIBs) are colloquially referred to as "banks" while actually acting as a Bank Holding Company owning multiple RSSD's. Because some entities even contain multiple layers of BHC's, we refer to the true parent bank holding company as the "high holder." Our empirical results use the date of the merger as the date reported in NIC that the RSSD of the acquired bank was merged into the acquirer.

We supplement this information on mergers with additional data from S&P Global Market Intelligence (SNL) deal filings. In particular, we retrieve data on announced deal values and consideration mix (cash versus equity) for a subsample of mergers, largely transactions involving at least one publicly traded company.

#### 2.2.2 Constructing the merger sample

Building an accurate dataset of bank mergers entails several challenges. First, many bank-to-bank mergers are not true changes in control of the sort we are interested in: in the wake of the pas-

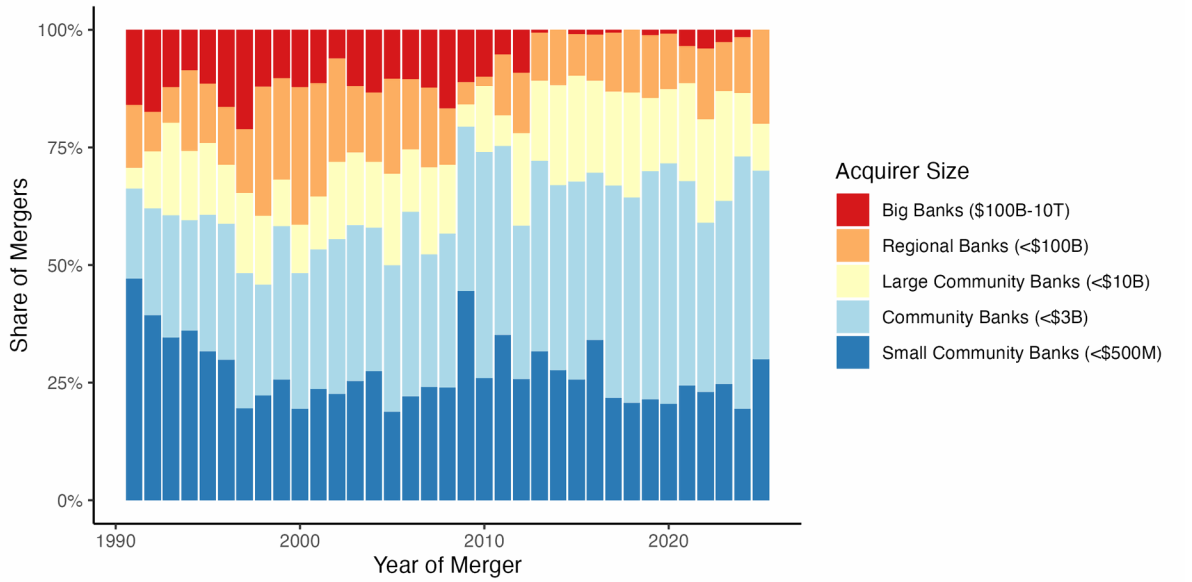
[13] See https://www.federalreserve.gov/apps/mdrm/data-dictionary/search/item?keyword=9364&show_short_title=False&show_conf=False&rep_status=All&rep_state=Opened&rep_period=Before&date_start=20190808&date_end=20190808; accessed 07/17/25.

sage of Riegle-Neal in 1994, many multi-bank holding companies merged subsidiary banks into one charter. This type of activity is more aptly characterized as a corporate reorganization. Second, many mergers are officially between holding companies, with the underlying subsidiary banks merged into a combined charter at some later date; this procedure should be considered a "true" merger, but on paper it resembles a corporate reorganization in the sense that, at the time the two banks are combined, they are already under the same holding company.

Our data construction procedure is intended to minimize the impact of issues of this sort by ensuring targets had a distinct high holder from its acquirer within two years prior to the merger date. To implement this algorithm, we begin with our dataset of commercial banks linked to their high holder at every point in the panel from NIC. Next, we merge on the RSSD ID of the commercial bank with the RSSD ID of targets in the NIC transformation file (subsetted to drop bank failures), keeping only matches on RSSD ID where the indicated high-holder relationship is either the current or penultimate link relative to the date of the merger. We store the current and previous high-holder RSSDs from this match, including the windows in which each is active, in our updated dataset. This population has a mix of true mergers and of internal corporate reorganizations. Finally, we merge this file back into the relationship file on the RSSD ID of the *acquiring* bank, again storing the date windows of the current and previous high-holders. We classify a true, inter-company bank-to-bank merger as a merger in the resulting dataset for which the target and acquirer banks have different high holders either at the time of the merger or at any point in the prior two years. The choice of window length is meant to account for the normal length of time that banks take, after merging holding companies, to consolidate the subsidiary bank charters.

Our resulting sample contains 4,940 mergers between commercial banks occurring between 1991 and 2024. The sample statistics reveal a distinguishing feature of the banking sector: the vast majority of M&A activity is between very small firms. Figure 1 plots the percent of mergers done by acquirer size bin. 80% of mergers are done by banks that have fewer than $10B in assets. Banks of this size, referred to by regulators as "community banks," comprise 90% of U.S. banks and provided just 20% of bank lending in 2024 (Figure A.2). Table 1 shows descriptive statistics of acquirers versus targets and uninvolved banks. Acquirers have larger return on equity (ROE) and are more cost efficient than their targets and than banks uninvolved in the merger market. The statistics are suggestive evidence that bank mergers are motivated to realize cost synergies. Table B.1 shows means by five-year cohorts of acquisitions between 1990 and 2024. The trend of efficient acquirers purchasing inefficient targets holds throughout the panel. Figures A.3 and D.8 show additional statistics about the merger sample. Table B.2 shows a difference in means table for county-year observations that have a presence of merging bank versus county-year observations that have not. Presence is defined as the combined bank, either the acquirer or target, operating at least one branch in the county between 2005-2024. County-year observations that have experienced mergers have lower unemployment rates, lower poverty rates, higher household income, higher populations, more highly educated individuals, fewer retired individuals, and are less rural.

Figure 1: Community banks in the merger market




Note: This figure shows the share of mergers done annually by the size of the acquiring bank. The majority of acquirers are community banks, banks have less than $10B in assets (in 2023 USD) in the quarter before the acquisition.

### 2.3 Measuring economic outcomes

In Section 5.7, we use economic outcomes of counties affected by bank mergers to evaluate the plausibility of the parallel trends assumption. We use the American Community Survey (ACS), available from 2005-2023 to measure the average unemployment rate, average poverty rate, average household income, and population by U.S. county. We use the publicly available Census Business Dynamics Statistics as our measure of establishment entry, exit, net job creation, job reallocation, and firm-death rates.

## 3 Profitable consolidation

Using matched difference-in-differences in order to estimate the causal effect on merging firms, we show that realized mergers since 1990 raised the profits of merging banks as a fraction of assets by 14% by increasing profits from offering deposits and lending (net-interest income) by 3% and reducing operational (non-interest) expenses by -3%.

### 3.1 Identification strategy

In order to estimate the causal effect on merging firms, we run difference-in-differences comparing the *pro forma* bank for every merger to a matched bank and assume parallel trends in outcomes absent treatment between the mergers and controls. With the sample construction and assumptions,

Table 1: Commercial bank merger sample

| Variable (pp) | Acquirer | Target | Uninvolved | Acq. - Tar. | Tar. - Unin. | Acq. - Unin. |
|---|---|---|---|---|---|---|
| Inefficiency Ratio | 64.2 | 73.8 | 73.3 | -9.5*** | 0.4 | -9.1*** |
| ROA: Net income / Assets | 0.29 | 0.19 | 0.96 | 0.10*** | -0.77 | -0.67 |
| ROA (adj): Economic profits / Assets | 0.30 | 0.20 | 0.97 | 0.10*** | -0.77 | -0.67 |
| ROE: Net income / Equity | 3.05 | 1.61 | 2.27 | 1.44*** | -0.66*** | 0.78*** |
| ROE (adj): Economic profits / Equity | 3.18 | 1.68 | 2.33 | 1.50*** | -0.65** | 0.85*** |
| Tier 1 Leverage Ratio | 9.3 | 9.9 | 11.5 | -0.6*** | -1.5*** | -2.1*** |
| Loans / Assets | 64.3 | 60.5 | 59.2 | 3.8*** | 1.3*** | 5.1*** |
| Deposits / Assets | 81.0 | 84.5 | 83.4 | -3.5*** | 1.1*** | -2.3*** |
| Observations | 4,191 | 4,940 | 1,144,752 | | | |

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Note: The first three columns show the means of profit and balance sheet variables for commercial bank acquirers, commercial bank targets, and all other uninvolved banks four quarters prior to the merger. The last three columns show the difference in means between acquirers and targets, targets and other banks uninvolved in a merger transaction, and acquirers and uninvolved banks. Acquirers have larger return on equity (ROE) and are more cost efficient than both their targets and banks uninvolved in the merger market in the year of their acquisition. Accounting terms are defined in Table C.1.

we estimate the causal effect of the merger on the *pro forma* bank using difference-in-differences.

The *pro forma* bank is a single entity, and every outcome variable for the *pro forma* bank is built from the acquirer and target in the merger of interest. Let entities involved or uninvolved in mergers be indexed by $i$. For outcomes measured in dollars, an entity's outcome $Y_{\mathcal{M}(k,j)t}$ is the sum of the buyer $k$ and seller $j$ outcomes. For instance, prior to the merger, the *pro forma* bank's net income is the sum of the acquirer and target's net incomes. For a given merger consummating at date $g$, we include it in our analysis sample if the acquirer has had no other merger within eight quarters before or after $g$ and the acquirer has a balanced panel around the merger date, filing a Call Report quarterly, 8 quarters before the merger date and 12 quarters afterward. This limits our regression sample to 1,464 unique mergers between 1991 and 2021.

The control observation for every merging bank is created via matching the *pro forma* merging banks on observables to firms geographically distant from the merger. Formally, we form a potential control group of banks with no branches in any state in which either merging bank operates and participating in no merger 8 quarters prior to the merger date and 12 quarters afterward. Then, from that potential group, we match each *pro forma* bank, without replacement, to the observably most similar bank on log deposits, log assets, inefficiency ratio (an industry measure of cost efficiency), and Tier 1 leverage ratio measured 8, 6, and 4 quarters prior to the merger date, using standardized Euclidean distance.[14] Table C.1 defines these accounting items. For any date $t$ that

[14]In the Appendix, we use every bank uninvolved in a merger between 1990-2024 as a control group for every merger cohort between 1990-2024. This population is referred to as a "stacked control group" and is useful for achieving statistical power and avoiding ad hoc assumptions for the comparability of the *pro forma* bank to any particular untreated banks. The results are quantitatively similar.

the control bank files a Call Report, they can act as a control bank at event time $s$ for cohort date $g = t - s$.

With this population of *pro forma* mergers with cohort dates $g(i)$, we run the following regression for outcome variables $Y_{ig(i)t}$

$$\mathrm{Y}_{ig(i)t} = \alpha_i + \gamma^{\text{DiD}} X_{it} + \sum_{s=-8,\neq -2}^{12} \beta_s \cdot \mathbf{1}(t - g(i) = s) \cdot \mathbf{1}(i = \text{Merging entity}) \tag{1}$$

with cohort time fixed effects and state-year fixed effects captured by $\gamma^{\text{DiD}} X_{it}$ and bank-cohort fixed effects $\alpha_{i,g(i)}$. To interpret the coefficient $\beta_s$ as the causal effect on the combined bank at quarter $s$ since the merger, we make two standard difference-in-difference assumptions. The Stable Unit Treatment Value Assumption (SUTVA) requires that potential outcomes of the control banks depend only on their own treatment status, not others around them nor the merged group. With the controls chosen, it amounts to assuming that the control bank's market operates independently of the merger occurring in the focal merger's state and that controls contaminated by a merger in their own state. Parallel Trends requires the control group of banks, post-merger date, would have moved in parallel to the *pro forma* banks counterfactually without the merger having occurred, conditional on controls, like conditional on state-year fixed effects. To make this explicit, one identification concern from this assumption could be that merger timing is systematically correlated with shocks to the business opportunities for the market of the merging banks versus the control banks in other markets. State-year fixed effects ameliorates this concern if the shocks occur at the level of the state. But, if the shock were targeted to banks focused explicitly on credit card lending, and the shock affected the profitability of credit card lending differently in one state versus another, then that shock would bias the identification of the merger's causal effect from $\beta_s$. $\beta_s$ is then the causal effect of the merger on the bank, compared to six months (two quarters) prior to the merger date. While both econometric assumptions are fundamentally untestable empirically, in Section 3.4, we show additional results that suggest they are reasonable.

### 3.2 Bank profitability

First, realized mergers improve the profitability of the combined firms. In order to document this fact, we estimate Equation (1) for profits as a fraction of assets of the *pro forma* entity.[15] The measure is nearly the accounting equivalent of "return on assets" (ROA), but we make two adjustments to create an estimate of *economic* profits. This profit is defined as

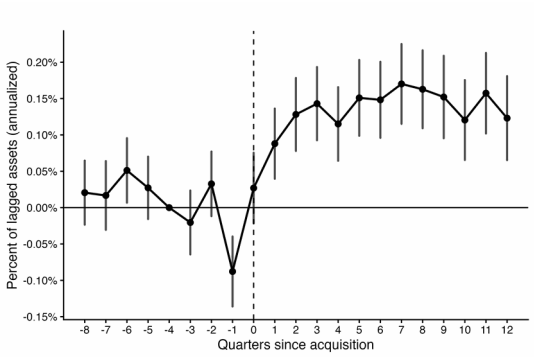
$$\begin{aligned}\text{Profit}_{it} \equiv \text{Net income}_{it} + \text{Goodwill impairment losses}_{it} + \\ \text{Amortization expense}_{it}.\end{aligned} \tag{2}$$

Figure 2a shows the resulting estimates of $\beta_s$ for growth in profits. In the quarter prior to

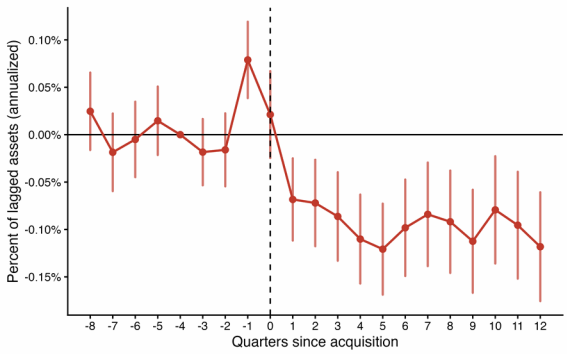
[15] Effects on bank profit and costs are robust to using the logarithm of profits and costs.

Figure 2: Bank-level effects of bank mergers

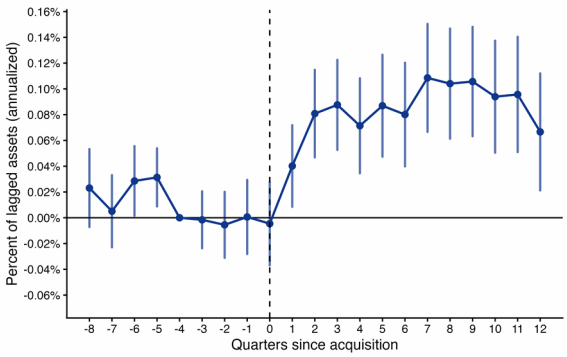
(a) Profits / Assets



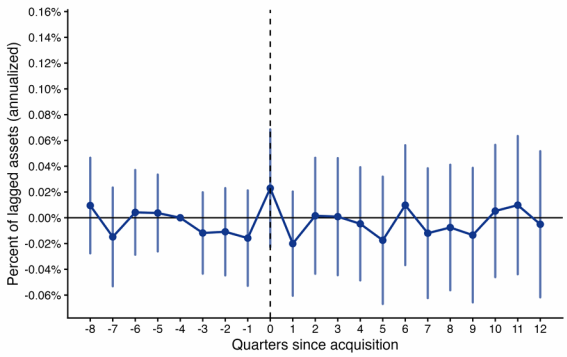

(b) Non-Interest Expense / Assets




(c) Net Interest Income / Assets




(d) Non-Interest Income / Assets




Note: This figure shows the effects of a merger on the revenue and costs of the merging bank. Each panel plots $\beta_s$ from Equation (1) estimated on bank, quarter observations. In the pre-period, the separate merging banks are treated as a single *pro forma* entity. In the post-period $(t \geq 0)$, the combined bank files only one profit and loss statement and balance sheet that is used to create the post-period. Merger date is defined as the quarter that the target's RSSD was absorbed by the acquirer, and, hence, the date that the merged bank files a Call Report under the auspices of a single bank charter. Standard errors are clustered at the bank level. Vertical bars show 95% confidence intervals. Panel 2a shows that profits increase immediately, flattening at 15bps higher two years after the merger date (a 14% increase). Profits decline temporarily in the quarter before acquisition, which corresponds with an increase in non-interest expenses in Panel 2b. That increase in profits comes from both an increase in revenues and a decrease in costs. On costs, Panel 2b plots operational expenses (measured by non-interest expenses). After expenses increase around the merger date, decline -9bps two years after acquisition (a -3% decrease). On revenue, panel 2c shows net interest income, which increases meaningfully after mergers to 12bp two years later (a 3% increase). Panel 2d shows that non-interest income does not increase at the 95% significance level. Outcome variables are defined in Table C.1.

combination, profit drops precipitously. Otherwise, there is no significant pre-trend in the outcome, consistent with our assumption of parallel trends between the merging and control banks. Then, in the quarter of combination, combined profits increase then grow in the two years following to 15bp of assets (a 14% increase compared to one year prior to the merger date). To demonstrate the importance for accounting profits as well, Figure D.1 shows the same specification run for net income. Merger timing could realistically be earlier than the date reported as the charter retirement date in NIC. We repeat the analysis at the date that bank holding companies were combined – an often simultaneous but sometimes earlier date – and the results are quantitatively similar (Figure D.3c).

### 3.3 Decomposing profit increase

The increase in profits is comprised of both increases in revenue and a decreases in costs of the merging banks. Both increases in net interest income, driven by increases in loan interest rates and decreases in deposit interest rates, and reductions in non-interest expense contribute.

Banks, by making money lending at a higher rate than they need to pay depositors, make profit from the spread between lending interest rates and deposit interest rates. Consider a simplified definition of bank profits as revenue minus costs, such that the components of Equation 2 is:

$$\text{Profit} \equiv \underbrace{\text{Net Interest Income} + \text{Non-Interest Income}}_{\text{“Revenue”}} - \underbrace{\text{Non-Interest Expense} - \text{Other items}}_{\text{“Costs”}}. \quad (3)$$

#### 3.3.1 Cost: Reductions in non-interest expenses

One channel by which profits rose was by increases in the cost efficiency of merging banks. We present results for several cost categories that together show that a meaningful portion of cost reduction occurred through non-interest expenses.[16]

Our first outcome is non-interest expense. Bank costs are typically divided into two categories: (1) interest expense, directly related to borrowing (from depositors and other creditors) and lending and (2) non-interest expense, which captures the remainder and encompasses the overhead rent on bank branches, IT, and salaries for loan officers and workers that allow banks to offer their products and services. The next section will consider (1) in the context of bank revenue. Figure 2b shows the estimates of $\beta_s$ addressing (2). The figure shows the logarithm of non-interest expense, where we then report the coefficient as the percent growth in the outcome. In the quarter prior to the merger date, non-interest expense spike by 8bp. After the merger, non-interest expenses decline steadily to a -9bp decline at the combined firm (-3% of the mean level of pro forma firms one year prior to the merger – 3.15%). The negative estimate implies that the total cost base declined. Breaking the cost items down further in Figure D.2, the decline persists for salaries, fixed expenses,

[16] In Section 5, we will use a model of depositor and borrower demand to distinguish whether profitability increases stem from marginal and fixed cost reductions, as well as whether mergers are harmful to depositor and borrower welfare.

and miscellaneous expenses. The stark effects are consistent with intuition about mergers that cost savings arise from eliminating redundancies in admin and legal.

### 3.3.2 Revenue: Higher loan interest rates and lower deposit spreads

The other channel by which profits rose could be a rise in revenue. Net interest income rises by 3%. The increase is driven by earning larger spreads between the interest rates charged borrowers versus deposit rates paid to depositors. Figure 2c plots the estimates of $\beta_s$ from Equation (1). As net interest income is akin to the revenue of a bank, deposit interest rates and loan interest rates are akin to the bank's prices. Net interest income is the amount of money earned lending at an interest rate that exceeds the cost of borrowing funds. Putting aside returns bank earn on other assets,

$$\begin{aligned} \text{Net Interest Income} = &\ (\text{Interest rate on loans}) \cdot (\text{Loans}) \\ &- (\text{Interest rate on deposits}) \cdot (\text{Deposits}) \end{aligned} \tag{4}$$

Figure 2c rises 12bp as a percent of assets two years after the merger. In magnitudes, that is a 3% increase from the average pre-merger level of NII as a percent of assets for the sample (3.8% of assets). That increase in revenue is consistent with negative but potentially modest effects on consumers due to the remaining banks having more market power. Figure 3 shows estimates from Equation (1) for each quantity in Equation (4). Immediately post-merger, deposit interest rates decrease by 2bp with confidence intervals that do not rule out no change. Loan interest increase immediately to 6bp but the effect vanishes two years after the merger. These small effects are consistent with prior work (Granja and Paixao, 2023; Brunetti et al., 2026).

## 3.4 Robustness

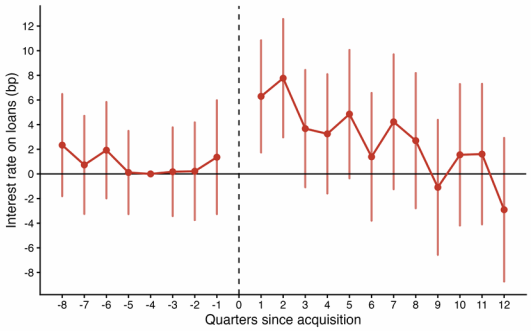
If the assumptions of parallel trends and SUTVA do not hold in this setting, the estimate of a causal effect of mergers would be biased. We discuss potential threats to identification and show that empirical evidence consistent with our assumptions.

### 3.4.1 Parallel trends

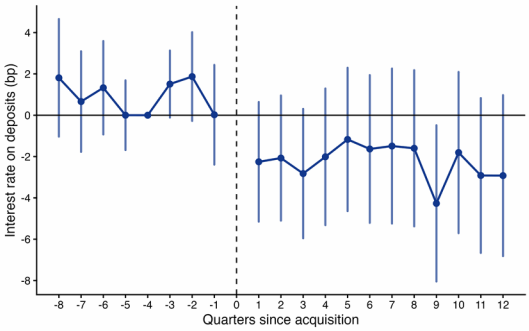
In evaluating the plausibility of the assumption, we discuss the ideal experiment. An ideal experiment to estimate the average treatment effect on treated banks would randomly select a pair of separately organized banks to run as a combined pair, then measure the change in $Y_{ig(i)t}$ experienced by that pair. *A priori*, parallel trends in pre and post outcomes could seem economically quite questionable. To become an acquirer, banks may be faring better, facing better financing conditions, or facing higher latent demand for their products that could be visible up prior to the merger and afterward. Sellers could be increasingly struggling with declining revenue before being willing to be purchased by a larger acquirer or run by management interested in ending their

Figure 3: Changes in prices and quantities at merging banks

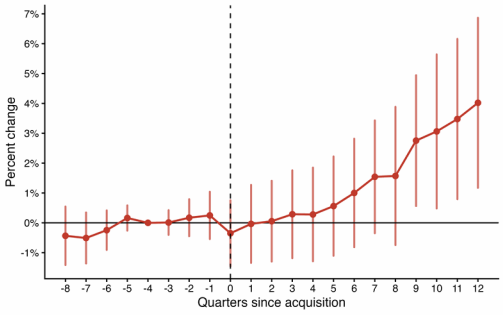
(a) Interest rate, loans



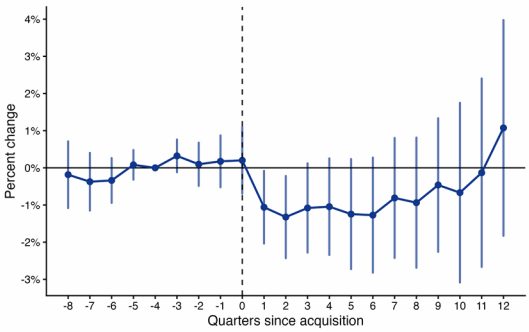

(b) Interest rate, deposits




(c) Loans (log, dollars)




(d) Deposits (log, dollars)




Note: This figure shows the estimates of $\beta_s$ from Equation (1) on the constructed population of bank mergers compared with a stacked population of banks that were never acquirers nor targets between 1990-2024. Standard errors are clustered at the bank level. Vertical bars show 95% confidence intervals. Panel 3a plots the estimates of $\beta_s$ for $s \in [-8, 12]$ for the average interest rate offered on loans (defined as net interest income divided by assets). Loan rates increase on impact but slowly revert to the level prior to merger. Panel 3b plots estimates for the average interest rate offered for deposits (defined as interest expense divided by assets). In any given quarter, deposit rates do not decline significantly. Panel 3c plots the estimates for the logarithm of dollars of loans, which increases steadily after the merger date. Panel 3d shows that deposits fall after mergers to roughly -2%. Deposit and loan rates are winsorized at the 1% and 99% level. Outcomes variables are defined in Table C.1.

involvement in the firm. Matching attempts to ameliorate that concern by finding banks similar on financial characteristics. However, matching hard-wires the parallel trends of banks prior to merger, and merging banks could still counterfactually have been susceptible to endogenous timing: a differential post-period trend, after accounting for state and year fixed effects and pre-period trend.

To mitigate this concern, we repeat the analysis using stacked-difference-in-differences. The results are nearly quantitatively identical and shown in Appendix D. These results have the convenient feature that, despite not imposing a pre-period match, merging banks move in parallel to non-merging banks. That fact makes clear that the stories of potential failures of parallel trends related to endogenous timing are limited: endogeneity must be unobservable in pre-period trends, and it must arise at the moment of merger or soon afterward.

We search for some proxies of this endogeneity with a negative control approach. Concretely, one component of that endogeneity could be time varying economic shocks, more local than at the state level, that very local banks may have differential access to. Then, banks are chosen as targets at precisely the moment their local area's economy will improve.[17]

First, Figure D.4 shows that consumer- and firm-related economic variables that could be associated with latent demand for deposits and loans appear parallel for merger and non-merger counties. In Appendix D.2.2, we take this even farther and run placebo tests on these outcomes, using the merger as a treatment on the outcomes. The argument for a parallel trends violation is that banks possess information not available to the econometrician that the county would perform better after the merger, and so the control used in stacked-difference-in-difference would be a biased counterfactual. Consequently, assuming the bank merger itself did not have a direct effect on such outcomes, the bank merger itself should not be predictive of economic improvements to the county. We find no positive effect of a county's first bank acquisition on economic measures or proxies for demand for deposits and loans. If anything, bank mergers are followed by declines in median household income and a transient spike in firm exits, and both effects are economically small. Hence, it is difficult to imagine that the causal, positive effect of mergers on bank profits could be overestimated because of targeting of mergers to geographic markets that about to fare particularly well.

### 3.4.2 SUTVA

Second, equilibrium logic threatens the assumption of SUTVA for the control banks. Banks in horizontal competition with the merging parties can be directly affected by the merger. For instance, the merging bank realizing cost synergies could charge lower prices, causing the merged entity to steal business from competitors in its market. On the other hand, a reduction in competition in a market could cause competitor banks to raise prices.

[17] For example, two plausible channels for a rapid post-period cost reduction that works through local geography if banks had remained standalone are 1) changes to the local customer demand that would allow bank to service them cheaper or 2) changes in technology available locally that would allow banks to offer the same services at lower cost.

The design of the control sample to be limited to banks outside of the merger state is valid under the assumption that the merger in the focal state has no equilibrium competition impact on banks outside of the state. That assumption seems reasonable based on the primary bank regulator – the Federal Reserve's – own published maps of deposit market boundaries used for merger evaluation being universally finer grained than the state level. Figure D.9 shows a map of Federal Reserve defined banking markets colored by their Herfindahl-Hirschman index. 92% of the U.S. population lived in a deposit banking markets with Herfindahl-Hirschman index below 1800 in 2022. Second, 77% of mergers occur between banks operating in the same state. Figure D.8 shows the statistic by U.S. state. Given mergers frequently cluster in this way, it would also be reasonable to infer that banks themselves see their markets as local within the state level.

# 4 Capital constrains bank mergers

This section presents evidence of a regulatory capital wedge (RCW) that reduces banks' utility of mergers, whereby binding capital regulation is incompatible with purchasing potential merger targets in cash. We proceed by explaining the mechanism then describing theoretical predictions that we test in data. We test both direct evidence of the mechanism and additional model predictions to validate our findings.

## 4.1 Mechanism

The illiquidity of equity stock for non-publicly traded firms is a well-known deterrent of financing new investment with newly issued equity. One salient rationale for the illiquidity is that asymmetric information between internal and potential external shareholders reduces their valuation of the stock and makes the market for new stock a market for lemons. However, in the case of M&A investment (Jovanovic and Rousseau, 2002), firms could theoretically overcome this failure of Modigliani and Miller (1958) by issuing new equity to their target's shareholders; financing the acquisition of the target in stock by directly paying the target in *stock*. Ostensibly, target shareholders could be more informed than other investors. They would see the value in the potential combination and would accept stock in the new firm as payment for their shares. But, if they do not, asymmetric information persists and attempting to issue equity to the target shareholders would be no better than attempting to issue outside shareholders. That bias falls particularly on buyers whose stock is not publicly traded, where the information gap would be the largest and the stock would be most illiquid currency for a potential target.

Firms could then, typically, overcome this additional friction by financing their investment in ways that increase firm leverage in order to pay their targets in *cash*. Two similar procedures are available. First, firms could procure new cash with debt. They could issue new debt, use that new cash to purchase the target and create a larger combined firm whose assets were supported by the same equity base plus far more debt. Alternatively, firms could use existing assets (cash) it has available to pay the target. In this way, firms cut into their existing asset base to pay the target, but

because fewer assets are now supported by the same amount of debt, the firm creates a smaller, but more levered, combined firm.

However, neither solution is readily available in industries when firm leverage is capped. When U.S. banks have already funded as many assets in deposits as they can and bind to their minimum equity capital ratio, all new investment must be equity financed. U.S. banks, who lever up to their minimum equity capital requirements by continually funding new loans with deposits, have little remaining wiggle room for cash- or debt-financing large investments such as the acquisition of a peer. Even if firms choose to lever to purchase their target, since the net effect of their transaction must keep them sufficiently capitalized, such debt- and cash-financed transactions could require re-raising equity to "recapitalize". We refer to this wedge between mergers one would expect to occur in stock than those that do in reality as the regulatory capital wedge (RCW).

## 4.2 Causal evidence of the Regulatory Capital Wedge

Bank regulatory capital ratios are computed differently after completed cash versus stock mergers, and that accounting process has varied over time. We test our mechanism with two instruments: 1) policy variation in how market values of bank assets are incorporated into one regulatory capital ratio – the Tier 1 Leverage Ratio and 2) variation in Treasury yields interacted with bank shares of long-term Treasury securities. The two approaches deliver quantitatively similar results: a 1 pp increase in regulatory capital ratios increase the 12-month probability of becoming a merger target by 15-20bp, or roughly 50%.

### 4.2.1 Cash mergers reduce the regulatory capital ratio of the combined bank

Bank equity capital is regulated to exceed several well-defined ratios. We will focus on the Tier 1 Leverage Ratio alone in our baseline specification. The Tier 1 Leverage Ratio must exceed 4% and is defined as

$$r_{it} \equiv \text{Tier 1 Leverage Ratio}_{it} = \frac{\text{Tier 1 Capital}_{it}}{\text{Average Total Consolidated Assets}_{it}} \equiv \frac{K_{it}}{A_{it}} \tag{5}$$

where Tier 1 Capital is reported on the Call Reports by banks but is almost entirely composed of common stock and retained earnings. Regulators (the Federal Reserve, OCC, and FDIC) require the maintenance of equity exceeding these ratios by promising immediate corrective action involving capping dividend payouts or mandating capital raises.

The Tier 1 Leverage Ratio disciplines mergers because the two methods of payment leave the pro forma bank at very different points relative to the 4% floor. Consider a buyer $i$ with Tier 1 capital $K_{it}$ and average total consolidated assets $A_{it}$ acquiring a target $j$ with $(K_{jt}, A_{jt})$, and normalize the purchase price to the target's book equity $K_{jt}$.[18]

[18]Normalizing the price to $K_{jt}$ therefore understates the regulatory cost of cash. Any premium above book, paid in cash, reduces both the numerator and denominator because Tier 1 capital deducts goodwill and other intangibles.

In a stock-for-stock merger, no cash leaves the pro forma bank and the seller exchanges her equity claim for the buyer's. Both banks' capital and assets survive the transaction:

$$R_{ijt}^{\text{Stock}} \equiv \frac{K_{it} + K_{jt}}{A_{it} + A_{jt}} \tag{6}$$

where we use uppercase $R_{ijt}$ to denote the *combined* bank's new capital ratio. The stock-financed ratio is simply the asset-weighted average of the two standalone ratios: acquiring a better-capitalized target *raises* the buyer's leverage ratio.

In a cash merger, the buyer pays $K_{jt}$ out of its own liquid assets and the target's equity is extinguished. Only the buyer's capital survives to support the combined asset base, and the combined asset base itself shrinks by the cash paid out:

$$R_{ijt}^{\text{Cash}} = \frac{K_{it}}{A_{it} + A_{jt} - K_{jt}}. \tag{7}$$

Comparing (6) and (7),

$$R_{ijt}^{\text{Stock}} - R_{ijt}^{\text{Cash}} \;\geq\; 0, \tag{8}$$

with strict inequality whenever the target has positive equity and the pro forma bank has any deposits or other debt. A cash merger therefore always leaves the pro forma bank closer to the regulatory minimum than the identical merger financed in stock, and the wedge is increasing in the size of the target and in the leverage of both banks. The comparison to the buyer's *standalone* ratio is equally stark: $R_{ijt}^{\text{Cash}} < r_{it} \equiv K_{it}/A_{it}$ whenever $A_{jt} > K_{jt}$, so a cash acquisition of any levered target mechanically consumes the buyer's capital buffer, while a stock acquisition need not.[19][20]

As a result, the simplest variation in RCW would simply be the distance a bank is from its regulatory equity minimum, where RCW is the largest when the distance is the smallest, $RCW \propto -(R_{ijt}^{\text{Cash}})$.

Quantifying the strength of RCW on banks introduces a risk of endogeneity. One risk of using contemporaneous Tier 1 Leverage Ratio as a proxy for a regulatory capital wedge is that, in a

[19] Consider a numerical example. A bank with $100 in assets and $10 in equity would like to purchase a bank with $20 in assets and $2 in equity. The buyer could pay management $2 in cash for their equity and the combined bank would have $118 in assets and $10 in equity. The transaction has increased leverage: it has reduced the Tier 1 Leverage Ratio from 10% to 8.5%. Only a stock-for-stock merger, where the seller willingly changes out her equity for the buyer's, retains a capital ratio of 10%.

[20] Equation (7) also caps how much a bank can buy without issuing equity. Writing $r_{jt} \equiv K_{jt}/A_{jt}$, the constraint $R_{ijt}^{\text{Cash}} \geq 4\%$ holds if banks choose a target smaller with ass,ets

$$A_{jt} \;\leq\; \frac{K_{it} - 0.04A_{it}}{0.04\,(1 - r_{jt})}. \tag{9}$$

In cash, acquisition capacity is proportional to the buyer's *excess* capital $K_{it} - 0.04A_{it}$: at a 4% floor and a target with a ratio of 10%, each dollar of excess Tier 1 capital supports roughly $28 of target assets. A buyer at 4% could only do a merger in stock.

dynamic sense, a banks could retain earnings and hold high Tier 1 Leverage Ratio *because* its cost of external equity issuance is large. Hence, banks could be very constrained but the correlation between capitalization and merger propensity could be negative. Another risk is confounding: more highly capitalized banks could be better managed banks. If acquirers were better managers, then a positive relationship between acquisition propensity and capitalization could be due entirely to productivity. Furthermore, the burden of proof is not just that capital structure predicts merger propensity – firm investment reasonably shifts with measures of how much debt or equity funds their assets. Instead, evidence of the RCW requires that regulatory capital ratios – not just the firm's capital structure – affects merger propensity. Consequently, the instrument will shift regulatory capital while leaving firm capital structures fixed.

#### 4.2.2 Instrument #1: The 2001 phase-out of Pooling of Interests Accounting

To isolate the causal effect of bank regulatory capital on merger propensity, we design an instrument for regulatory capital ratio using policy variation in the accounting of how market values of bank assets are incorporated into the target bank's regulatory ratio in the event of a merger.

**Book versus Market Values.** Importantly for the construction of our instrument, banks are regulated on the *book* value of their ratios. Consequently, the frequent changes in the *market* value of their assets are not used to compute bank regulatory capital minima. Although bank capital structure according to market values could determine its investment and M&A decisions through any number of violations of Modigliani and Miller (1958), independent variation in the book value of bank leverage holding market value fixed should indicate that the regulatory capital ratio, in and of itself, affects bank decisionmaking.

**Pooling of Interests Accounting Policy Change.** The 2001 elimination of Pooling of Interests (POI) accounting yields exactly this policy variation for bank merger targets. The change in accounting policy, if independent of the merger synergies achievable by banks before and after the policy change, changed how much bank regulatory capital constrained bank mergers, holding fixed, for a given merger, the market values of the banks' assets and liabilities and the method of financing the transaction. Banks filing Call Reports must follow U.S. GAAP, a requirement effective since March 31, 1997 (Federal Deposit Insurance Corporation, 1997). Under GAAP, before the change, mergers were accounted for using either Purchase accounting or Pooling of Interests (POI) accounting depending on the form of the transaction. FASB Statement #141 eliminated POI for business combinations within its scope initiated after June 30, 2001 (Financial Accounting Standards Board, 2001). While POI allowed firms to add up the book values of their assets and liabilities, Purchase accounting requires allocating the acquisition cost to the target firm's identifiable assets and liabilities based on their estimated fair (market) values at the acquisition date (Financial Accounting Standards Board, 2001).

Consequently, in 2002, mergers became a unique event where bank market values had be incorporated into bank regulatory capital ratios. The identification argument is that, after controls, pre- and post-2002 differences in sensitivity to regulatory capital shifted by accounting are independent of potential endogeneity of market values of bank assets and bank operations endogeneity of regulatory capital with merger synergies. The ability that the accounting policy provides in distinguishing any relationship of bank market values and M&A from the causal effect of regulatory capital on bank M&A will inform construction of the empirical strategy.

**Approximating post-merger capital pre- and post-POI.** We approximate the post-merger regulatory capital ratio using the available market values of a portion of bank assets – bank securities. In the post-2001 regime, true post-merger Tier 1 Leverage Ratio would be computed by marking of the entire book asset portfolio to market. However, HTM and AFS securities are the only components of the bank's portfolio for which the market value is observable. As a result, we construct an approximation for the post merger ratio incorporating HTM and AFS into Equation (6) as follows. For a given bank as a potential target $i$ in quarter $t$,

$$\tilde{r}_{it} \equiv \begin{cases} \frac{K_{it}+\text{Unmarked HTM/AFS assets}_{it}}{A_{it}+\text{Unmarked HTM/AFS assets}_{it}} & t > 2001 \\ \frac{K_{it}}{A_{it}} & t \leq 2001 \end{cases} \tag{10}$$

Hence, target with unrealized gains would bring a larger leverage ratio post-merger than their pre-merger regulatory ratio constructed using book values. In contrast, targets with unrealized losses would bring a smaller leverage ratio.

We calculate a measure of unrealized gains on these securities as a fraction of a bank's regulatory capital.

$$\text{Unrealized Gains}_{it} \equiv \frac{\text{Unmarked HTM/AFS assets}_{it}}{\text{Assets}_{it}}.$$

Dividing the numerator by total assets allows unrealized gains to measure the relative share of capital that is revealed specifically in a merger.

We then evaluate the importance of unrealized gains and post-merger leverage ratio on the extensive margin using this regression specification, for the panel 1994-2024 when fair-value reporting in the Call Reports began:

$$\begin{aligned} \mathbf{1}(\text{Acquired within four quarters}) = \alpha_i + \tau_t + \beta X_{it} + \gamma_{\text{Post 2001}} \left(\text{Unrealized Gains}\right)_{it} \mathbf{1}(t > 2001)_t \\ + \gamma_{\text{Pre-2001}} \left(\text{Unrealized Gains}\right)_{it} \mathbf{1}(t \leq 2001)_t + \varepsilon_{it} \end{aligned} \tag{11}$$

Table F.1 shows the results of an OLS regression of being acquired on unrealized gains of a seller, for $h = 4$. As expected, $\gamma_{\text{Post-2001}} > 0$. All mergers use Purchase accounting, and so every merger

marks target asset fair values to market. Further consistent with this being a causal association, the association is statistically insignificant prior to 2002 at the 95% confidence level. In the post-2001 regime, the effect of unrealized gains are also sizable, economically. The standard deviation post-2001 of unrealized gains was 1% of assets, in the full panel banks had a 2.1% chance of being targets. So, a standard deviation increase in unrealized gains shifted the likelihood of a bank becoming a merger target by 19bps (8.8%).

The results are robust to using other covariates that may shift the quality of the bank as a potential merger target, such as being a less efficient or lower return on assets bank with room to improve. Importantly, the regression coefficient on Tier 1 (Leverage) Ratio is negative: times when banks have more capital are *less* likely to be acquired in the next year. This shows why forthcoming causal identification to shift regulatory capital ratios independent of merger synergies is helpful for this setting.

**Instrumenting for post-merger Tier 1 Leverage Ratio** Next, the effect of unrealized gains can be translated into the causal effect of bank regulatory capital on merger propensity using unrealized gains as an instrument. The first and second stage of the model to identify the causal effect of regulatory capital on merger propensity is

We instrument the post-merger leverage ratio $\tilde{r}_{it}$ with unrealized gains in the post-2001 regime, for $h = 4$:

$$\tilde{r}_{it} = \pi \left(\text{Unrealized Gains}\right)_{it} \mathbf{1}(t > 2001)_t + W_{it}'\delta + \alpha_{i,\mathbf{1}(t>2001)} + \tau_{s(i),t} + v_{it}, \tag{12}$$

$$\mathbf{1}(i \text{ acquired between } (t, t+h]) = \beta\, \tilde{r}_{it} + W_{it}'\theta + \alpha_{i,\mathbf{1}(t>2001)} + \tau_{s(i),t} + \varepsilon_{it}, \tag{13}$$

where $\alpha_{i,\mathbf{1}(t>2001)}$ are bank $\times$ regime fixed effects, $\tau_{s(i),t}$ are size-quantile $\times$ quarter fixed effects, and $W_{it}$ contains return on assets, the inefficiency ratio and securities / assets. Standard errors are two-way clustered by bank and quarter.

The identification argument is that unrealized gains are relevant to the constructed regulatory capital and excluded from the outcome of becoming a merger target, except through regulatory capital. Relevance for this instrument is nearly mechanical. Exclusion requires more assumptions about the reasons banks are and are not targets. For unrealized gains to affect merger propensity only through the regulatory capital ratio, banks must not be more or less desirable targets for factors that also control why banks might have unrealized gains on securities, including duration exposure or interest rate risk. One mitigating force for this issue is controlling for bank fixed effects, granular size by time fixed effects, and the share of bank assets comprised of securities, where banks must have variation in duration or interest rate risk exposure conditional on the securities portfolio. The goal of interacting with $\mathbf{1}(t > 2001)_t$ is to limit the endogeneity stories to mechanisms that could reasonable vary over time. Banks can be selected on unrealized gains, but if that selection, even after controls, is constant in magnitude pre- and post-2001, then the policy variation identifies a causal effect of regulatory capital on merger propensity.

Table 2: Merger propensity versus Tier 1 Leverage Ratio; Pooling of Interests Accounting

| | $\tilde{r}_{it}$ (pp) | | $\mathbf{1}$(Acquired within 4 quarters) (bps) | | | |
|---|---|---|---|---|---|---|
| | First stage | | OLS | | 2SLS | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| $UG_{it} \times \mathbf{1}(t > 2001)_t$ | 1.159*** | 1.167*** | | | | |
| | (0.030) | (0.030) | | | | |
| $\tilde{r}_{it}$ (pp) | | | -4.738*** | -4.842*** | 15.737*** | 14.585*** |
| | | | (0.931) | (0.932) | (4.824) | (4.660) |
| Return on Assets (pp) | | -0.039 | | -3.375*** | | -2.727*** |
| | | (0.024) | | (1.013) | | (0.802) |
| Efficiency Ratio (pp) | | 0.000 | | 0.012 | | 0.006 |
| | | (0.000) | | (0.010) | | (0.005) |
| Securities / Assets (pp) | | 0.005 | | -2.141*** | | -2.111*** |
| | | (0.004) | | (0.506) | | (0.506) |
| +1 SD of $\tilde{r}_{it}$, post-2001 (bps) | | | | | +100.9 | +93.5 |
| Bank $\times$ (Post 2001) FE | X | X | X | X | X | X |
| Size $\times$ quarter-year FE | X | X | X | X | X | X |
| Observations | 804,409 | 804,409 | 804,409 | 804,409 | 804,409 | 804,409 |
| $R^2$ | 0.731 | 0.731 | 0.241 | 0.241 | | |

* p < 0.1, ** p < 0.05, *** p < 0.01

Note: This table shows the first stage and second stage of instrumenting for regulatory capital, with standard errors clustered at the bank-quarter-year level. Columns 1 and 2 shows the estimates of Equation (12), with and without covariates. Column 3 and 4 show the OLS regressions. Columns 5 and 6 show the second stage, Equation (13). Using the instrument for Tier 1 Leverage Ratio from the Pooling of Interests accounting change, a 1 standard deviation increase in unrealized gains increases the probability of being sold by 94bps, or 44% from baseline.

Table 2 shows the result of estimating Equations 12 and 13. The first stage is mechanically positive and extremely precise. The $R^2$ is not equal to 1 because there is independent variation in the level of $\tilde{r}_{it}$ that we intentionally do not wish to use in identifying a causal effect of regulatory capital on merger propensity. Between columns (3)-(4) and (5)-(6), the usefulness of the instrument is clear. In the cross-section, when banks are less capitalized, even controlling for bank and granular time fixed effects, banks are more likely to be acquired. Two stories could be that banks that are about to fail or banks with particularly strong deposit franchises could be particularly good merger targets. Importantly, the instrument flips this sign. Columns (5) implies that banks have a 15.7 bp increase in 12-month merger propensity for a 1pp increase in regulatory capital. Adding covariates to control for observables that could make banks a stronger merger target reduce the estimate to 14.6bps. With a baseline merger rate of 2.1%, the increase in both specifications is large.

#### 4.2.3 Instrument #2: Changes in Treasury yields

Since the wedge between book and market values of securities could reflect bank-time-specific securities management strategies, we design another instrument to validate the mechanism. This instrument shows that increasing bank leverage, as measured by their post-merger Tier 1 Leverage

Table 3: Merger propensity versus Tier 1 Leverage Ratio, using Bartik instrument

| | $\tilde{r}_{it}$ (pp) | | **1**(Acquired within 4 quarters) (bps) | | | |
|---|---|---|---|---|---|---|
| | First stage | | OLS | | 2SLS | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| $w_{i,t-8} \times \Delta y_t^{5\text{-}yr}$ | -2.207*** | -2.068*** | | | | |
| | (0.389) | (0.477) | | | | |
| $\tilde{r}_{it}$ (pp) | | | -4.174*** | -3.086*** | 17.132* | 21.554** |
| | | | (0.605) | (0.637) | (9.797) | (10.367) |
| Return on Assets (pp) | | 0.209** | | -4.941 | | -10.116* |
| | | (0.094) | | (3.127) | | (5.952) |
| Efficiency Ratio (pp) | | -0.000 | | 0.037 | | 0.042 |
| | | (0.000) | | (0.031) | | (0.035) |
| Securities / Assets (pp) | | 0.026*** | | -2.425*** | | -3.064*** |
| | | (0.005) | | (0.232) | | (0.389) |
| First-stage $F$ (Kleibergen–Paap) | 32.3 | 18.8 | | | | |
| +1 SD of $\tilde{r}_{it}$, post-2001 (bps) | | | | | +98.7 | +124.2 |
| Size × quarter-year FE | X | X | X | X | X | X |
| Observations | 786,373 | 786,373 | 786,373 | 786,373 | 786,373 | 786,373 |
| $R^2$ | 0.060 | 0.088 | 0.005 | 0.006 | | |

* p < 0.1, ** p < 0.05, *** p < 0.01

Note: This table shows the first stage and second stage of instrumenting for regulatory capital with the Bartik instrument, with standard errors clustered at the bank-quarter-year level. Columns 1 and 2 shows the estimates of Equation (14), with and without covariates. Column 3 and 4 show the OLS regressions. Columns 5 and 6 show the second stage, Equation (15). Using the instrument for Tier 1 Leverage Ratio built from yield curve changes and long-maturity securities, a 1 pp increase in regulatory capital increases the probability of being sold by 17.1bps. With covariates controlling for the bank's attractiveness as a merger target, this rises to 21.6bps. The estimate in Column 6 corresponds implies that a standard deviation increase in regulatory capital corresponds to a 1.2pp rise in the 1-year likelihood of acquisition (mean: 2.2%), or 57%.

Ratio, lowers the propensity for banks to become targets in the near-term.

While unrealized losses in aggregate could be reflective of poor choices made in the securities portfolio, changes in the prices of a bank's Treasury securities, which have no default risk, are driven entirely by movements in the Treasury yield curve. Importantly, while the shape of the Treasury yield curve has many potential economic explanations, the explanations like interest rate policy and borrowing conditions are typically considered outside of the firm's control. As a result, the instrument uses the interaction of a bank's exposure to the yield curve and changes in the yield curve as an instrument. Assuming exposure is exogenous, that creates a Bartik instrument for bank post-merger regulatory capital by the market value of bank assets. The two equation model instrumenting for $\tilde{r}_{it}$ is

$$\mathbf{1}(\text{Acquired within 4 quarters})_{it} = \tau \tilde{r}_{it} + \beta X_{it} + \varepsilon_{it} \tag{14}$$

$$\tilde{r}_{it} = \gamma W_{it-8} \times \Delta y_t^{5-year} + \beta X_{it} + u_{it} \tag{15}$$

Table 3 shows the estimates for Equation (14). The first stage is weaker than instrumenting for unrealized gains, understandably, due to the larger number of confounders discussed above that determine equilibrium Tier 1 Leverage Ratio choice, but the F statistic of 32 is still large. Despite the additional noise, The covariates strengthen the effect such that a 1 pp increase in post-merger Tier 1 Leverage Ratio corresponds to a 21.6bp increase in probability of acquisition. However, given a standard deviation of leverage ratio of 5.76pp, the economic magnitude of the estimate grows. The estimates imply that a standard deviation increase in $\tilde{r}_{it}$ raises the 1-year likelihood of being acquired by 46%. The effect is larger when adding covariates intended to proxy for the attractiveness of the bank as a merger target (57%).

The endogeneity concern in this design is a mechanism by which exposure to the yield curve and the desirability of a bank as a merger target are intertwined, conditional on controls. If the yield curves changes were predictable, than exposure two years prior to merger to the current period would be endogenous – managers could optimize the maturity of their Treasury security portfolio holdings to ensure that their securities had the highest value possible. Having knowledge of forward looking interest rate policy is not enough for the changes in yield to be predictable, because the current yield curve itself changes immediately upon announcement of the future interest rate policy. So, arguments that bank managers could have predicted changes require additional skill or information, and for that skill to threaten the regression, it would have to be systematically to the quality of the bank as a merger target for an acquirer.

## 4.3 Additional predictions of RCW

We have shown directly that higher capital ratio banks have a higher propensity to become merger targets. The presence of RCW is also consistent with the use of cash for mergers and that realized mergers create lower profits. Appendix E presents a model with derivations of each prediction, and Appendix E shows empirical test of each prediction.

### 4.3.1 Validating results with $R_{ijt}$, the regulatory capital of merger pairs

Despite the potential endogeneity of capitalization the desirability of a bank as a merger target or acquirer, correlations in the cross-section of merger *pairs*, more highly capitalized pairs are more likely to combine.

To show this fact, we construct potential buyer-seller pairs as our unit of observation. The potential buyer-seller pairs for 4,000 commercial banks just in 2024 is nearly 16 million ordered pairs. In order to keep the analysis tractable, we limit combinations to pairs headquartered in the same state and where the seller is smaller than the buyer, as we will in the empirical model in Section 5. Constructing observations in this way takes into account how constrained a particular merger pair is. The merger pair can be constrained because of the buyer's standalone leverage ratio, the seller's standalone leverage ratio, *or*, the relative size of the seller to the buyer.

Consistent with our mechanism, Figure F.1 shows that the selection of the pairing is increasing

in the capital that the target would have post-merger. Figure F.1 separates combinations by the combinations value of $R_{ijt}^{Cash}$, then whether the combination has a privately held or a publicly traded buyer.

### 4.3.2 Intensive margin variation in merger profitability

In this section, we show evidence of RCW by documenting a gradient in merger profitability. Profitability is highest for the most constrained mergers. In doing so, we show that the capital constraint on bank mergers has meaningful economic effects, which we will have a framework to quantify formally in Section 5.

The argument for finding a gradient is as follows. If RCW were indeed a shadow cost of mergers, then mergers transacted despite a high RCW should provide larger gains to the merging parties than mergers transacted with a lower RCW. So, given Tier 1 Leverage Ratio, a proxy for RCW, mergers should be more profitable when the combining firm has a lower Tier 1 Leverage Ratio.

Our results align with this prediction. Figure F.3 reports the coefficients $\beta_s$ from Equation (1) for two-years after merger for bank mergers in our sample, interacted with the quantile of pro forma Tier 1 ratio two years prior to combination. We find that profits increased for pro forma banks with the lowest quantile Tier 1 leverage ratio by 17%. Increases were smaller as the combined bank was more capitalized and as low as 2% for combinations that were the most capitalized prior to the transaction. The gradient in merger synergies between the lowest capitalized banks and highest capitalized banks is consistent with our mechanism. While the figure provides an intuitive proxy for the regulatory cost on the x-axis, a positive association between merger synergies between banks with regulatory cost could contribute to the trend and justifies the Bartik instrument design we presented above.

Together, the evidence is consistent with these regulations being a meaningful deterrent to high value mergers by low capitalized banks. However, the evidence presented does not provide machinery for aggregating up the effect of changing such regulatory costs. To quantify the aggregate effect and account for heterogeneity between banks and heterogeneous choices, we present a structural model of endogenous mergers subject to such frictions. The model will deliver estimates for merger synergies created heterogeneously by different pairs and subject to capital regulation that can match moments like those presented in Figure F.3.

### 4.3.3 Cash merger preference of privately traded firms

We show that within realized mergers, the use of cash as payment is larger when a proxy for the cost of external equity issuance is larger. The easiest proxy is whether one's stock is publicly or privately traded. The liquid method of payment for acquisitions theory of going public (Brau and Fawcett, 2006) makes the claim that public and privates face this difference.

To ascertain the transaction medium of bank mergers, we turn to the reported mergers available in S&P SNL. This bank merger dataset is not exhaustive in its coverage of bank mergers, as it relies

on public announcements. Availability of RSSD numbers that could match the dataset to the Call Reports is also imperfect. As a result, the analysis is limited to 6,757 bank mergers occurring between 1991 and 2024, but not only commercial bank mergers defined in the same way as in Section 5.7.

Methods of payment reported since 1991 show that indeed, publicly traded firms are more likely to use stock. Figure F.2 shows the breakdown of mergers by financing medium and shows that while 62% of deals by private buyers are done exclusively in cash, only 25% of deals done by publicly traded buyers are. The difference is highly statistically significant. We can show this in a deal-level regression on percent of the transaction total transacted in cash. The regression we run, for deals $i$ closed in years $t(i)$,

$$\text{(Percent of deal consideration in cash)}_{it(i)} = \tau_t + \beta \mathbf{1}\text{(Private buyer)}_{it(i)} + \varepsilon_{it} \tag{16}$$

Table F.3 shows the relative frequency of cash by estimating the regression. Being a private buyer is associated with funding more than 30pp more of the deal value in cash. The gap between public and private buyers is robust to time fixed effects, the size of the buyer (measured in number of branches), and the size of the target.

# 5 Empirical model

In order to simulate the effects of counterfactual policy interventions, we develop and structurally estimate a model of the bank merger market. Vertically- and horizontally-differentiated banks compete à la Bertrand-Nash in both the deposit and loan markets, incurring heterogeneous fixed and variable operating costs. Each period, prior to product market competition, there is a "merger market": banks can bid in simultaneous auctions to buy another bank in their market and/or buy themselves and continue operating in their current capacity. Business combinations may produce profit synergies, but combinations also incur integration costs. Importantly, but complicating the model significantly, we account for selection on levels of merging versus nonmerging banks. In doing so, we estimate parameters of the model while limiting the bias that could arise from selection of merging banks' innovations in fixed costs separate from mergers.

## 5.1 Setup

At the beginning of a period $t$ in market $k$, there exist $\mathcal{J}_{kt}$ banks. Each market is a U.S. state, and each U.S. state has a deposit market and a loan market. The markets have sizes $(Q^D_{kt}, Q^L_{kt})$, The *ex-ante* state vector (i.e., prior to any merger activity) is given by:

$$\boldsymbol{\Omega}_{kt} \equiv (\boldsymbol{A}_{kt}, \boldsymbol{B}_{kt}, \boldsymbol{K}_{kt}, \boldsymbol{\xi}^D_{kt}, \boldsymbol{\xi}^L_{kt}, \boldsymbol{c}^D_{kt}, \boldsymbol{c}^L_{kt}, \boldsymbol{\kappa}_{kt})$$

where $\boldsymbol{A}_{kt}$ is the vector of bank assets in market $k$ at the beginning of year $t$, $\boldsymbol{B}_{kt}$ is the branch network of the banks, and $\boldsymbol{K}_{kt}$ is the vector if Tier 1 capital of every bank. We suppress market subscripts $k$ for ease of exposition.

**Bank profit fundamentals.** We define the bank flow profit function with several parameters in order to accommodate banks targeting cost synergies, market power, or both in their choice of merger partner. Banks indexed by $j$ offer two vertically- and horizontally-differentiated products, loans and deposits, indexed by $d = \{L, D\}$. Banks earn profits from lending at a spread above the risk-free rate, $\rho^L_{jt} \equiv r^L_{jt} - R^F_t$, and borrowing at a spread below the risk-free rate, $\rho^D_{jt} \equiv R^F_t - r^D_{jt}$. Vertical differences stem from non-price product quality ($\xi^d_{jt}$) and number of physical branch locations $B_{jt}$. As a result, they face demand for their products, market shares $s^d_{jt}(.)$, that are functions of their prices, quality differences, branches, and the characteristics of their competitors. For clarity, we will denote $s^d_{jt}(\rho^d, \xi^d, B_{jt}, \boldsymbol{\Omega}_t)$, where the inclusion of the first three terms is technically redundant. Banks incur heterogeneous marginal costs in servicing their deposits and loans ($c^d_{jt}$), as well as heterogeneous fixed costs of operating, $\kappa_{jt}$. Operating profits are thus given by:

$$\pi_j(\boldsymbol{\Omega}_t) \equiv Q^L_t s^L(\rho^L_{jt}, \xi^L_{jt}, B_{jt}, \boldsymbol{\Omega}_t)(\rho^L_{jt} - c^L_{jt}) + Q^D_t s^D(\rho^D_{jt}, \xi^D_{jt}, B_{jt}, \boldsymbol{\Omega}_t)(\rho^D_{jt} - c^D_{jt}) - \kappa_{jt} \tag{17}$$

**Merger technology.** With this definition of profit, mergers can produce profit synergies by realizing synergies in bank fundamentals. Namely, the merger can increase quality, decrease marginal costs, or decrease fixed costs. We will assume only fixed cost synergies are created in our baseline results. So, if a bank $i$ purchases bank $j$, an event we denote by $\mathcal{M}(i,j)$, their fundamentals evolve as:

$$\kappa_{\mathcal{M}(i,j),t} = \kappa_{it} + \kappa_{jt} + A_{jt}\left[a^\kappa \mathcal{O}(B_{it}, B_{jt}) - \zeta_{ijt}\right], \qquad \zeta_{ijt} \sim \text{T1EV}(0, \sigma_\zeta) \tag{18}$$

$$c^d_{\mathcal{M}(i,j),t} = \frac{s_{it}c^d_{it} + s_{jt}c^d_{jt}}{s_{it} + s_{jt}} \tag{19}$$

The fixed cost expression includes one pair-specific unobservable, $\zeta_{ijt}$. This pair-specific unobservable scales with seller assets. The shock, $\zeta_{ijt}$ follows a Type-1 Extreme Value distribution.

Finally, quality $\xi^d_{\mathcal{M}(i,j)}$ is set so that, absent depositors or borrowers facing new prices due to the merger, the market share of the resulting firm would be equal to sum of the predecessors. Mathematically, the assumption is,

$$s^d\left(\frac{s^d_{it}\rho^d_{it} + s^d_{jt}\rho^d_{jt}}{s^d_{it} + s^d_{jt}}, \xi^d_{\mathcal{M}(i,j)t}, B_{\mathcal{M}(i,j)t}, \boldsymbol{\Omega}_t\right) = s^d_{it} + s^d_{jt} \tag{20}$$

where branches $B_{\mathcal{M}(i,j)t} = B_{it} + B_{jt}$. In this way, holding prices fixed, consumers do not have higher demand for the combined firm, marginal costs are not lower, but fixed costs can decrease via branch overlap $\mathcal{O}(.)$, and a pair-specific synergy between the acquirer and target.

**Selection.** To correct for potential selection on levels between merged and unmerged banks, we account for the standalone $\psi_{it}$ of merged banks to be unobserved. Bank fundamentals $\psi \equiv (\xi_{it}^D, \xi_{it}^L, c_{it}^D, c_{it}^L, \frac{\kappa_{it}}{A_{it}})$ are parameterized as functions of their lagged values a time-specific intercept, and mean-zero, unobservable shocks that are i.i.d. Gaussian across banks, markets, and time. So,

$$\psi_{it} = m_t^{\psi} + \eta^{\psi}\psi_{it-1} + \tilde{\psi}, \qquad \tilde{\psi}_{it} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}). \tag{21}$$

where $\mathbf{\Sigma}$ is a diagonal matrix encoding the variances of the unobserved shocks determining profits and $m_t^{\psi}$ is an intercept that varies over time.

**Timing.** The timing of the game for a period $t$ is as follows.

0. Bank fundamentals $\boldsymbol{\psi}_t$ and merger-payoff shocks $\boldsymbol{\zeta}_t$ are drawn from their distributions.
1. Banks place simultaneous bids on themselves and other banks.
2. Remaining banks post loan and deposit rates.
3. Depositors and borrowers choose banks, and both markets clear. Time progresses to period $t+1$.

The remainder of this section describes the model step by step.

### 5.2 Step 0: Information

All banks, as well as the econometrician, observe the realizations of $(\boldsymbol{A}_{kt}, \boldsymbol{B}_{kt}, \boldsymbol{K}_{kt})$. All banks, but not the econometrician, observe $\boldsymbol{\psi}_t$, the shocks to fundamentals.

### 5.3 Step 1: Merger formation

The expected surplus to merging is the change in the buyer $i$ and the seller $j$'s sum of the discounted present value of utility induced by a merger in period $t$. Denoting the present values of utility by $U_j$:

$$\mathbb{E}[U_{\mathcal{M}(i,j)} - U_i - U_j \mid \mathbf{\Omega}_t] = \underbrace{\mathbb{E}[\sum_{s=0}^{\infty} \beta^s \Delta\pi_{\mathcal{M}(i,j)} \mid \mathbf{\Omega}_t]}_{\text{Merger surplus}} \underbrace{-\mathcal{C}^B(\frac{K_{it}}{A_{it} + A_{jt} - K_{jt}}) - \tau A_{jt}}_{\text{Merger frictions}}. \tag{22}$$

**Merger surplus.** $\Delta\pi(\cdot)$ captures the difference in profits under the newly-combined bank relative to the sum of utilities of the two stand-alone banks in a counterfactual in which they do not merge. The precise function is given by Equations (18)-(19) through Equation (17). The expectation is taken with respect to the bank's information set, which, in the model, is the full market's $\mathbf{\Omega}_t$.

**Merger frictions.** Equation (22) makes explicit that merger surplus will be decreasing in merger frictions: the seller's reserve value and the one-time costs of the merger. The one-time costs of the merger will be the focus of RCW, presented in Section **??**.

The seller's reserve value is derived from the present value of the utility of a bank's owner. This quantity is the sum of the bank's expected future profits.

$$\mathbb{E}[U_j \mid \mathbf{\Omega}_t] \equiv \mathbb{E}[\sum_{s=0}^{\infty} \beta^s \pi_{js} \mid \mathbf{\Omega}_t] \tag{23}$$

where $\beta$ parametrizes the rate of time preference. One possible source for transaction costs $\tau$ is a raising of the seller's reserve price from the benefits of keeping the bank "in the family." [21]

The one-time costs of the merger holds the bank's financing friction and a transaction cost $\tau$. $\mathcal{C}^B$is a function of the Tier 1 Leverage ratio of the combined firm, if the buyer were to pay cash for the seller.

**Simultaneous auctions.** Each bank can sell to a bank that is larger, in assets, within its market. Each bank can be sold once, to one another bank, but each bank can potentially buy multiple of its competitors in a given period. Equivalently, the choice set of a seller $j$ of buyers is the set of banks in the market (self-inclusive) who have at least as many assets as seller $j$, which we symbolize with $\mathcal{B}_{jt} \subseteq \mathcal{J}_t$. In one microfoundation, the allocation of selling banks to buying banks happens via simultaneous auctions for every bank. (Appendix G.2 provides additional microfoundations.) Every bank sets a reserve price for itself by bidding on itself, then bids on every smaller bank within its market. Banks sit out of the merger market by "selling" to themselves, realizing no merger synergies but paying no merger frictions. This means that for a market of size $\mathcal{J}_t$ in a given year, $\frac{\mathcal{J}_t(\mathcal{J}_t-1)}{2}$ potential mergers could occur. We now describe how this choice occurs.

Both to capture industry practices regarding the projection of merger returns better and to greatly simplify computational burden, we adopt a reduced-form approach to characterizing merger formation. We assume that the buyers bids are formed as in a delegated agent model. In particular, a given bid is constructed as if no other mergers will occur in the pair's market and as if no mergers will occur in the future, and that primitives after the merger will remain fixed. That assumption still allows the quantity $\Delta\pi_{\mathcal{M}(i,j)}$ to take into account that the combined bank from buyer and seller, has additional market power to realize cost synergies using Equations (18)-(20) and reprice deposits and loans with additional market power. We define the surplus quantity for a given pair $i, j$ as $\Phi_{\mathcal{M}(i,j)}(\mathbf{\Omega}_t)$.[22] Given the data generating process specified, the sum of future profits can also

[21] This is not separately identified from a one-time cost of performing an acquisition, where purchasing a larger target incurs larger one-time costs of hiring supporting services to assess and implement the integration. Whether this cost that scales with seller assets is related to the buyer or seller side is not separately identified Agarwal, Li, and Somaini (Agarwal et al.) and so we will not interpret it. This is another advantage of our model setup that does not require observing prices in order to estimate parameters.

[22] In general, true expected surplus with rational expectations and the expected surplus used in this model for choices need not be equivalent. The realization of a merger between $i, j$ could affect the surplus of another potential merger between $\ell, m$, $\mathbb{E}_{-\mathcal{M}(\ell,m)}[\frac{\Delta u(\mathbf{\Omega}_t)}{1-\beta}]$. For instance, one consequence of such an externality is *free-riding*: that mergers

be simplified into a discounted sum. As a result, a given bid by buyer $i$ for seller $j$ will be equal to:

$$\Phi_{\mathcal{M}(i,j),}(\mathbf{\Omega}_t) \equiv \begin{cases} \frac{\zeta_{ijt} A_{jt}}{1-\beta} & i = j \\ \frac{\Delta\pi_{\mathcal{M}(i,j)}(\mathbf{\Omega}_t)}{1-\beta} - \mathcal{C}^B(\frac{K_{it}}{A_{it}+A_{jt}-K_{jt}}) - \tau A_{jt}. & i \neq j \end{cases} \tag{24}$$

In other words, $\zeta_{ijt}$ can reduce the fixed cost of the combined firm if $ij$ choose to merge. Otherwise, $\zeta_{ijt}$ is a shock to the surplus of a "self-acquire."

Formally, for a hypothetical merger between $i$ and $j$, $\mathcal{M}(i,j)$, the winning buyer is:

$$i = \arg \max_{\ell \in \mathcal{B}_{jt} \cup \{j\}} \Phi_{\mathcal{M}(\ell,j)}(\mathbf{\Omega}_t) \tag{25}$$

Despite our simplifying assumption that allocation uses myopic surplus, Equation (24) is still a complicated, nonlinear function of all banks' characteristics, primitives, and equilibrium strategies of acquisition and sale. For example, the computation of profit synergies $\Delta\pi$ requires performing a merger simulation of the surplus created for a seller selling to each potential buyer – computing post-merger quality, marginal cost, and fixed cost for every partner and solving the fixed point problem of Bertrand-Nash pricing given competitor prices for every partner. In assuming a mechanism that matches on highest surplus, we can use allocation to identify parameters without taking a stance on equilibrium prices, which are often unobserved.[23]

### 5.4 Step 2: Price-setting in deposit and lending markets

After the merger market has cleared, variables for banks that merged are updated to their combined values. Then, banks set prices to maximize profits. For deposits and loans, this price is the spread – the amount above the risk free rate that banks are willing to lend at, and the amount below the risk free rate that banks are willing to offer depositors. Banks set prices in the deposit and loan markets ($d \in \{D, L\}$) following the standard Bertrand-Nash pricing equation:

$$\rho^d_{jt} = \arg\max_r \left\{ (r - c_{jt}) M^d_t s^d_{jt}(r, B_{jt}, \xi^d_{jt}, \mathbf{\Omega}_t) \right\}. \tag{26}$$

by other pairs in your market that realize sufficiently limited efficiency gains encourage firms remaining independent because the smaller number of firms increases market power and hence increases the value of remaining as a standalone firm (Stigler, 1968; Gowrisankaran and Holmes, 2004), Such free-riding incentives shrink with the number of firms in the market. Given the competitiveness of the banking sector and the importance of cost efficiencies in merging established, our modeled merger surplus excludes this force for tractability.

[23] This centralized allocation can also be micro-founded by separate ascending auctions, separate second price auctions, or a centralized planner, as we discuss here. In the auction setup, banks must send bids for each other bank separately, without taking into account how their valuation of two banks may be larger (or smaller) than twice their valuation for holding each bank separately. We discuss the observational equivalence of these microfoundations in Appendix G.2.

### 5.5 Step 3: Depositor and borrowers select banks

We will now make explicit that demand for deposits and loans $(s_{jt}^D, s_{jt}^L)$ come from the utility maximizing decisions of consumers. Deposit and lending markets are structured identically; the exposition here, laid out in terms of the deposit market, extends without alteration to borrowers in the lending market.

Households demand deposit accounts and can choose one bank $j$ from the remaining set $\mathcal{J}_t$ available in market $k$ to deposit their funds. Alternatively, as an outside option, households can elect to place their funds in an investment vehicle paying the market return (e.g., a money market fund). Depositors derive utility from promised deposit rates and other bank services. The indirect utility derived by depositor $i$ from bank $j$ in market $k$ at time $t$ is given by:

$$u_{ijt}^d = \alpha^d \rho_{jt} + \beta^d \log(B_{jt}) + \xi_{jt}^d + \varepsilon_{ijt}^d. \tag{27}$$

The utility of the outside option is normalized to $0$. Without loss of generality and for ease of exposition, we write the utility of a depositor in terms of deposit spreads ($\rho_{jt}$) rather than deposit rates. The spread then reflects the "price" that depositors pay for banking services. The parameter $\alpha < 0$ measures the disutility depositors experience from paying higher spreads (i.e., prices).

Depositors also derive utility from banking services not captured by the price, captured by $\beta \log(B_{jt}) + \xi_{jt} + \varepsilon_{ijkt}$. We use the logarithm of number of branches as the product characteristic. The term $\xi_{jt}$ appears in the merger market: it reflects additional unobserved quality differences at the bank-by-market-by-time level. For example, Bank of America may offer superior services relative to Citibank in a specific market, for instance due its online user interface. The term $\varepsilon_{ijkt}$ captures horizontal differentiation in the marketplace.

All depositors select the bank that maximizes their indirect utility. Following the demand estimation literature, we assume that the depositor-bank-market-time specific utility shocks ($\varepsilon_{ijkt}$) are distributed as Type-1 Extreme Value (Berry, 1994; Berry et al., 1995). This depositor optimization implies that the market share of bank $j$ in market $k$ at time $t$ (i.e., the probability a depositor selects bank $j$ in market $k$ at time $t$) is given by:

$$s^d(\rho_{jt}^d, B_{jt}, \mathbf{\Omega}_t^d) = \frac{\exp(\alpha^d \rho_{jt}^d + \beta^d \log(B_{jt}) + \xi_{jt}^d)}{1 + \sum_{\ell \in \mathcal{J}_t} \exp(\alpha^d \rho_{\ell t}^d + \beta^d \log(B_{\ell t}) + \xi_{\ell t}^d)} \tag{28}$$

### 5.6 Solution concept

The allocation of banks to merger partners and subsequent prices and allocations of deposits and loans to consumers proceeds as follows.

**Step 1: Merger formation.** Bank ownerships are reassigned to the highest bidders. Denote the resulting assignments of every bank $j$ over identities of buyers $i$ with a matrix $\boldsymbol{P}_{kt}$ (of dimension $\mathcal{J}_{kt} \times \mathcal{J}_{kt}$), where for bank $j$ with a maximum surplus buyer $i$, $P_{ij,kt} = 1$ and $P_{\ell j,kt} = 0$ for all other

banks in the market. In the case where a bank $i$ wins an auction for a bank $j$ but is also purchased by a bank $\ell$, ownership of $i$ and $j$ are both transferred to $\ell$. That clearing of "chains" continues until each bank is re-assigned to a buyer that is standalone at the conclusion of the merger market. That transformation is equivalent to raising $\boldsymbol{P}_{kt}$ to the $\mathcal{J}_{kt} - 1$ power. Denote the final assignments, the ownership matrix, by $\boldsymbol{M}_{kt} \equiv \boldsymbol{P}_{kt}^{\mathcal{J}_{kt}-1}$.

**Step 2: Price-setting.** Remaining standalone banks compete horizontally within market Bertrand-Nash and accordingly set profit-maximizing interest rates for deposits and loans.

**Step 3: Depositors and borrowers select banks.** Depositors and borrowers choose banks to offer deposit accounts and loans by maximizing utility.

This allocation is unique. The statement is formalized in Theorem 1 with a proof in Appendix G.2.

**Theorem 1.** *With bank merger choices Equation* (25)*, price setting Equation* (26)*, and consumer choice Equation* (27) *with primitives* $\boldsymbol{\Omega}_t$*, there is a unique matching (allocation) of sellers to buyers* $\boldsymbol{M}$*, surviving banks* $\mathcal{J}'_{kt}$*, and set of prices* $\boldsymbol{\rho}^D, \boldsymbol{\rho}^L$*.*

In Appendix G.2, we present a simultaneous auction microfoundation. We also prove that the uniqueness from Equation (25) can be recovered from the auction microfoundation of the merger market.

### 5.7 Discussion of model set-up and assumptions

Our approach offers two clear advantages for empirical work. First, this assumption of merger surplus from Equation (24) is realistic how firms consider the return on investment of M&A. In their quantitative assessment of targets, acquirers compute the return on investment of a potential acquisition by computing the increase in cash flows that would be created by the acquisition. However, they are typically done under the assumption that no competitors also take simultaneous activities and do not build in potential complementarities or substitutions arising from the acquirers future acquisition plans, if those plans occur after the focal target evaluation.

Second, this modeling choice is parsimonious. We exposit the model using simultaneous ascending auctions, but at least three extensive form games can microfound the resulting allocation. Rather than choosing an extensive form game whose details we do not observe and may vary across merger negotiations, in assuming that sellers are matched to their highest surplus buyer (before resolving chains), we impose that assignment of sellers to buyers efficient given the assumptions about surplus established (Shapley and Shubik, 1972; Chiappori et al., 2012). In the two-sided case, the allocation would be the result of the "competitive assignment game." Limiting how externalities can affect merger choices greatly simplifies the model, but through modeling merger

synergies, a bank merger still generates equilibrium spillovers on other banks through product market competition with new $\xi^D, c^D, \kappa$.

Lastly, complementary to the difference-in-difference evidence in Section , our model explicitly accounts for some econometric endogeneity of mergers. There can be *selection on gains*, or "essential heterogeneity," in the spirit of Roy (1951): banks are more likely to pursue mergers that are more profitable. Difference-in-differences identifies the average effect on the treated, and extrapolating this estimate to become the average treatment effect for *any* bank pair would be to assume away selection on gains. To capture selection on gains, we allow merger synergies between prospective bank pairings to differ on both observables as well as an unobservable shock that is additively separable in the profit function and independent of all other shocks.[24] In this way, if more mergers occurred, the fixed cost synergies of the marginal merger would decline, and since mergers created less consumer choice, the detriment of the marginal merger consumers would increase. We turn now to estimation of the model.

# 6 Estimation

Estimation first uses the approach of Berry (1994) to estimate consumer demand and parameters of bank profit functions. Then, the central econometric challenge is the joint endogeneity of merger decisions with unobserved shocks to bank fundamentals (selection on levels) and to pair-specific synergies (selection on gains). We address both by constructing the joint likelihood of merger outcomes and post-merger primitives under the model's distributional assumptions, and integrating over the unobservables that drive selection. In this way, banks select merger partners in a way akin to product choice with logit demand and random coefficients. That "first stage" selection of merger partners can then be corrected for in the "second stage" in a manner similar to a Heckman (1979) selection correction.

## 6.1 Step 3: Depositors and borrowers select banks

The selection of depositors and borrowers into banks identifies their demand parameters, namely the price elasticities, their demand for branches, and the latent quality of each firm. As before, the deposit and loan markets proceed analogously. In order to account for the endogeneity of price and the quality of a given bank's deposits or loans, we instrument $\rho^d_{jkt}$ for $d = \{D, L\}$ with operating cost intensity, defined as 1-year lagged fixed expenses as a share of bank assets. The two equation model becomes:

$$\log s^d_{jkt} = \alpha^d \rho^d_{jkt} + \beta^d \log(B_{jkt}) + \nu^{d,2}_{kt} + \varepsilon^d_{jkt} \tag{29}$$

$$\rho^d_{jkt} = \gamma^d \text{Operating cost intensity}_{jkt} + \nu^{d,1}_{kt} + \varepsilon^d_{ijt} \tag{30}$$

[24] It is possible to allow for more general forms of selection on gains, though at greatly-increased computational cost in estimation – we discuss the contours of such an approach in the estimation section.

for market and time fixed effects $\nu_{kt}^{d,1}, \nu_{kt}^{d,2}$, where $\nu_{kt}^{d,2}$ absorbs the outside option. The assumption is that this rolling average is excluded from latent demand and so can act as an instrument for $\rho$ and can identify $\alpha^d, \beta^d$. Next, inverting the multinomial-logit demand system at observed shares and prices from Equation (28) yields the unobserved quality residuals for each bank $j$ in market $k$ at year $t$.

$$\xi_{jkt}^d \;=\; \log s_{jkt}^d - \alpha\rho_{jkt}^d - \beta^d \log(B_{jkt}) - \nu_{kt}^{d,2}, \qquad d \in \{D, L\}. \tag{31}$$

### 6.2 Step 2: Price-setting in deposit and lending markets

Given $(\alpha^d, \beta^d, \xi_{jkt}^d)$, the Bertrand-Nash first-order condition in Equation (26) gives variable costs for bank $j$ at year $t$ in market $k$ as a function of data and parameters:

$$c_{jkt}^d \;=\; \rho_{jkt}^d + \frac{1}{\alpha^d\big(1 - s_{jkt}^d\big)}. \tag{32}$$

Finally, the net fixed cost is the residual between observed total profits and the sum of model-implied variable profits:

$$\kappa_{jkt} \;=\; Q_{kt}^D s_{jkt}^D\big(\rho_{jkt}^D - c_{jkt}^D\big) + Q_{kt}^L s_{jkt}^L\big(\rho_{jkt}^L - c_{jkt}^L\big) - \pi_{jkt}^{\text{obs}}. \tag{33}$$

The parameters from Steps 2 and 3 are then used as data in Step 1.[25]

### 6.3 Step 1: Merger market

The core economic importance of the likelihood function is selection of the merged pairs on gains and levels. In estimating the fixed cost synergies created by realized mergers and hence extrapolating from estimates to counterfactual changes in industry structure merged banks, accounting for the selection of the merged pairs makes extrapolation credible. Without selection on gains, extrapolating the effect of realized mergers would predict too large an increase in bank efficiency for yet unrealized mergers through pair-specific synergies that made the merger attractive. Without selection on levels of banks into their optimal merging partners, the recovered AR innovations would be unconditionally distributed as $\mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$ – but observing that bank $j$ merged is a selection event, so the conditional distribution of the shocks given the realized assignment in $(k, t)$ is mean-shifted and would bias all model estimates. To solve this problem, we absorb this shift with a Mills correction analogous to the classical inverse-Mills correction (Heckman, 1979). For merger pairs, we observe only the post-merger combination $\kappa_{\mathcal{M}(i,j),kt}$, not the standalone $(\kappa_{ikt}, \kappa_{jkt})$; the residual after subtracting the deterministic prediction and a control-function shift. Refer to the vector of these shocks for merged firms as $\tilde{\boldsymbol{\kappa}}_{kt}^m$, separately from the realized period-t vector of unmerged

[25] In doing so, we assume that the demand parameters are not subject to large selection created by the positively censored distribution of demand parameters of surviving banks. That issue is dealt with in Ciliberto and Tamer (2009) by jointly estimating demand and the "selection equation," which, in our case is the merger market assignment of sellers to buyers. That exercise could show the robustness of the demand parameter estimates.

banks' fixed costs, $\boldsymbol{\kappa}^u_{kt}$, which can be taken as given.

Before proceeding, it will be useful to denote some of the bank-market-time observables with $\boldsymbol{y}_{kt}$, where $\boldsymbol{y}_{kt} \equiv (\boldsymbol{A}_{kt}, \boldsymbol{B}_{kt}, \boldsymbol{K}_{kt}, \boldsymbol{\Omega}_{kt-1})$. Under the model's distributional assumptions, the joint density of observed primitives and merger assignments factors into an *outcome-stage* term and a *selection-stage* term. For estimation, we will use just fixed cost primitives for the estimation of the likelihood, as the merger synergies are concentrated on fixed costs alone. Defining the likelihood $\mathcal{L}$ as the probability of that data given a value of the parameters $\theta$, this likelihood is,

$$\begin{aligned}
\mathcal{L}(\theta) &= \prod_{k,t} \underbrace{\Pr(\boldsymbol{M}_{kt} \mid \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta)}_{L^{sel}_{kt}(\boldsymbol{M}_{kt}|\theta)} \underbrace{f_{\kappa^m|M,\kappa^u}(\kappa^m_{kt}|\boldsymbol{M}_{kt}, \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta) f_{\kappa^u}(\boldsymbol{\kappa}^u_{kt}|\boldsymbol{y}_{kt}, \theta)}_{L^{out}_{kt}(\boldsymbol{\kappa}_{kt}|\boldsymbol{M}_{kt},\theta)} \\
\log \mathcal{L}(\theta) &= \sum_{k,t} \Big[ \log L^{\text{sel}}_{kt}(\mathbf{M}_{kt} \mid \theta) + \log L^{\text{out}}_{kt}(\boldsymbol{\kappa}_{kt} \mid \mathbf{M}_{kt}, \theta) \Big]. \qquad (34)
\end{aligned}$$

The parameters of the synergy function as well as the AR(1) persistence and variance for $\kappa$ are embedded in $\theta \equiv (c_S, c_4, c_5, c_6, c_8, c_7, c_8, c_{10}, \delta_\kappa, \delta_\mathcal{O}, \rho, \sigma_\kappa)$ . The selection stage uses data on selection of sellers into potential buyers or in staying alone given their level of fixed costs $\kappa_{ikt}$, comprised in an ownership matrix $\boldsymbol{M}_{kt}$. The outcome stage uses data of the level of fixed costs in $\kappa_{ikt}$ for all remaining banks.

We will estimate the likelihood by simulation. Derivations of $L^{sel}_{kt}(\boldsymbol{M}_{kt} \mid \theta)$ and $L^{out}_{kt}(\boldsymbol{\kappa} \mid \boldsymbol{M}_{kt}, \theta)$ and more details about the computing the likelihood function as function of the model parameters then estimating by simulation can be found in Appendix H.

# 7 Model estimates

This section describes the demand parameters and consequential changes in marginal costs, fixed costs, and quality of merging banks and depositor and borrower welfare from bank mergers. Using those estimated parameters in our model of endogenous mergers, we verify significant pair-specific motivations for mergers and find meaningful costs of merging on both the seller and buyer side.

## 7.1 Stage 1: Deposit and loan demand estimates

Here we describe the demand estimates for bank-specific deposits and loans. The demand estimation is standard, intended to proceed similarly to prior work (Egan et al., 2022; Koont, 2023; Wang et al., 2022).

Consumers value branches and are elastic to deposit rates and loan rates. Estimates from Equation (29) and of the Berry (1994) model are shown in Table 4. The instrument "operating cost intensity" is significant at the 1% level. Columns 1 and 2 show the first stage of the 2SLS demand system. The dependent variables are the endogenous deposit and loan rates. The instrument, by shifting cost of operating the bank, corresponds to a lower interest rate paid on deposits and a

Table 4: Deposit and loan demand

| | Rate (pp) | | Log market share | | | |
|---|---|---|---|---|---|---|
| | First stage | | OLS | | 2SLS | |
| | Deposits | Loans | Deposits | Loans | Deposits | Loans |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Fixed expenses$_{t-1}$ / Assets$_{t-1}$ (pp) | -0.598*** | 0.603*** | | | | |
| | (0.021) | (0.042) | | | | |
| Deposit rate ($\alpha_d$; pp) | | | 0.111*** | | 2.194*** | |
| | | | (0.011) | | (0.106) | |
| Loan rate ($\alpha_l$; pp) | | | | -0.157*** | | -1.886*** |
| | | | | (0.012) | | (0.144) |
| Log(# branches) | -0.068*** | -0.191*** | 0.989*** | 0.994*** | 1.154*** | 0.683*** |
| | (0.003) | (0.008) | (0.008) | (0.008) | (0.013) | (0.031) |
| First-stage $F$ (Kleibergen–Paap) | 791.3 | 209.6 | | | | |
| State $\times$ year FE | X | X | X | X | X | X |
| Observations | 250,298 | 249,660 | 250,298 | 249,660 | 250,298 | 249,660 |
| $R^2$ | 0.858 | 0.757 | 0.758 | 0.754 | 0.778 | 0.759 |

+ p $<$ 0.1, * p $<$ 0.05, ** p $<$ 0.01, *** p $<$ 0.001

Note: Berry (1994) logit demand estimated by 2SLS. Columns 1 and 2 show the first stage of the 2SLS demand system. Dependent variable is the (endogenous) interest rate. The interest rate (deposit rate in column 1, loan rate in column 2) is instrumented with operating-cost intensity (1-year lagged fixed expenses / assets). Kleibergen-Paap is the cluster-robust first-stage $F$. Columns 3 and 4 show the OLS regressions. As is typically found in demand regressions, without instruments for prices, demand seems fairly price inelastic. However, Columns 5 and 6 show the second stage of the 2SLS system. Using the operating cost intensity as the instrument for deposit rate and loan rate, banks command higher market shares when they offer higher deposit rates and lower loan rates. Annual data, where each observations is the Q4 Call Reports filing. Standard errors clustered by bank.

Table 5: Difference in means, acquirers and targets

| Variable | Acquirer | Target | Uninvolved | Acq. - Tar. | Tar. - Unin. | Acq. - Unin. |
|---|---|---|---|---|---|---|
| $\xi^D$ (deposit unobserved quality) | 0.289 | 0.082 | -0.002 | 0.207*** | 0.083*** | 0.290*** |
| $\xi^L$ (loan unobserved quality) | 0.043 | 0.154 | -0.001 | -0.111*** | 0.155*** | 0.044 |
| $mc^D$ (deposit marginal cost, pp) | 0.417 | 0.476 | 0.280 | -0.059** | 0.195*** | 0.136*** |
| $mc^L$ (loan marginal cost, pp) | 3.376 | 3.906 | 3.833 | -0.530*** | 0.073*** | -0.457*** |
| $\kappa$ (fixed cost, $) | 144211.0 | 12821.4 | 19511.9 | 131389.7*** | -6690.6** | 124699.1*** |
| $\kappa$ / Assets | 0.0049 | 0.0066 | 0.0230 | -0.0017*** | -0.0164 | -0.0181* |
| Observations | 4,056 | 4,731 | 992,405 | | | |

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Note: This table shows the difference means for acquirers, targets, and uninvolved banks in the merger sample for the estimated profit parameters in Q4 prior to the year of merger. $\xi^D, \xi^L$ are non-price, unobserved quality residuals for every bank estimated from the Berry (1994) demand system for deposits and loans, respectively. The price instrument requires an additional lag of bank financial data and hence profit parameters were derived only for sample mergers occurring in 1992 onward. Marginal costs $c^D, c^L$, and fixed cost $\kappa$ are backed out from observed prices using the assumption that banks compete Bertrand-Nash.

higher loan rate charged. The F statistics for both exceed 10. Columns 3 and 4 show the OLS regressions. As is typically found in demand regressions, without instruments for prices, demand seems fairly price inelastic. However, Columns 5 and 6 show the second stage of the 2SLS system. Using the operating cost intensity as the instrument for deposit rate and loan rate, banks command higher market shares when they offer higher deposit rates and lower loan rates.

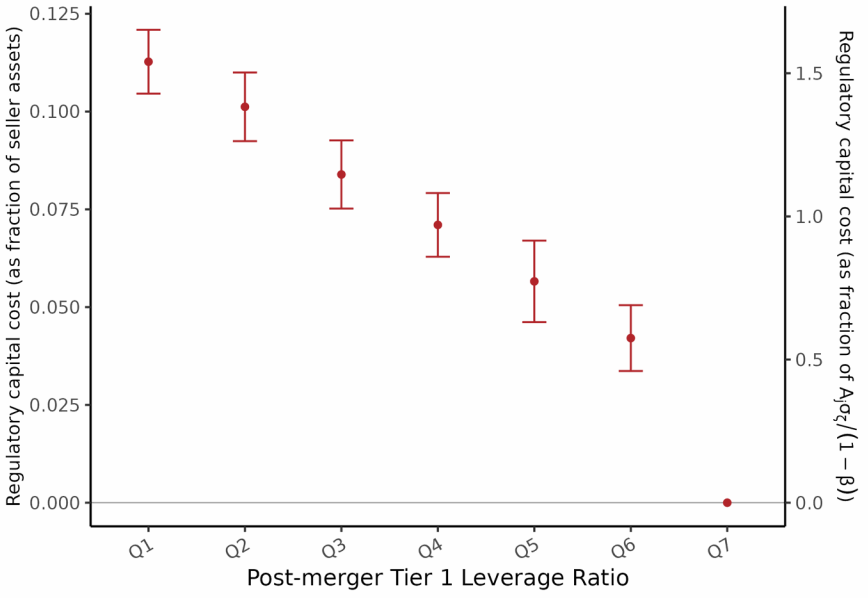
Differences in the median level of the profit parameters between acquirers and borrowers is significant. Table 5 shows the difference in means table for acquirers, targets, and uninvolved banks between 1990-2024. As was reported with accounting data in Table 1, acquirers are more cost efficient than merger targets. Compared to targets, acquirers have a 6bp advantage in the year prior to merger in deposit marginal costs and a 53bp advantage in loan marginal costs. Fixed costs are larger in dollar terms for acquirers, but this difference reflects their scale. When normalizing by assets, acquirers have significantly smaller fixed costs than targets. Acquirers have higher unobserved quality in providing deposits, but lower unobserved loan quality, insignificant at the 90% confidence level.

### 7.2 Stage 2: Merger market

Estimates of the merger market parameter suggest that regulatory capital constraints for private firms create large one-time merger costs. The model estimates suggest mergers create value through fixed cost synergies.

Table 6 presents estimates of the structural model parameters. First, incidental to quantifying merger synergies and merger frictions with selection, we estimate parameters related to the data-generating process of fixed costs, $\kappa$. Row 1 reports the persistence of the AR(1) process for $\kappa/A_i$.

Figure 4: Model estimates of regulatory capital wedge




Note: This plot shows the estimates of $\boldsymbol{c}_{RCW}$. Each of seven quantiles of post-merger capital ratios for privately held acquirers has a separately estimated RCW. The estimate is plotted in red with 95% confidence intervals in bars. The results show that costs increase approaching the 4% minimum Tier 1 Leverage Ratio. At its peak, combinations that would have a first quantile post-merger Tier 1 Leverage Ratio face an implicit cost of 11% of seller assets. For reference, this statistic would be equivalent to approximately 11 times the capitalized value of the seller's annual profits. The cost then declines as the pair would be more capitalized. The most capitalized quantile (Q7) is set to zero to have no regulatory capital wedge. The second axis normalizes the estimate by the Gumbel scale parameter of the full present value of $\zeta$ shocks. The estimates range, by capitalization, from being 1.5 one scale (1.2 standard deviations) of pair-specific merger surplus for the most constrained pairs to 0.4.

From this persistence we also recover, row 2, the variance of shocks to fixed cost $\kappa/A$. Variance in these shocks mediates selection on levels: the merged pair could have received particularly negative fixed cost shocks, meaning the banks poised to become more cost efficient merged and fixed cost synergies would be over-estimated. Rows 3 and 4 list the parameters $\delta$. $\delta_\kappa$ is actually a fixed dissynergy: between all merger pairs, holding branch overlap and pair-specific adjustments fixed, the $\kappa$ per assets of the sum of a buyer and seller is actually larger than what they would counterfactually be apart. As expected, buyers and sellers having more branch overlap increases the fixed cost synergy. The remainder of fixed cost synergies are loaded into pair-specific shocks $\zeta$. The scale parameter of the pair-specific Gumbel synergies, $\zeta_{ij}$ is 0.7% of seller assets annually in profits, which is large – just smaller than a typical bank's annual return on assets. The parameter implies that fixed cost synergies from bank mergers, should they arise are from branch overlap and pair-specific unobservables outside of the model.

Table 6: Merger market model estimates

| Category | Parameter | Estimate | Units | SE (Hessian) |
|---|---|---|---|---|
| Profit parameters | $\eta^{\kappa}$ | 0.718 | – | (0.0004) |
| | $\sigma_{\tilde{\kappa}}$ | 0.00739 | Assets | (1.60e-06) |
| Merger synergies | $\delta_{\mathcal{O}}$ | -0.0191 | Seller assets | (0.0002) |
| | $\delta_{\kappa}$ | 0.00297 | Seller assets | (0.0002) |
| | $\sigma_{\zeta}$ | 0.00732 | Seller assets | (3.20e-05) |
| Merger selection | $\tau$ | 0.621 | Seller assets | (0.0036) |
| | Private $\times$ Q1 ($c_1$) | 0.113 | Seller assets | (0.0042) |
| | Private $\times$ Q2 ($c_2$) | 0.101 | Seller assets | (0.0045) |
| | Private $\times$ Q3 ($c_3$) | 0.0839 | Seller assets | (0.0044) |
| | Private $\times$ Q4 ($c_4$) | 0.0710 | Seller assets | (0.0042) |
| | Private $\times$ Q5 ($c_5$) | 0.0566 | Seller assets | (0.0053) |
| | Private $\times$ Q6 ($c_6$) | 0.0421 | Seller assets | (0.0043) |
| | Private $\times$ Q7 (ref.) | – | – | – |

Note: This table shows the estimates for the merger friction and merger synergy parameters, where $\zeta$ is a shock that mediates merger selection through merger surplus (the buyer's bid for a given target) but does not affect realized profits. One-time merger costs related to being a private buyer for a combination close to the Tier 1 Leverage ratio minimum are larger than costs for more capitalized combinations. Fixed cost $\kappa$ is fairly persistent over time, with $\rho^{\kappa} = 0.7$. The negative estimate of $\delta_{\mathcal{O}}$ implies that branch overlap drives reductions in fixed costs between potential buyers and sellers that could amount to 1.9% of assets if seller and buyer branch networks overlapped counties completely. That said, without additional shifts in $\zeta_{ijt}$ or branch overlap, any given merger pair has a dissynergy of 0.3% of seller assets ($\delta_{\kappa}$). The scale parameter of the pair-specific surplus shocks Gumbel $\zeta_{ijt}$ is 0.7% of seller assets.Importantly, estimates of RCW through $\boldsymbol{a}$ are significant. Costs decline as potential pairs become more capitalized, ranging from 11% to 0% for the reference group, pairs whose post-merger Tier 1 Leverage Ratio would exceed 10%.

With merger synergies mediated primarily by pair-specific shocks, we turn to costs that affect merger selection but are excluded from the real merger outcomes used in estimation. Most important for policy, mergers involving pro forma banks close to the Tier 1 Leverage Ratio are significantly more expensive. Figure 4 plots the coefficients for the piecewise function of Tier 1 Leverage ratio. Consistent with theory, the mergers closest to the threshold have the highest cost. The regulatory capital wedge is 11% of seller assets for the first quantile of mergers. As expected, moving up toward larger capital ratios, estimates fall, consistent with empirical results in Section 4.

In sum, the results suggest that temporary capital relief around mergers or permanent adjustments would have large effects on the aggregate merger market. We now use the model estimates to quantify the effect of alternative policy using data re-simulated from model parameters compared to data re-estimated from alternative parameters. Appendix I shows how the re-simulated data fit against empirical merger data.

# 8 Counterfactual capital regulation

We have established that the realized mergers between U.S. banks between 1990 suggest a large RCW from reduced form and estimates of the structural model. Using counterfactual simulations of banking sector mergers and competition from the model parameters, we now document how adjustments to capital regulation would shape industry structure. First, temporary relief from capital regulation around mergers would increase consolidation drastically among the smallest banks because capital regulation makes acquisitions by privately traded banks costly. Such relief would increase banking sector efficiency while reducing consumer welfare. Second, we consider the pairing of temporary capital relief with merger review on antitrust grounds.

## 8.1 Temporary capital regulation relief

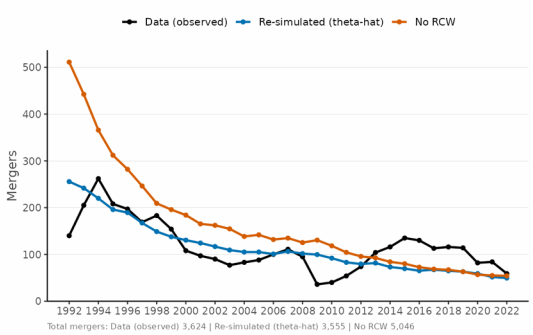
The counterfactual exercise tests a policy that allows banks to breach their Tier 1 Leverage ratio temporarily. Temporary relief from capital regulation around mergers would increase consolidation drastically among the smallest banks because capital regulation makes acquisitions by privately traded banks costly. Such relief would increase allocative efficiency by reducing fixed costs through more mergers, but decrease consumer welfare.

We implement this by setting the coefficients on post-merger Tier 1 Leverage Ratio to zero, or assuming there is no regulatory capital wedge, then measuring how the 1992-2023 world would have progressed. To isolate just the change in industry structure of the regulatory capital wedge, we first repeatedly simulate counterfactual banking sectors by state and year from our parameter values to use as a baseline. Appendix J describes the procedure. The simulations begin with the 1992 population of U.S. commercial banks, then play the merger market and horizontal competition given the opponents available every year, then uses the newly consolidated sector for the next year. We do not use an outside option in the computation of optimal rates set in the deposit and loan markets, so the estimates of consumer welfare harm are likely high upper bounds. However, the model's current specification shutting down marginal cost and quality synergies mean that consumer welfare could never rise with additional consolidation and efficiency gains, since efficiency gains would add only to fixed costs not used for pricing. We then compare these simulations with simulations from the same parameter values, but RCW parameters set to zero. The results below show the comparison of 300 simulations.

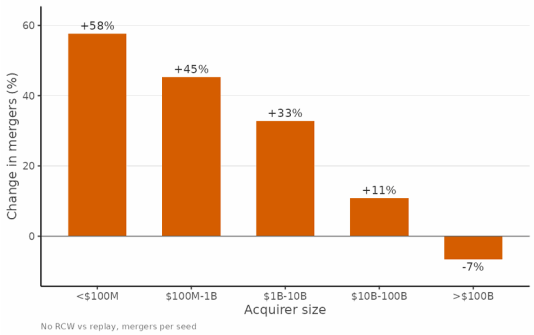
Removing capital constraints changes the number and also type of mergers that happen than in the data. Figure 5 shows the mean number of mergers occurring annually in the baseline versus counterfactual. The policy makes merging less costly and, as expected, increases the rate of consolidation drastically in the smallest banks, the vast majority of which are privately held. Setting the RCW to zero does almost nothing to the regional and large national banks, as shown in the last two panels, but for the smallest banks, mergers increased by 58% over the period. In sum, the increase in consolidation among the smallest banks corresponded to a 19% decline in banks operating in the U.S. in 2022.

Figure 5: Removing RCW for private firms

(a) Merger counts between 1992-2022




(b) Percent increase in mergers by size of acquirer




Note: Panel 5 shows the effect on the merger rate if RCW were set to zero. Without a wedge from capital regulation making mergers costly, private firms would have consolidated more, especially through the 1990's. The number of mergers with very small acquirers, with assets <$100M would have increased by 58%, but there would smaller effects on regional banks (10-100B) and even slightly negative for the largest banks who are nearly all publicly traded and would have been crowded out from additional purchases.

We can then assess the welfare of this counterfactual industry between 1992 and 2022. Figure 6 shows the resulting total welfare, broken down by producer surplus via fixed cost synergies, increases in variable profits from market power and the reallocation of consumers to more cost efficient banks, and decreases in depositor and borrower welfare. The figures sum over changing market conditions. While temporary capital relief was useful for a time for increasing the cost efficiency of the sector, its usefulness was front-loaded to early years, and after realizing high synergy mergers, every additional merger harms consumer welfare more because fewer firms are left. In total over the thirty year period, surplus would have been $9B larger.[26]

## 8.2 Combining temporary capital relief with additional merger screening

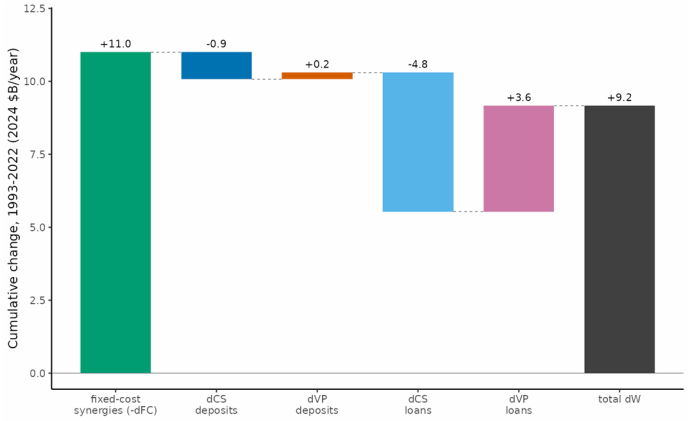
Section 8.1 suggests that temporary regulatory capital regulation relief, in isolation, would be detrimental to present-day welfare. However, the evidence thus far shows that RCW was a stand-in for other merger screening, so it stands to reason that using additional screening tools could have implemented a higher welfare outcome.[27]

Using the estimated model, we show that providing temporary capital relief combined with deposit caps would retain most of the cumulative of fixed cost synergies while also increasing

[26]The estimates assume that bank fixed costs evolve with an AR(1), estimated to be 0.7. As a result, synergies realized from mergers are transient, as fixed costs revert to the market's average. Under the alternative assumption that the synergies component of fixed cost were perfectly persistent, counterfactual fixed cost savings would be 4x larger due to the greater number of years that the merger's cost savings would be applied. In this way, the counterfactual understates surplus gains in order to compare consumer welfare against a conservative producer surplus estimate.

[27]While we will study the effect of such tools using the "backward looking" counterfactual world, estimation of the model on recent years and projecting forward could validate the findings to inform policy design.

Figure 6: Counterfactual welfare from removing RCW



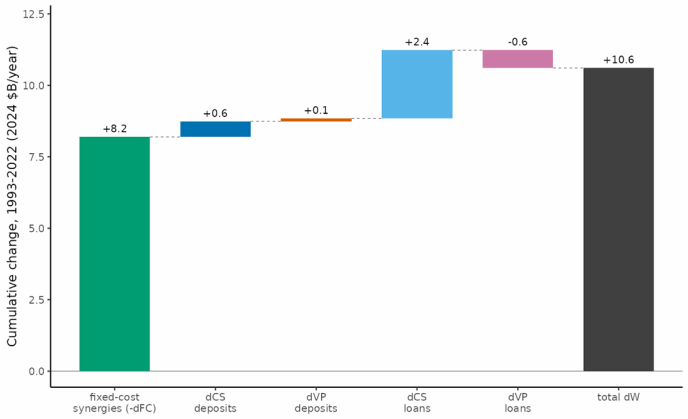

Note: This figure shows the estimated cumulative welfare effect of removing the regulatory capital wedge (RCW) from every merger decision from 1992 onward, relative to the re-simulated sector with the RCW. The figures add up the welfare realized in every year, undiscounted. The welfare impacts of no RCW compound over time as additional mergers are realized. The counterfactual compares a sector walked forward from 1992 using the model parameters to a sector walked forward with the model parameters with no RCW cost imposed on privately held acquirers. Each year, the sector could be more efficient, reducing the fixed cost base (green) or the reallocation of consumers to lower marginal cost firms. But, each year, depositors and borrowers were on average worse off through higher prices (dark blue, light blue). Aggregating consumer and producer surplus, a banking sector without RCW would have created larger surplus than the status quo.

consumer welfare. While merger review typically takes into account HHI thresholds, strict caps on market share or number of subsidiaries have been implemented in some industries (Jeziorski, 2023). Banking is one such industry. The modern incarnation of ownership caps for banking is the default 30% deposit cap implemented by Riegle-Neal in 1994, which states could choose to either raise or lower. The counterfactual imposes a 25% deposit cap starting in 1992 alongside a cost of the regulatory capital wedge of zero. The results are shown in Figure 7. Instead of the tradeoff shown in Panel 6, the cumulative effect since 1992 on the sector now was one of both cost efficiencies and higher consumer welfare.

## 9 Conclusion

Despite thousands of mergers between commercial banks since 1990, the United States still has 4,000 operating commercial banks. That fragmentation is puzzling when, in measuring the outcomes of realized mergers since 1992, mergers seem to be profitable. We show this with difference-in-differences measuring the effect on the merging banks as a combination. Along the way, the difference-in-differences estimates suggest that bank profits may come at the cost of higher prices by way of higher interest rates charged to borrowers of bank loans and lower interest rates offered

Figure 7: Counterfactual welfare from removing RCW alongside a state-level 25% deposit cap




Note: This figure shows the estimated effect of temporary capital relief, if, alongside relief, mergers could not be consummated if they created banks whose combined market shares in the state deposit market summed to greater than 25% before combining. Comparing with Figure 6, the negative effects of removing RCW on consumer welfare would largely disappear. Fixed cost synergies collectively amount to $8B and consumer surplus realized over the period rises by $3B between the deposit and lending markets.

to bank depositors.

This paper demonstrates that capital regulation, by creating a shadow cost of cash mergers, makes financing M&A challenging and consequently dissuades mergers. The mechanism starts with the premise that financing new investment with equity is more costly than financing with debt, for either asymmetric information problems in equity issuance or the absence of a liquid market for the equity of small, privately held firms. The consequence is that for small, privately held acquirers with illiquid stock, cash mergers are preferable. They do not require the firm to raise new equity, cash is readily available from the firm's own balance sheet or from raising new debt. However, in the banking sector, doing a cash merger conflicts with microprudential regulation.

We test this mechanism with two instruments designed to vary the regulatory capital of a bank, in the event it were a merger target, separately from the appeal of the potential profit synergies that purchasing the bank could offer an acquirer. The incidence of this regulatory capital wedge is greatest for firms who cannot easily finance mergers by using their own stock as payment, making this wedge largest for smaller bank acquirers.

Estimating a structural model of endogenous mergers arising between heterogeneous banks, we find that this financing friction is quantitatively important in shaping aggregate industry structure. The endogenous mergers model allows profitable mergers to occur by realizing fixed cost synergies but impose heavy costs on combinations that would create especially less capitalized new merged entities.

With those estimates, we evaluate how a regulator could improve the efficiency of the banking sector by adjusting the shadow cost of capital regulation around M&A. We estimate that fewer

fixed costs would be paid by the banking sector annually today if affected acquirers were allowed to breach capital ratios in order to purchase another bank, but such benefits would be of the same order of magnitude as reductions in consumer welfare from the greater pricing power of remaining banks. However, with this model where mergers are proposed and executed endogenously, we can show that temporary capital relief with adjustments in antitrust policy could potentially raise total welfare. A 25% deposit cap retains most of fixed cost synergies while blunting consumer welfare reductions.

At a broader level, while banks are only one of a couple industries with active or proposed capital regulation (namely insurers) that could be targeted alongside antitrust, financing frictions have been shown to be an important driver of M&A investment across industries. By embedding such frictions in models of endogenous M&A, the effect of the extant financing environment on industry structure can be measured.

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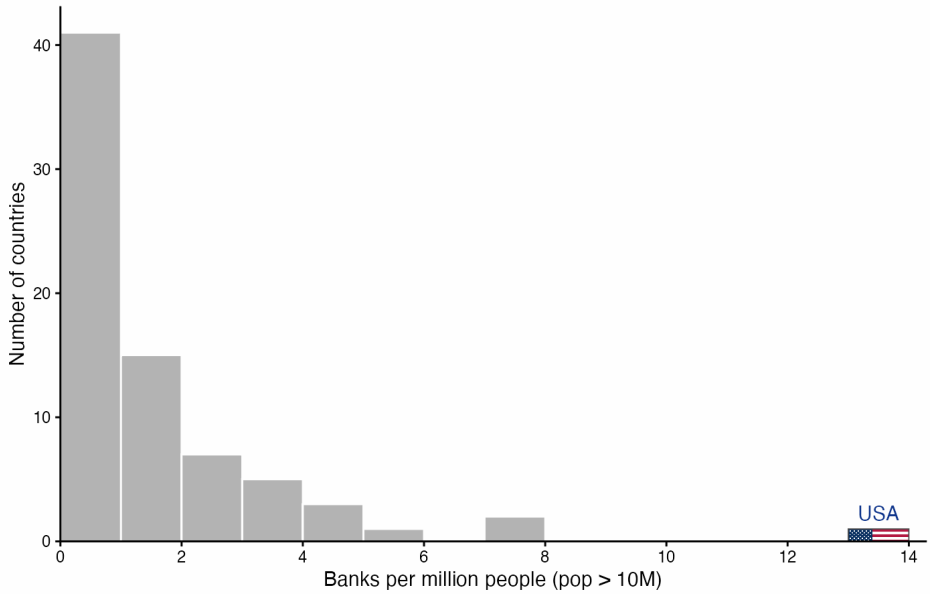
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Wollmann, T. G. (2024). How to Get Away with Merger: Stealth Consolidation and Its Effects on US Dialysis.

Yim, S. (2013, April). The acquisitiveness of youth: CEO age and acquisition behavior. *Journal of Financial Economics 108*(1), 250–273.

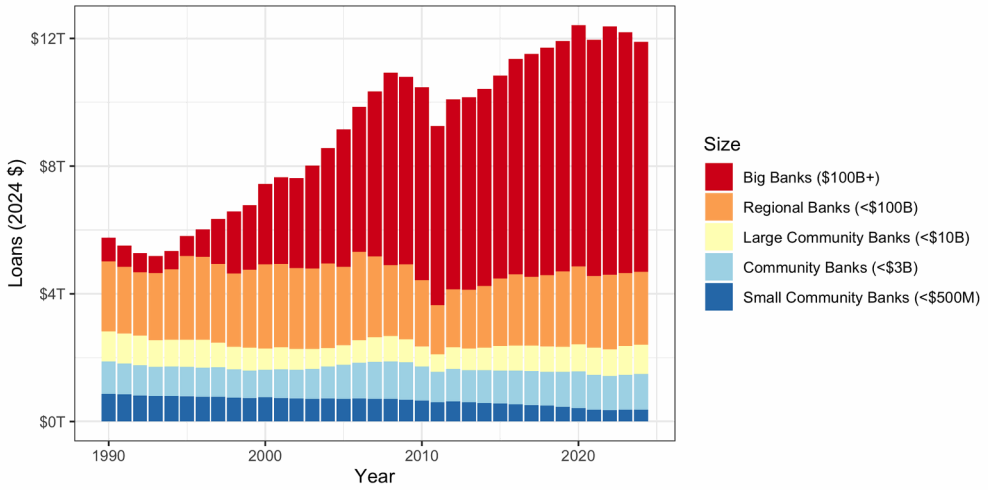
# A Additional figures

Figure A.1: Banks per capita, by country



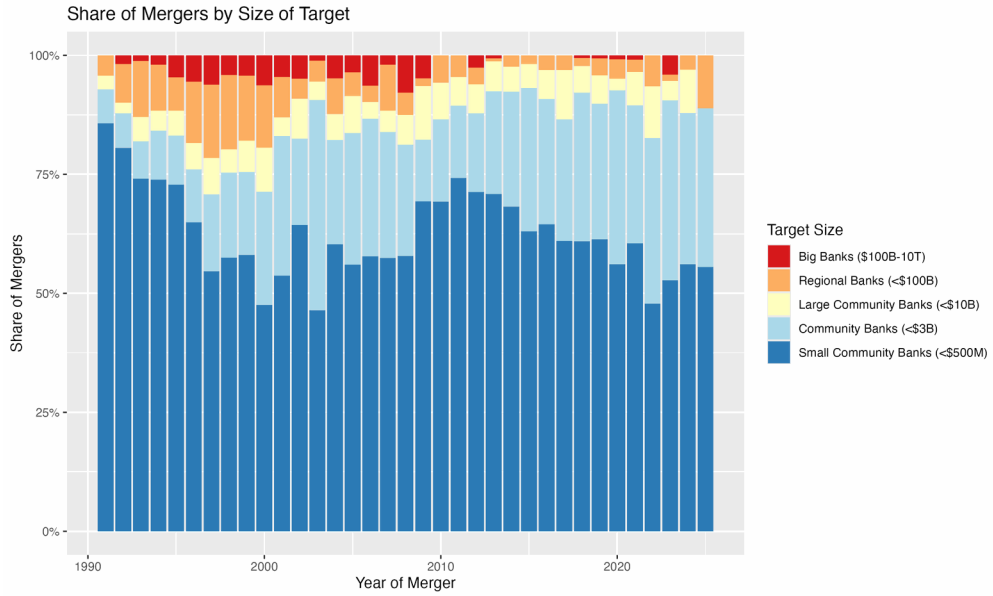

Note: This figure shows the number of banks per one million people where each observation is a country. The population are countries with populations exceeding 10 million people. The United States, as marked, has the largest number of banks per capita in this population.

Figure A.2: Loan balance by bank size



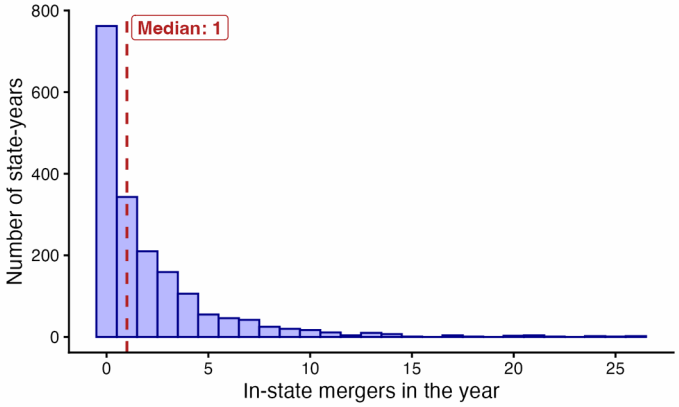

Note: This figure shows the dollars of loans on the balance sheet of banks filing Call Reports, summed by size of the bank. Only 20% of loans in 2024 were given by community banks (banks holding <$10B in assets, are shown in blue and yellow).

Figure A.3: Target size (2023 USD)



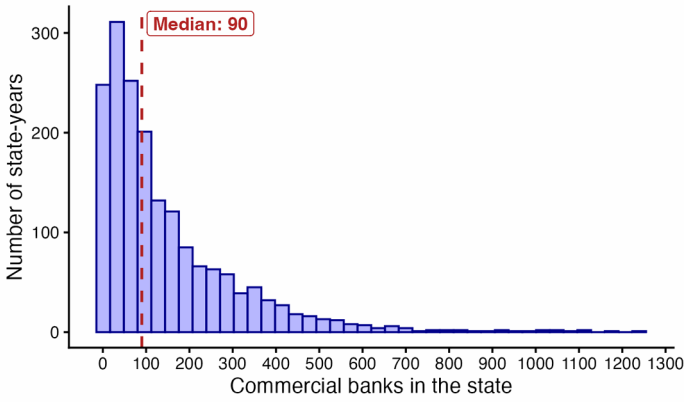

Note: This figure shows the share of transactions in the merger sample cut by the size of the target bank. The majority of target banks have less than $500M in assets (in real terms, 2023 USD) in the quarter before they are acquired.

Figure A.4: Merger frequency by market




Note: This histogram shows the distribution number of commercial bank mergers occurring between 1995-2024 in a given year between banks headquartered in the same state, where each observation is a state-year cell. Mergers between banks headquartered in different states are excluded.

Figure A.5: Market size




Note: This histogram shows the distribution number of commercial banks in each market, where each observation is a state-year cell between 1990 and 2024.

# B Additional tables

Table B.1: Commercial bank merger sample: Descriptive statistics by cohort

| **1990-1994** | | | | **1995-1999** | | | |
|---|---|---|---|---|---|---|---|
| | Acq. | Tar. | Unin. | | Acq. | Tar. | Unin. |
| Inefficiency Ratio | 67.2 | 77.5 | 70.5 | Inefficiency Ratio | 61.5 | 65.6 | 66.1 |
| ROA: Net income / Assets | 0.28 | 0.18 | 0.22 | ROA: Net income / Assets | 0.32 | 0.26 | 4.39 |
| ROA (adj): Economic profits / Assets | 0.29 | 0.19 | 0.23 | ROA (adj): Economic profits / Assets | 0.34 | 0.27 | 4.40 |
| ROE: Net income / Equity | 3.22 | 1.10 | 1.76 | ROE: Net income / Equity | 3.72 | 2.80 | 2.80 |
| ROE (adj): Economic profits / Equity | 3.36 | 1.17 | 1.81 | ROE (adj): Economic profits / Equity | 3.94 | 2.90 | 2.88 |
| Tier 1 Leverage Ratio | 8.0 | 8.6 | 9.6 | Tier 1 Leverage Ratio | 8.6 | 9.4 | 11.4 |
| Loans / Assets | 56.9 | 53.8 | 53.3 | Loans / Assets | 61.7 | 59.2 | 57.6 |
| Deposits / Assets | 84.8 | 87.8 | 86.8 | Deposits / Assets | 80.4 | 85.0 | 84.1 |
| **2000-2004** | | | | **2005 – 2007-11** | | | |
| | Acq. | Tar. | Unin. | | Acq. | Tar. | Unin. |
| Inefficiency Ratio | 62.1 | 75.5 | 69.2 | Inefficiency Ratio | 61.6 | 68.8 | 71.0 |
| ROA: Net income / Assets | 0.31 | 0.20 | 0.24 | ROA: Net income / Assets | 0.31 | 0.27 | 0.23 |
| ROA (adj): Economic profits / Assets | 0.33 | 0.21 | 0.25 | ROA (adj): Economic profits / Assets | 0.32 | 0.28 | 0.23 |
| ROE: Net income / Equity | 3.47 | 1.67 | 2.76 | ROE: Net income / Equity | 3.22 | 2.43 | 4.72 |
| ROE (adj): Economic profits / Equity | 3.62 | 1.75 | 2.81 | ROE (adj): Economic profits / Equity | 3.32 | 2.47 | 4.76 |
| Tier 1 Leverage Ratio | 8.7 | 10.0 | 11.6 | Tier 1 Leverage Ratio | 9.3 | 10.8 | 13.1 |
| Loans / Assets | 66.4 | 63.0 | 60.8 | Loans / Assets | 68.0 | 66.6 | 63.4 |
| Deposits / Assets | 77.6 | 81.8 | 82.0 | Deposits / Assets | 76.3 | 81.3 | 80.4 |
| **Recession (2007-12 – 2009-06)** | | | | **2009-07 – 2014** | | | |
| | Acq. | Tar. | Unin. | | Acq. | Tar. | Unin. |
| Inefficiency Ratio | 67.5 | 96.9 | 119.9 | Inefficiency Ratio | 68.3 | 89.9 | 83.1 |
| ROA: Net income / Assets | 0.27 | -0.00 | 0.09 | ROA: Net income / Assets | 0.25 | 0.04 | 0.12 |
| ROA (adj): Economic profits / Assets | 0.28 | 0.01 | 0.11 | ROA (adj): Economic profits / Assets | 0.26 | 0.05 | 0.13 |
| ROE: Net income / Equity | 2.82 | -0.08 | 1.88 | ROE: Net income / Equity | 2.30 | -0.24 | 0.43 |
| ROE (adj): Economic profits / Equity | 2.91 | -0.02 | 2.06 | ROE (adj): Economic profits / Equity | 2.38 | -0.11 | 0.50 |
| Tier 1 Leverage Ratio | 9.1 | 9.7 | 12.2 | Tier 1 Leverage Ratio | 10.6 | 10.1 | 11.6 |
| Loans / Assets | 70.9 | 73.3 | 64.6 | Loans / Assets | 64.2 | 59.1 | 60.5 |
| Deposits / Assets | 76.0 | 81.8 | 79.7 | Deposits / Assets | 81.8 | 85.1 | 82.6 |
| **2015-2019** | | | | **2020-2024** | | | |
| | Acq. | Tar. | Unin. | | Acq. | Tar. | Unin. |
| Inefficiency Ratio | 64.6 | 74.6 | 71.5 | Inefficiency Ratio | 62.8 | 70.8 | 68.6 |
| ROA: Net income / Assets | 0.27 | 0.18 | 0.30 | ROA: Net income / Assets | 0.29 | 0.24 | 0.33 |
| ROA (adj): Economic profits / Assets | 0.28 | 0.18 | 0.30 | ROA (adj): Economic profits / Assets | 0.30 | 0.24 | 0.33 |
| ROE: Net income / Equity | 2.46 | 1.22 | 2.09 | ROE: Net income / Equity | 2.93 | 1.80 | 2.59 |
| ROE (adj): Economic profits / Equity | 2.53 | 1.26 | 2.12 | ROE (adj): Economic profits / Equity | 3.00 | 1.83 | 2.63 |
| Tier 1 Leverage Ratio | 10.4 | 10.8 | 12.6 | Tier 1 Leverage Ratio | 10.2 | 11.0 | 12.6 |
| Loans / Assets | 69.1 | 64.1 | 64.0 | Loans / Assets | 65.1 | 58.6 | 60.8 |
| Deposits / Assets | 81.7 | 84.3 | 82.2 | Deposits / Assets | 84.4 | 84.7 | 83.3 |

Note: This table shows the means of profit and balance sheet variables for commercial bank acquirers, commercial bank targets, and banks uninvolved in the merger market. Means for the acquirer and target are evaluated for each bank four quarters prior to their merger date. Means for the uninvolved banks are taken for the sample period.

Table B.2: County economic characteristics: merger-exposed vs. non-exposed county-years

| Variable | Merger county-yrs | Non-merger county-yrs | Diff. |
|---|---|---|---|
| Unemployment rate (%) | 5.81 | 6.11 | -0.30*** |
| Poverty rate (%) | 14.05 | 15.84 | -1.79*** |
| Median HH income ($) | 55,463 | 49,431 | 6,032*** |
| Log population | 11.52 | 10.04 | 1.48*** |
| Share age 65+ | 0.163 | 0.181 | -0.018*** |
| Share bachelor's+ (%) | 24.51 | 18.57 | 5.94*** |
| Nonmetro (RUCC $\geq 4$) | 0.353 | 0.663 | -0.310*** |
| N county-years | 9,698 | 54,722 | |

$^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

Note: An observation is a county $\times$ year. A county-year $(c, t)$ is merger-exposed if a bank acquired in year $t$ (2005–2024) operated a branch in county $c$ (footprint at the merger's SOD year).

# C Data Appendix

## C.1 Call Reports

Table C.1: Glossary of Call Reports variables

| Quantity | Call Reports item |
|---|---|
| Assets | RCFD 2170 |
| Risk-weighted assets | RCFA A223 |
| Tier 1 Capital | RCFA 8274 |
| Tier 1 leverage denominator | RCFA A224 |
| Tier 1 leverage ratio | RCFA 8274 / RCFA A224 |
| Securities of greater than five year maturity | RCFD A553 + RCFD A554 + RCFD A559 + RCFD A560 |
| (pre-1997) | RCFD 0346 + RCFD 4552 |
| Net income | RIAD 4340 |
| Goodwill impairment losses | RIAD C216 |
| Amortization expense | RIAD C232 |
| Goodwill amortization (pre-2002) | RIAD 4531 |
| Non-interest expenses | RIAD 4093 |
| Salaries and employee benefits | RIAD 4135 |
| Fixed expenses | RIAD 4217 |
| Misc. expenses | RIAD 4092 |
| Total interest income | RIAD 4107 |
| Total interest expense | RIAD 4073 |
| Net interest income | RIAD 4074 |
| Non-interest income | RIAD 4079 |
| Economic profits ("Profit") | RIAD 4340 + RIAD C216 + RIAD C232 |
| (pre-2002) | RIAD 4340 + RIAD 4531 |
| Economic costs | RIAD 4093 − RIAD C216 − RIAD C232 |
| (pre-2002) | RIAD 4093 − RIAD 4531 |
| Inefficiency Ratio | RIAD 4093 / (RIAD 4074 + RIAD 4079) |
| Unmarked HTM/AFS assets | (RCFD 1773 + RCFD 1771) − (RCFD 1772 + RCFD 1754) |
| Domestic deposits | RCON 2200 |
| Deposit interest expense | RIAD 4508 + RIAD 0093 + RIAD HK03 + RIAD HK04 |
| (pre-2017) | RIAD 4508 + RIAD 0093 + RIAD A517 + RIAD A518 |
| (pre-2001) | RIAD 4170 |
| Deposit interest rate | Deposit interest expense / RCON 2200 |
| Loans | RCFD 2122 |
| Loan interest and fee income | RIAD 4010 |
| Loan interest rate | Loan interest and fee income / RCFD 2122 |
| Total liabilities | RCFD 2948 |
| Total equity capital | RCFD 3210 |
| Funding cost | RIAD 4073 / RCFD 2948 |
| Securities share | (RCFD 1773 + RCFD 1754) / RCFD 2170 |
| Return on assets (ROA) | RIAD 4340 / RCFD $2170_{t-1}$ |

Note: Item numbers are from the June 2026 Call Report (FFIEC 031) unless a row gives an earlier date range. For banks that file domestic-office reports only (FFIEC 041/051), each RCFD item is replaced by the corresponding RCON item and each RCFA item by RCOA. Before 2015, regulatory capital items come from RCFD/RCON 8274, A223, and A224. Deposit and loan interest rates are the sum of four quarterly flows divided by the end-of-year balance. Funding cost and ROA are quarterly flows multiplied by four.

Call Report income-statement items are reported year-to-date. Financial variables used throughout the paper are defined in Table C.1.

Mergers sometimes lead to inconsistent accounting of the profits of acquirers and targets. In using the charter subsumation date in the baseline specification, the data construction already allows the pro forma bank to be well-defined by summing Call Reports filings in the pre-period and using the buyer's Call Reports filing in the post-period. We recover quarterly flows by first-differencing raw Call Reports fields within bank-year. Sometimes, this differencing is distorted by mid-year ownership changes, which we correct in two places. First, in the panel build, when a bank changes hands mid-year and continues to file, purchase/push-down accounting restarts its year-to-date income at the deal date. Differencing quarter to quarter would create a spurious large negative income for that quarter. We classify a bank-quarter as a confirmed reset when it both (i) coincides with a documented ownership change (NIC transformations or a high-holder change in our relationship panel, where we also check for one additional quarter differences between the first Call Report date and the merger date) (ii) shows a material within-year decline in at least one series that should be monotone by construction (we use interest income, interest expense, salaries, or non-interest expense, where the balance of salaries paid year to date cannot reasonably decline as the year continues). At a confirmed reset date, the flow is set to the post-deal year-to-date stub, rather than the difference from last quarter.

Second, if and only if the acquirer and target were already under the same top holder before the charter combination, the surviving entity may restate its year-to-date income as if the two banks had been combined since the start of the year. That discretion in accounting creates a spurious spike at the event quarter (t=0) that double-counts the target's income earlier in the same year and reverts the following quarter. For same-holder combinations occurring after the first quarter, and only for the quarter of the merger date, we subtract the target's own within-year (pre-event) year-to-date income from each affected income/expense line at t=0, applying the correction only to events whose noninterest-income pattern matches this pooling signature to within a tolerance (15% of the expected combined level) so that non-pooling deals are left untouched.

## C.2 Market shares

In the structural model, the banks active in each state, year merger market are banks headquartered in the state, in that year. Banks with deposits in other states are present for deposit competition within that state but not involved in the merger market. Aligning with that norm deposit market shares and deposit market sizes for a given state are created using the sum of a bank's deposits at all branches in the state using Summary of Deposits. Because we cannot allocate the locations of loans to states like the Summary of Deposits allows us to do for the deposit market, we assume banks operate in the loan market only in their state and create market shares and market sizes for the lending market accordingly. Model results treating the deposit market like the lending market are nearly identical to model results with the Summary of Deposits correction, but the proper adjustment of the market shares of the largest banks meaningfully improves precision calculations

of counterfactual depositor welfare. Banks without a market presence in the year prior do not have estimated pre-merger parameters and are hence removed from the merger choice set for other players. However, their market shares in the period after the merger, when they have entered, are taken into account as part of the outside option product in the deposit markets and loan market when potential merging parties compute the expected future profit synergies from combination.

# D Additional results and robustness for Section 5.7

This section contains additional results and robustness checks for the baseline difference-in-differences estimates presented in Section 5.7. Such checks broadly evaluate 1) concerns about parallel trends that come from potentially endogenous merger timing and 2) concerns about the Stable Unit Treatment Value Assumption related to the geographic scope of bank competition.

## D.1 Additional results

Figure D.2 shows that non-interest expense cost synergies are driven by declines in all cost categories within non-interest expense.

## D.2 Merger timing robustness

We show robustness to merger timing date, and violations of parallel trends coming from the control banks used to estimate the causal effect on pro forma banks or simultaneous changes to macroeconomic shocks in places mergers were targeted.

### D.2.1 Bank Holding Company merge date

Figure D.3c shows the result of performing analysis where merger date is the date that the bank changed bank holding company ownership. If the bank was previously not held by a bank holding company, it is the quarter that it is acquired by the bank holding company. This date can be simultaneous to the date reported to the NIC that the target's charter was merged into the acquirers charter. As is expected given the dates are generally close together, the results are robust. Inefficiency ratio remains elevated, likely related to merger costs, in $t = 0$ relative to the baseline specification. The difference is consistent with the charter merge date sometimes being lagged by one quarter from the bank holding company acquisition date.

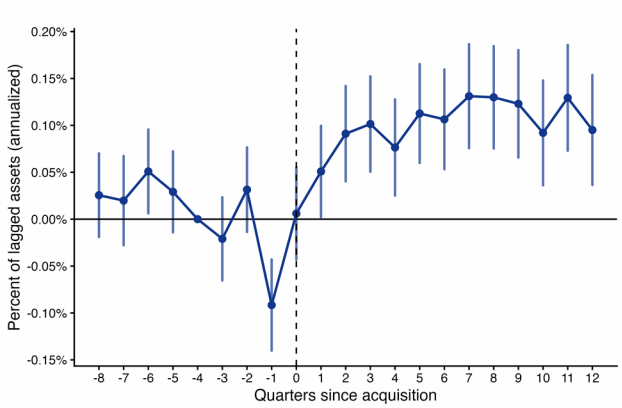
### D.2.2 Placebo test on county economic variables

Profits from mergers could also seem large because they are simulatenous with windfalls to the economic conditions of places where the mergers took place. We account for this by performing event study analysis at the county level. Using the date a county $c$ experienced its first bank acquisition post 1990 in year $g(c)$, we estimate the regression for years $t$

$$Y_{ct} = \alpha_c + \tau_t + \sum_{s=-5}^{5} \beta_s \mathbf{1}(t - g(c) = s) \tag{35}$$

The specification uses variation in merger timing between counties experiencing a merger versus those that had not yet been treated. We use Sun and Abraham (2021) to address concerns about this

Figure D.1: Effect of merger on accounting profits




Note: This graph shows estimates $\beta_s$ from Equation (1) where the outcome is *pro forma* bank net income as a fraction of *pro forma* assets. Net income rises 12bp as a fraction of assets.

staggered design. With random merger timing, the coefficient $\beta_s$ is the causal effect on county $c$ of the merger. Figure D.5 shows the results for the ten economic variables described in the main text. Effects are largely insignificant at the 95% confidence level five years pre- and post-merger date. As further confirmatory evidence, Figure D.4 shows that U.S. counties that experienced mergers move in parallel with counties that did not.

### D.2.3 Narrative evidence

The merger date coincides with a change in outcomes that seem specifically merger related. Specifically, while one salient confounding story violating parallel trends could be that mergers coincide with a change in the latent demand for deposits or loans, the merger affects cost items at the merger date that are fundamentally merger-specific, like consulting and legal. Figure D.6 shows the estimates of Equation (1).

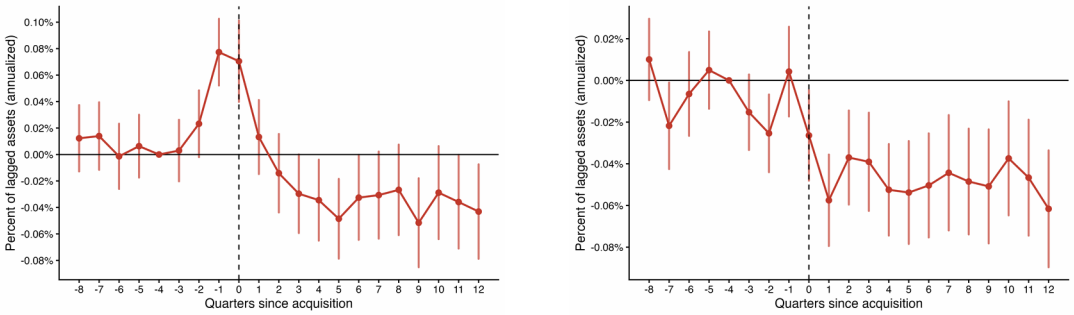
## D.3 Mitigating concerns of inter-state bank merger spillovers

Figure D.8 shows that the majority of mergers are within state. Second, the Federal Reserve uses bank market definition maps to evaluate mergers (Federal Reserve Bank of St. Louis, Federal Reserve Bank of St. Louis). Figure D.9 shows that Federal Reserve bank market definition maps.

Figure D.2: Matched difference-in-difference evidence of cost synergies

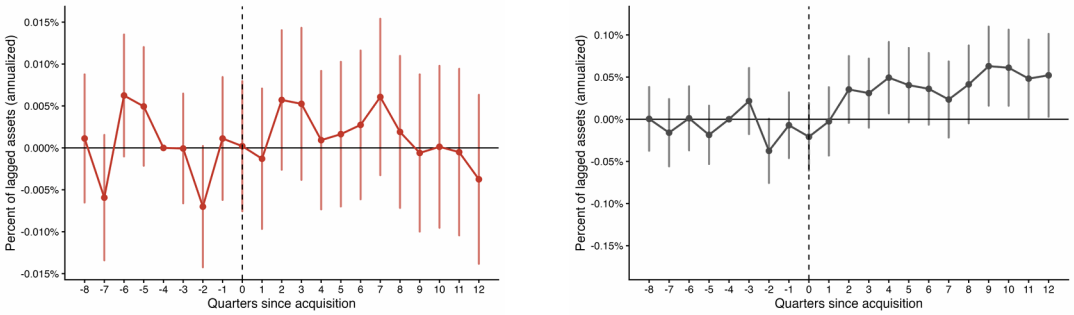
(a) Misc. expenses / Assets

(b) Salaries / Assets



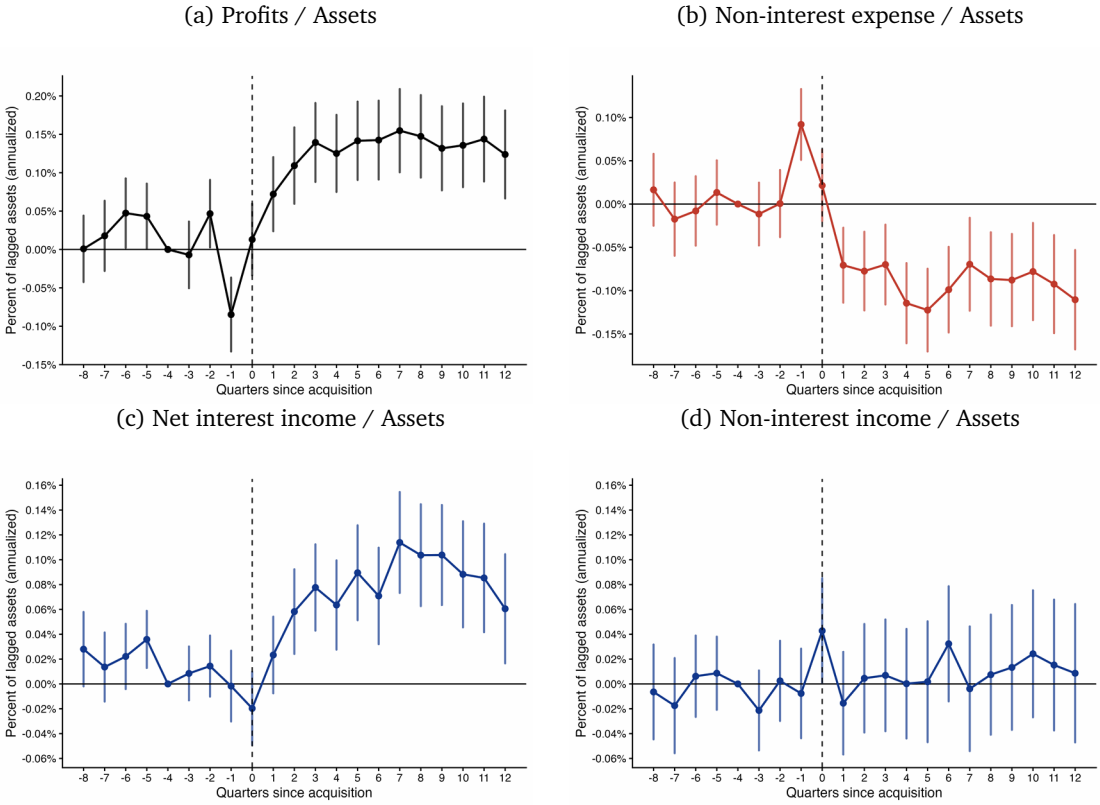

(c) Fixed expenses / Assets

(d) Other items / Assets




Note: This figure shows the estimates of $\beta_s$ from Equation (1) on the constructed population of bank mergers compared to a matched control sample of banks who where neither acquirers nor targets in the event window and operated no branches in the same state as the acquirer or target at the time of merger. Other items is the set of costs "below the line" not included in non-interest expense, such as taxes and depreciation.

Figure D.3: Matched difference-in-differences around merger date of Bank Holding Company



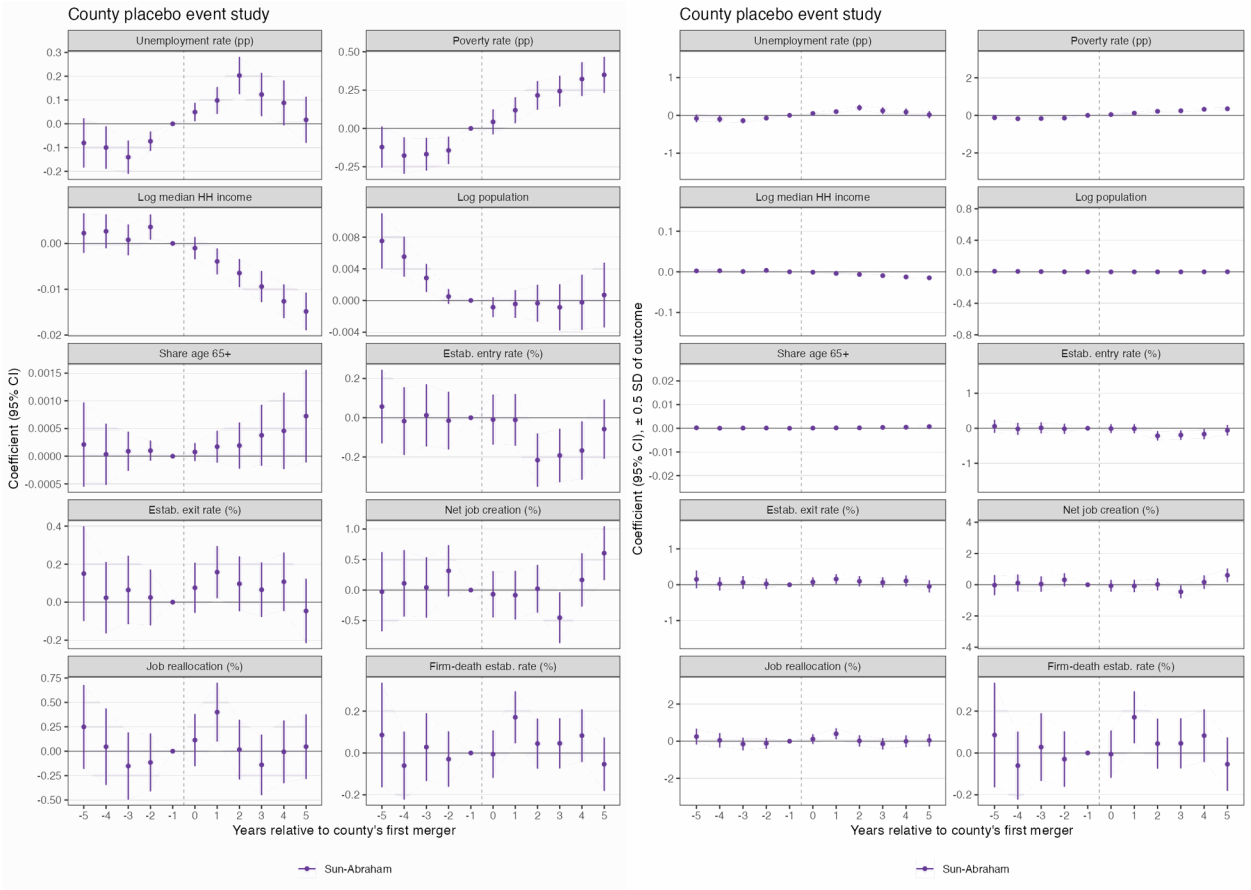

Note: This figure shows the estimates of $\beta_s$ from Equation (1) using the same merger sample. The estimate is equivalent to Figure 2a and similar in every way, except that the merger date is defined as the date that the target's bank charter changed ownership to the acquirer Bank Holding Company. If the acquiring bank did not have a bank holding company, the date is equivalent to the date the charter was subsumed by the parent.

Figure D.4: Economic outcomes of merger versus non merger counties




Note: This figure plots unweighted cross-county means by year for counties ever exposed to a bank merger (blue; a bank with a branch in the county was acquired in some year, $N = 2{,}569$) versus counties never exposed (red, $N = 640$), pooling the two years before and after each county's own reference period. Panels show, from top-left: unemployment rate, poverty rate, nominal median household income, population, the population share aged 65+, and Census Business Dynamics Statistics (BDS) establishment entry, exit, net job creation, job reallocation, and firm-death rates (all in percent). Income and population figures are reported in nominal (not inflation-adjusted) dollars/counts. Sample restricted to ACS availability, 2005–2023.

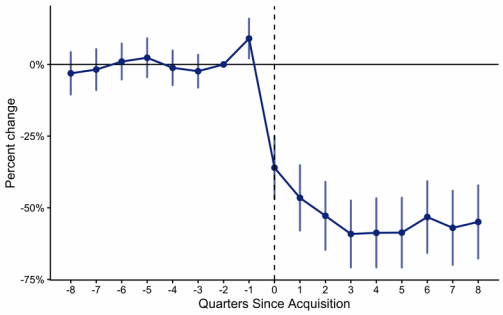
Figure D.5: Placebo: The effect of first bank acquisition on counties



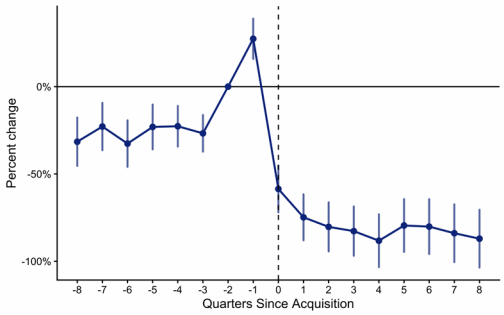

Note: The panels show the result of estimation Equation (35) on counties around the earliest year post-1990 that the county had at least one branch of an acquired bank. The left panel shows the results with 95% confidence intervals. The right panel rescales the left panel so that the y-axis for every graph has the length of the standard deviation of the variable in sample. When rescaling, it is clear that statistically significant effects, while observable especially for household median income, are economically very small. The results are consistent with one aspect of parallel trends, that absent acquisition, a bank would have been performing at least as well as before, hence biasing estimates of profit increases up in difference-in-differences. If anything, merger target banks are located in counties that are about to degrade on economic indicators, including decreasing population and increasing poverty and unemployment rates.

Figure D.6: Professional services

(a) Consultants



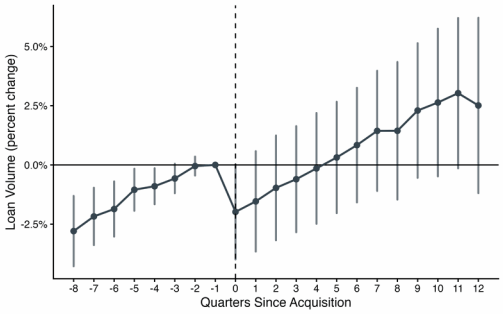

(b) Lawyers



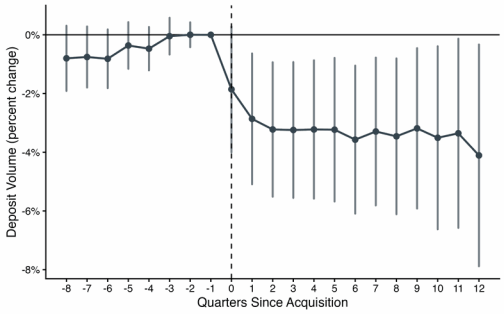

Note: This figure shows the estimates of $\beta_s$ from Equation (1) for professional services. The spike in professional services at $t = -1$ and consistent lack of pre-trend is consistent with the merger date being correctly identified as the RSSD subsumption date. Furthermore, the reduction in these estimates mirror the non-interest-expense reductions and strengthen the case that cost reductions after mergers in the difference-in-difference design were indeed due to the mergers themselves.

Figure D.7: Prices and quantities: stacked difference-in-differences

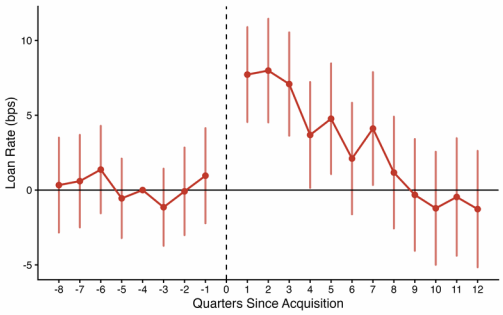
(a) Loans (log, dollars)



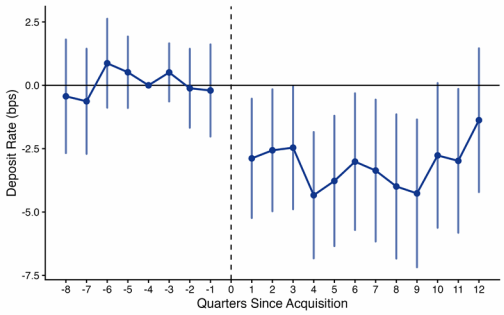

(b) Deposits (log, dollars)




(c) Loan rate

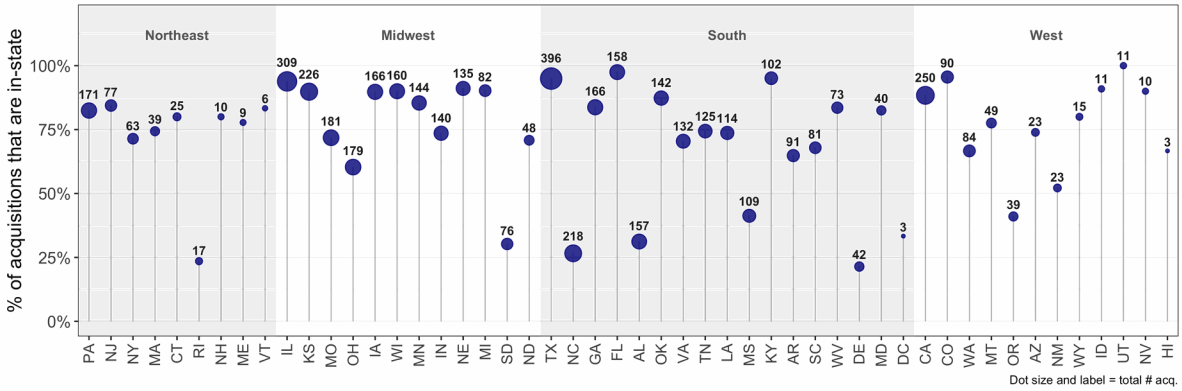



(d) Deposit rate




Note: This figure shows the estimates of $\beta_s$ from Equation (1) on the constructed population of bank mergers compared with a stacked population of banks that were never acquirers nor targets between 1990-2024. Panel (a) plots the estimates of $\beta_s$ for growth in dollars of loans. The graph shows a visible pre-trend prior to the merger where merging banks had higher growth in loans than control banks. Panel (b) plots the estimates of $\beta_s$ for $s \in [-8, 12]$ for growth in dollars of deposits. Panel (c) and (d) plot the estimates for loan and deposit rates. Deposit and loan rates are winsorized at the 1% and 99% level. Standard errors are clustered at the bank level. Vertical bars show 95% confidence intervals. The reduction in deposits after mergers compared to the matched sample is potential evidence that SUTVA violations are important to consider in constructing the treatment effect of a merger on banks. The decline, for instance, is consistent with competitors within the same state gaining share simultaneous with the merger, because using within-state competitors as controls suggests a decline in deposits while matching does not.

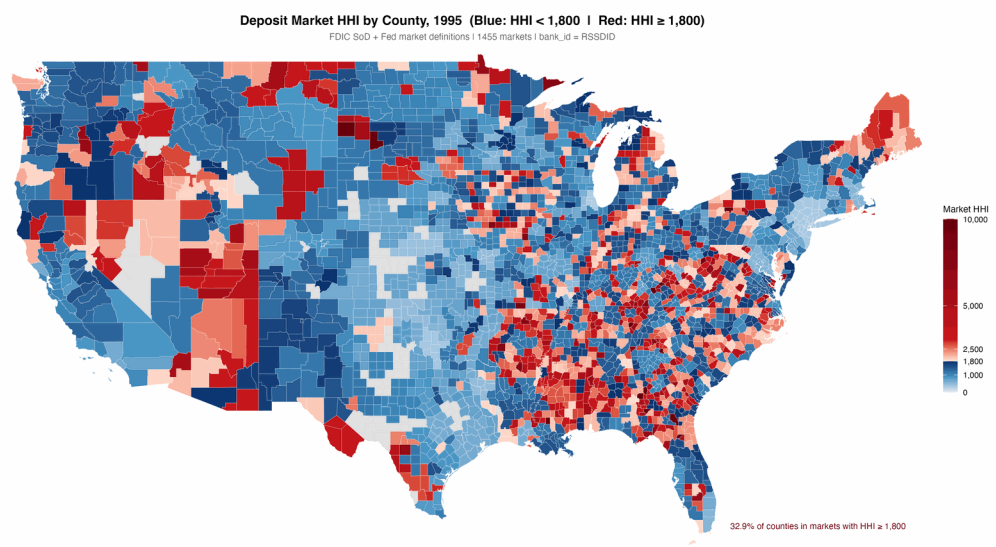
Figure D.8: In-state acquisitions



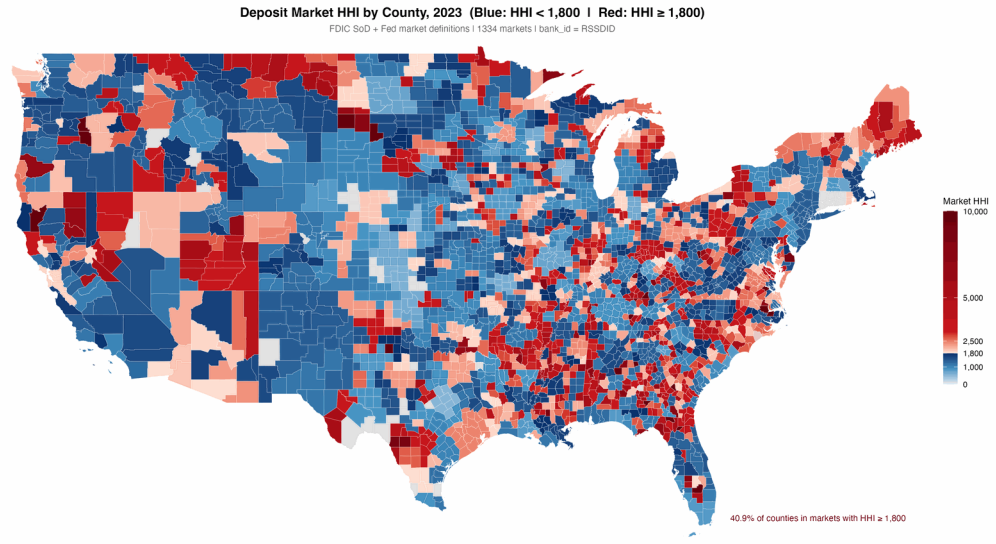

Note: This graph shows the percent of mergers occurring between 1995-2024 between banks headquartered in the same state. The size of the circle for a given state is the total number of mergers occurring between 1995-2024.

Figure D.9: Federal Reserve bank market definitions

(a) 1995




(b) 2023




Note: Panels (a) and (b) show a heatmap of Federal Reserve banking market maps, colored by their deposit market HHI. HHI's are computed with Call Reports filing institutions and do not include credit unions. 1995 market maps are assumed to be equivalent to 2009 maps provided by CASSIDI. 92% of the U.S. population lived in a competitive banking market as classified by the DOJ as moderately concentrated ($HHI < 1800$) in 2022, using population estimates from the ACS.

# E Proofs and theoretical results for Section 4

## E.1 Toy model with testable empirical predictions

In order to discipline our reduced form results, we formalize the logic of this constraint mathematically. A bank $i$ wishes to evaluate a target $j$ with a price of $\pi_j$. A merged bank of $i$ and $j$, indexed by $\mathcal{M}(i,j)$ has an idiosyncratic value of future profit synergies of $\Delta\pi_{\mathcal{M}(i,j)} \equiv \pi_{ij} - \pi_i - \pi_j$. The bank will finance their investment in the cheapest possible manner. In the event of a cash-financed merger, the firm can choose the lesser of two evils: keeping a permanently lower equity capital ratio and incurring a shadow cost $c^{Reg}$, or facing a marginal cost $c^{Ext}$ to raise external equity to recapitalize. To perform a stock merger, the firm incurs a marginal cost $c^{Ext}$ representing the inducements or premium required to entice the seller to accept buyer stock as opposed to cash that, as discussed, is roughly equivalent to the ability to solicit equity from external shareholders.[28] The buyer's choice financing choice becomes:

$$\min\left\{\underbrace{\min\left\{\underbrace{c^{\text{Reg}}}_{\text{Without recapitalizing}}, \underbrace{c^{\text{Ext}}}_{\substack{\text{Issuing equity}\\ \text{to recapitalize}}}\right\}}_{\text{Cash merger}}, \underbrace{c^{\text{Ext}}}_{\text{Stock merger}}\right\} \tag{36}$$

Or, simply, the firm chooses its cheapest financing, net of the method of payment and net of its *ex post* choice of whether to recapitalize. In other words, the buyer bank $i$'s valuation of bank $j$ is:

$$V_{ij} \equiv \Delta\pi_{\mathcal{M}(i,j)} - \underbrace{\min\left\{\underbrace{c^{\text{Reg}}}_{\text{Not issuing equity}}, \underbrace{c^{\text{Ext}}}_{\text{Issuing equity}}\right\}}_{\textit{regulatory capital wedge (RCW)}}. \tag{37}$$

In industries without capital regulation, $c^{Reg} \approx 0$, and the value of a potential merger is simply the value of its potential profit synergies, $V_{it} \approx \Delta\pi_{ij}$. In contrast, the bank proceeds with the merger if $V_i > 0$ (so merger $M_{ij} = \mathbf{1}(V_i > 0)$), taking the shadow cost of capital regulation into account. We refer to this wedge in the merger incentives as the *regulatory capital wedge* (RCW).

The setup delivers empirical predictions that Section 4 tests empirically. For simplicity of exposition, the following predictions assume that $c_i^{Reg}$, $\pi_i$, and $c_i^{Ext}$ are mutually statistically independent and $\Delta\pi \sim \mathcal{N}(1,1)$.

[28] We leave out the possibility of purchasing another bank in debt, which would further increase leverage. We implicitly assume that any cost of increasing leverage (i.e., increasing the probability of incurring bankruptcy cost by decreasing distance to default) is weakly smaller than $c^{Reg}$.

**Prediction 1.** *If $c_1 > c_2$, then*

$$\Pr(\textit{Not issuing equity} \mid M_{ij} = 1, c_i^{Ext} = c_1) \geq \Pr(\textit{Not issuing equity} \mid M_{ij} = 1, c_i^{Ext} = c_2). \quad (38)$$

Prediction 1 relies on variation in $c_i^{Ext}$ independent of variation in regulatory cost and merger synergies $\pi_{ij}$. It states that, given that the buyer has chosen to acquire the target, the use of cash is increasing in the cost of external equity issuance faced by the buyer.

**Prediction 2.**

$$\frac{dPr(M_{ij} = 1 \mid c^{Reg} = r, c^{Ext} = \bar{c})}{dr} < 0 \quad \textit{When } r < c_i^{Ext} \quad (39)$$

$$\frac{dPr(M_{ij} = 1 \mid c^{Reg} = r, c^{Ext} = \bar{c})}{dr} = 0 \quad \textit{When } r > c_i^{Ext} \quad (40)$$

Prediction 2 states that the inverse relationship between regulatory cost and merger propensity is conditioned on the cost of issuing external equity. When the cost of equity issuance is high, the RCW is active. The propensity of a combination consummating is decreasing in the regulatory cost that the combination would enforce. The prediction implicitly holds fixed $c_i^{Ext}$ and the value of the merger $\Delta\pi_i$ while varying $c_i^{Reg}$.

**Prediction 3.**

$$\frac{d\mathbb{E}[\Delta\pi_{ij} | c^{Reg} = r, c^{Ext} = \bar{c}, M_{ij} = 1]}{dr} \geq 0 \quad (41)$$

Prediction 3 is another corollary of the setup; if $c_i^{Reg}$ is a constraint (RCW is active), the mergers that do consummate are censored to be of higher value. The choice to acquire is a discrete choice, and so, with the assumptions established, $V_{ij}$ is nonincreasing in $c_i^{Reg}$.

Write $G$ for the CDF of $c^{\mathrm{Reg}}$, and $\varphi$, $\Phi$ for the standard normal density and CDF, so that $\Pr(\Delta\pi > x) = \Phi(1 - x)$. The proofs then follow:

***Proof, Prediction 1.*** It suffices to show that $P(c) \equiv \Pr(\text{Not issuing equity} \mid M_{ij} = 1, c_i^{Ext} = c)$ is nondecreasing in $c$. Given $c^{\mathrm{Ext}} = c$, not issuing is the event $\{c^{\mathrm{Reg}} \leq c\}$. By independence,

$$N(c) \equiv \Pr\big(M = 1,\, c^{\mathrm{Reg}} \leq c \,\big|\, c^{\mathrm{Ext}} = c\big) = \int_{[0,c]} \Phi(1-r)\, dG(r), \qquad T(c) \equiv \Phi(1-c)\big(1-G(c)\big), \quad (42)$$

with $T(c) = \Pr(M = 1, c^{\mathrm{Reg}} > c \mid c^{\mathrm{Ext}} = c)$. $N$ is nondecreasing and $T$ is nonincreasing (a product of nonnegative nonincreasing functions), so $P(c) = N(c)/\big(N(c) + T(c)\big) = 1/\big(1 + T(c)/N(c)\big)$ is nondecreasing wherever $N(c) > 0$, and $P(c) = 0$ where $N(c) = 0$. □

***Proof, Prediction 2.*** By independence, conditioning on $(c^{\mathrm{Reg}} = r, c^{\mathrm{Ext}} = \bar{c})$ leaves the distribution of $\Delta\pi$ unchanged, and by (45) the merger occurs iff $\Delta\pi > \min\{r, \bar{c}\}$. Hence, $p(r) = \Phi(1 -$

$\min\{r, \bar{c}\}$); taking derivatives in the two regimes of r yields

$$p'(r) \;=\; \begin{cases} -\varphi(1-r) \;<\; 0, & r < \bar{c} \quad \text{(wedge active)}, \\ 0, & r > \bar{c} \quad \text{(wedge slack)}. \end{cases} \tag{43}$$

□

***Proof, Prediction 3.*** $m(r) \equiv \mathbb{E}[\Delta\pi \mid M = 1,\, c^{\text{Reg}} = r,\, c^{\text{Ext}} = \bar{c}] = 1 + \lambda\big(\min\{r, \bar{c}\} - 1\big)$, where $\lambda(z) \equiv \varphi(z)/\big(1 - \Phi(z)\big)$, so

$$m'(r) \;=\; \begin{cases} \lambda(r-1)\left[\lambda(r-1) - (r-1)\right] \;>\; 0, & r < \bar{c} \quad \text{(wedge active)}, \\ 0, & r > \bar{c} \quad \text{(wedge slack)}. \end{cases} \tag{44}$$

□

## E.2 General forms of Predictions 1–3

**Definitions.** Suppress buyer-seller subscripts. The random variables assigned to are $(\Delta\pi, c^{\text{Reg}}, c^{\text{Ext}})$: the merger synergy, the shadow cost of operating closer to the regulatory capital floor, and the cost of inducing shareholders to absorb equity. Define

$$w \equiv \min\{c^{\text{Reg}}, c^{\text{Ext}}\}, \qquad M = \mathbf{1}\{\Delta\pi > w\}. \tag{45}$$

Let $F_c$ denote the probability distribution of $c^{\text{Reg}}$ among consummated mergers at $c^{\text{Ext}} = c$:

$$F_c(s) \;\equiv\; \Pr\big(c^{\text{Reg}} \le s \,\big|\, M = 1,\; c^{\text{Ext}} = c\big), \qquad s \ge 0, \tag{46}$$

"Not issuing equity" is the event $\{c^{\text{Reg}} \le c^{\text{Ext}}\}$ and $M = 1$.

To interpret method of payment as meaningful, we need that among consummated mergers, the buyer's facing a higher $c^{Ext}$ have weakly lower regulatory cost $c^{Reg}$ stochastically. To use differences in profit synergies as indicative of meaningful $c^{Reg}$, profit synergies must move independently from $c^{Reg}$. These two statements motivate Assumptions 1 and 2.

**Assumption 1** (First order stochastic dominance)**.** *For $c_1 > c_2$ and every $s \ge 0$, $F_{c_1}(s) \ge F_{c_2}(s)$.*

**Remark 1.** *$(c^{Reg}, \Delta\pi) \perp c^{Ext}$ is sufficient to satisfy Assumption 1.*

**Assumption 2.** $\forall \bar{c},\ \Delta\pi \;\perp\; c^{\text{Reg}} \,\big|\, c^{\text{Ext}} = \bar{c}$

**Remark 2.** *Mutual independence of $(c^{Reg}, \Delta\pi, c^{Ext})$ (the simplifying assumption in the main text) is sufficient to satisfy both Assumption 1 and Assumption 2.*

**Proposition 1.** *Under Assumption 1, if $c_1 > c_2$, then*

$$P(\textit{Not issuing equity} \mid M = 1, c^{Ext} = c_1) \ge Pr(\textit{Not issuing equity} \mid M = 1, c^{Ext} = c_1)$$

*Proof.* First, notice

$$F_{c_1}(c_1) = P(\text{Not issuing equity} \mid M = 1, c^{Ext} = c_1)$$

and

$$F_{c_2}(c_2) = P(\text{Not issuing equity} \mid M = 1, c^{Ext} = c_2).$$

Then,

$$F_{c_1}(c_1) \underbrace{\geq}_{(1)} F_{c_1}(c_2) \underbrace{\geq}_{(2)} F_{c_2}(c_2)$$

(1) holds because $F_{c_1}$, being a CDF, is nondecreasing in its argument and $c_1 > c_2$.
(2) is Assumption 1 evaluated at $s = c_2$. □

**Proposition 2.** *Fix $\bar{c}$. Let $p(r) \equiv \Pr(M = 1 \mid c^{\text{Reg}} = r,\ c^{\text{Ext}} = \bar{c})$. Under Assumption 2:*

*(a) $p$ is nonincreasing on $[0, \bar{c}]$ and constant on $[\bar{c}, \infty)$.*

*(b) If in addition $\Delta\pi$ given $c^{\text{Ext}} = \bar{c}$ admits a density $f_{\bar{c}}$, then*

$$p'(r) \;=\; \begin{cases} -\,f_{\bar{c}}(r) \;\leq\; 0, & r < \bar{c} \quad (\textit{wedge active}), \\ 0, & r > \bar{c} \quad (\textit{wedge slack}), \end{cases} \tag{47}$$

*with $p'(r) < 0$ wherever $f_{\bar{c}}(r) > 0$.*

*Proof.* Write $S_{\bar{c}}(x) \equiv \Pr(\Delta\pi > x \mid c^{\text{Ext}} = \bar{c})$.

By Assumption 2, conditioning on $c^{\text{Reg}} = r$ leaves this probability unchanged. Since $M = \mathbf{1}\{\Delta\pi > \min\{r, \bar{c}\}\}$ by (45), $p(r) = S_{\bar{c}}\big(\min\{r, \bar{c}\}\big)$.
(a) $x \mapsto S_{\bar{c}}(x)$ is nonincreasing, and the threshold $\min\{r, \bar{c}\}$ is nondecreasing in $r$ below $\bar{c}$ and constant above.
(b) Below $\bar{c}$ the threshold is $r$ itself, so $p(r) = S_{\bar{c}}(r)$ and $S'_{\bar{c}} = -f_{\bar{c}}$; above $\bar{c}$ the threshold is fixed. □

**Proposition 3.** *Suppose $\mathbb{E}[|\Delta\pi| \mid c^{\text{Ext}} = \bar{c}] < \infty$, and let $m(r) \equiv \mathbb{E}[\Delta\pi \mid M = 1,\ c^{\text{Reg}} = r,\ c^{\text{Ext}} = \bar{c}]$, defined wherever $p(r) > 0$. Under Assumption 2:*

*(a) $m$ is nondecreasing on $[0, \bar{c}]$ and constant on $[\bar{c}, \infty)$.*

*(b) If in addition $\Delta\pi$ given $c^{\text{Ext}} = \bar{c}$ admits a density $f_{\bar{c}}$, then with $h(w) \equiv f_{\bar{c}}(w) \,/\, \Pr(\Delta\pi > w \mid c^{\text{Ext}} = \bar{c})$,*

$$m'(r) \;=\; \begin{cases} h(r)\,\big(m(r) - r\big) \;\geq\; 0, & r < \bar{c} \quad (\textit{wedge active}), \\ 0, & r > \bar{c} \quad (\textit{wedge slack}), \end{cases} \tag{48}$$

*with $m'(r) > 0$ wherever $f_{\bar{c}}(r) > 0$.*

*Proof.* Notice that under Assumption 2, $m(r)$ can be rewritten as

$$m(r) = \mathbb{E}\big[\Delta\pi \,\big|\, \Delta\pi > \min\{r, \bar{c}\},\ c^{\text{Ext}} = \bar{c}\big].$$

(a) The threshold $w = \min\{r, \bar{c}\}$ is nondecreasing in $r$ and constant above $\bar{c}$, so it suffices that the truncated mean is nondecreasing in $w$.
Take $w_1 \geq w_2$ with $\Pr(\Delta\pi > w_1 \mid \bar{c}) > 0$ (all probabilities and expectations below condition on $c^{\text{Ext}} = \bar{c}$). For every $t$, $\Pr(\Delta\pi > t \mid \Delta\pi > w_1) \geq \Pr(\Delta\pi > t \mid \Delta\pi > w_2)$: for $t \leq w_1$ the left side is 1; for $t > w_1 \geq w_2$ the two sides are $\Pr(\Delta\pi > t)/\Pr(\Delta\pi > w_1)$ and $\Pr(\Delta\pi > t)/\Pr(\Delta\pi > w_2)$, and $\Pr(\Delta\pi > w_1) \leq \Pr(\Delta\pi > w_2)$.

So the censored distribution of $\Delta\pi$ above $w_1$ first order stochastically dominates the censored distribution of $\Delta\pi$ above $w_2$. So, $\mathbb{E}[\Delta\pi \mid \Delta\pi > w_1] \geq \mathbb{E}[\Delta\pi \mid \Delta\pi > w_2]$.
(b) With $S_{\bar{c}}$ as in the proof of Proposition 2, write $m(w) = \big(\int_w^\infty x f_{\bar{c}}(x)\,dx\big)/S_{\bar{c}}(w)$ and differentiate:

$$m'(w) = \frac{-\,w f_{\bar{c}}(w)\, S_{\bar{c}}(w) + f_{\bar{c}}(w) \int_w^\infty x f_{\bar{c}}(x)\,dx}{S_{\bar{c}}(w)^2} = h(w)\,\big(m(w) - w\big); \tag{49}$$

below $\bar{c}$ the truncation point is $w = r$, above it is fixed. $f_{\bar{c}}(w) > 0$ implies $h(w) > 0$ and implies $S_{\bar{c}}(w) > 0$, and $m(w) > w$ by construction. So, $m'(r)$ is weakly positive. □

**Remark 3.** *Where the wedge is active, the slope of merger propensity is (minus) the density of marginal deals extinguished by regulatory cost, and the selection gradient is the exit rate of those marginal deals, $h(r)$, times the markup of the average selected synergy over the marginal one, $m(r) - r$.*

# F Additional empirical results for Section 4

Table F.1: Merger propensity versus unrealized gains

| | (1) | (2) |
|---|---|---|
| Unrealized Gains (pp) x Pre-2001 | 3.956 | 1.045 |
| | (11.015) | (11.018) |
| Unrealized Gains (pp) x Post-2001 | 17.806*** | 17.403*** |
| | (4.064) | (4.069) |
| Return on Assets (pp) | | -3.760*** |
| | | (1.068) |
| Efficiency Ratio (pp) | | 0.014 |
| | | (0.011) |
| Tier 1 Ratio (pp) | | -4.442*** |
| | | (0.730) |
| Securities / Assets (pp) | | -1.999*** |
| | | (0.427) |
| Bank $\times$ (Post 2001) FE | X | X |
| quarter-year FE | X | X |
| Observations | 806,102 | 804,410 |
| $R^2$ | 0.240 | 0.239 |

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: This table shows the estimates from Equation (11), with standard errors clustered at the bank level. The results show that, with bank and time fixed effects, a 1 pp increase in unrealized gains corresponds to a 17 bp increase in the propensity to become a merger target in the next 12 months. With a 2.1% merger rate, a standard deviation increase in unrealized gains shifted the likelihood of a bank becoming a merger target by 19bps (8.8%). Standard erorrs are clustered at the bank-quarter-year level.

Table F.2: Merger propensity versus unrealized gains, pre- and post-2001

| | Unrealized Gains (pp) | | $\mathbf{1}$(Seller) (bps) | | | |
|---|---|---|---|---|---|---|
| | First stage | | OLS | | 2SLS | |
| | Pre-2001 | Post-2001 | Pre-2001 | Post-2001 | Pre-2001 | Post-2001 |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| $w_{i,t-8} \times \Delta y_t^{5\text{-}yr}$ | -0.649*** | -2.348*** | | | | |
| | (0.021) | (0.046) | | | | |
| Unrealized Gains (pp) | | | -22.152** | -4.190 | -31.414 | 20.744*** |
| | | | (9.598) | (3.892) | (68.873) | (7.861) |
| First-stage $F$ (Kleibergen–Paap) | 989.4 | 2577.1 | | | | |
| Size $\times$ quarter-year FE | X | X | X | X | X | X |
| Observations | 267,155 | 519,219 | 267,155 | 519,219 | 267,155 | 519,219 |
| $R^2$ | 0.395 | 0.600 | 0.004 | 0.005 | 0.004 | 0.005 |

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: This table shows the estimates from Equation (**??**) and (**??**) for the 1994-2001 and 2002-2024 subpopulations separately. As expected, only the 2002-2024 population yields a statistically significant second stage coefficient $\tau$ on unrealized gains. The differential effect of unrealized gains 2002-2024 is consistent with unrealized gains shifting capital and thereby increasing merger propensity.

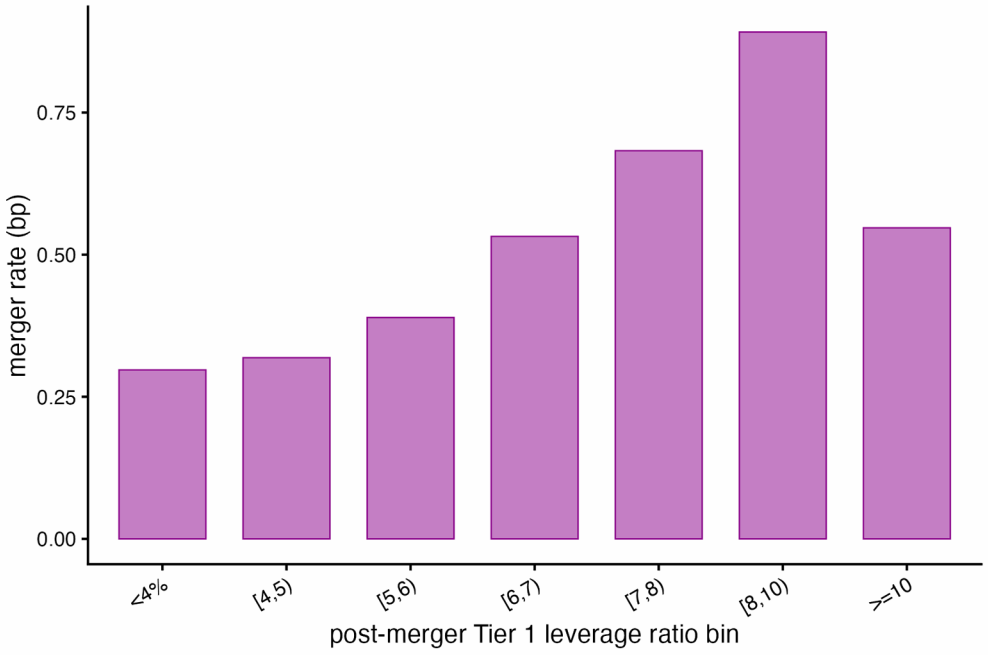
Table F.3: Use of cash in merger transactions, by listing status of buyer

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| Private Buyer | 0.305*** | 0.368*** | 0.340*** | 0.326*** | 0.326*** |
| | (0.014) | (0.014) | (0.015) | (0.015) | (0.015) |
| Private Target | | | 0.169*** | 0.163*** | 0.158*** |
| | | | (0.020) | (0.020) | (0.020) |
| Buyer Branches (std.) | | | | -0.045*** | -0.038*** |
| | | | | (0.007) | (0.007) |
| Target Branches (std.) | | | | | -0.018* |
| | | | | | (0.007) |
| Num.Obs. | 4495 | 4495 | 4495 | 4495 | 4495 |
| R2 | 0.095 | 0.185 | 0.198 | 0.206 | 0.207 |
| FE: ANNOUNCE_YEAR | | X | X | X | X |

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

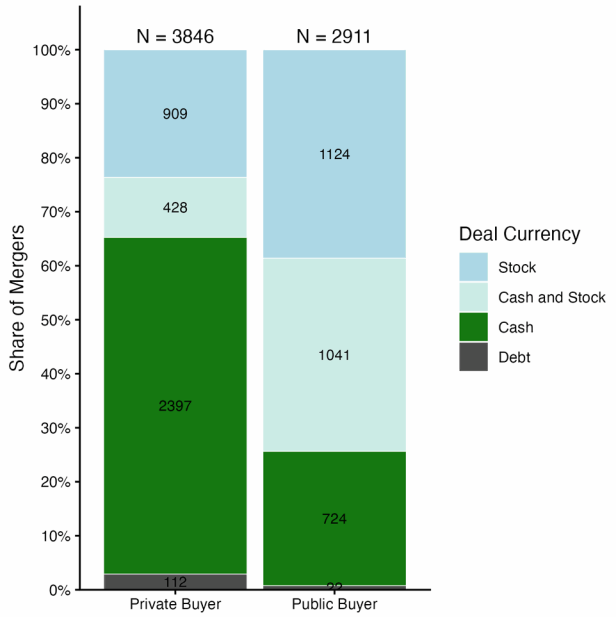
Note: This table shows the result of estimating Equation (16) with increasing covariates using S&P SNL data between 1991-2024 who list the breakdown of the deal value by currency (stock, cash, or debt). We limit the population of commercial bank mergers to those with a reported number of branches for the target and buyer and those who report a breakdown of the deal value. The coefficient on private buyer is interpreted as the additional share of the acquisition funded in cash, as opposed to stock or debt, that is associated with being a privately held (rather than publicly traded) buyer. With an increasing number of fixed effects controlling for the size of the acquirer, the target, and the year of announcement, the relationship between being a publicly traded acquirer and using more cash to finance the merger is stable. In the rightmost specification, being a privately held buyer is associated with funding 32.6pp more of the acquisition's value in cash.

Figure F.1: Merger propensity by Tier 1 Leverage Ratio



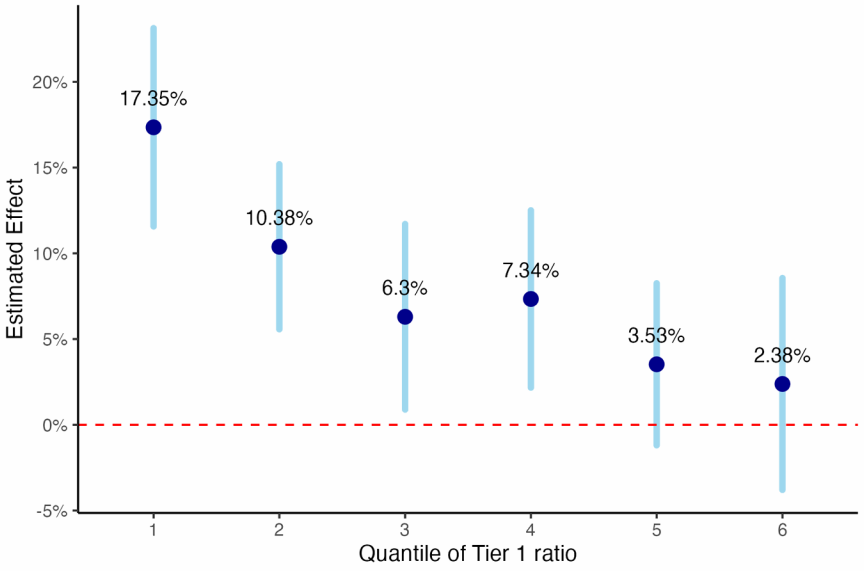

Note: This graph shows the merger rate for potential pairs of banks, cut by the post-merger leverage ratio. The leverage ratio is the Tier 1 Leverage Ratio, computed using reported Tier 1 Capital and Tier 1 Assets in Call Reports. Post-merger Tier 1 Leverage ratio is computed assuming that the buyer buys the target in cash, reducing the pro forma company's assets by the value of the target's equity. The equation is given in Equation (7). Each observation is a potential merger combination, where potential merger combinations are within-state mergers between 1990-2024 where the seller is smaller than the buyer bank in the year prior to the merger.

Figure F.2: Transaction medium, SNL bank merger data



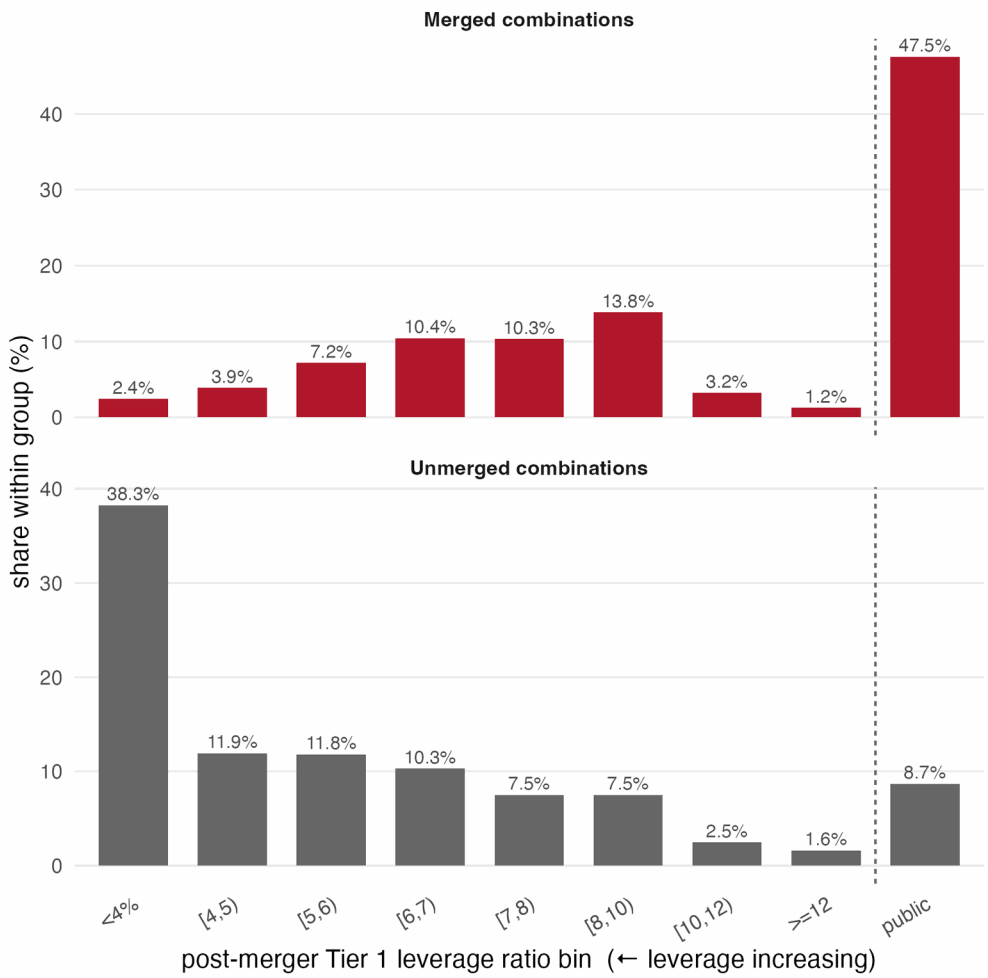

Note: This graph shows the breakdown of transaction medium for bank-to-bank acquisitions reported in the S&P SNL data for bank mergers announced between 1991-2024. 62% of private buyer acquisitions use cash alone to pay for the acquiring bank, compared to just 25% of public buyer acquisitions. 11% of private acquisitions use a mix of cash and stock, and 24% use stock alone. In contrast, stock is part of the deal value for 74% of public buyer transactions.

Figure F.3: Net income gains by Tier 1 Leverage Ratio




Note: This figure reports the coefficients $\beta_s$ for net income from Equation (1) for two-years after merger for bank mergers in our sample, interacted with the quantile of pro forma Tier 1 ratio two years prior to combination. The regression implies that net income increased for pro forma banks with the lowest quantile Tier 1 ratio by 17%. Increases were smaller as the combined bank was more capitalized and as low as 2% for combinations that were the most capitalized prior to the transaction. Standard errors are clustered at the bank level. Vertical bars show 95% confidence intervals.

Figure F.4: Merger counts by Tier 1 Leverage Ratio




Note: This graph shows a histogram of number of combinations in each post-merger Tier 1 Leverage Ratio bin. The relative counts of combinations in each bin that merge and do not merge create Figure F.1.

# G Proofs and microfoundations for Section 5

## G.1 Proof for Theorem 1

*Proof.* Given a set of firms $\mathcal{J}'_{kt}$ posting a set of prices $\boldsymbol{\rho}^D, \boldsymbol{\rho}^L$, the allocation of consumers to firms in horizontal Bertrand-Nash competition is generically unique because consumer utilities functions specified by $\alpha, \rho^D, \xi^D, \rho^L, \xi^L$ have unique maxima (Equation (27)).

Given a set of firms, each firm has a single optimal choice of $\rho^D_{jkt}$ and a single optimal choice of $\rho^L_{jkt}$. (Firm problem has unique maximum, Equation (26)).

The most novel part of the proof is showing a unique allocation in the merger market governed by Equation (25). The number of banks in a market $k$ at time $t$ $\mathcal{J}_{kt}$ is an integer, $\mathcal{J}_{kt} \in \mathbb{Z}$. To prove that the allocation is unique, represent Step 1 of the game using a directed graph. The adjacency matrix to that directed graph, $P$ is a square $\mathcal{J}_{kt} \times \mathcal{J}_{kt}$ matrix that captures the outcomes of Step 1.

This graph is an upper triangular matrix where $P_{ji} = 1$ indicates that bank $j$ chooses to sell to bank $i$ and $P_{j\ell} = 0 \; \forall$ elements $\ell \neq i$. $P$ represents a unique allocation given a set of surpluses because every bank will generically have a highest surplus option to sell to. Furthermore, the mapping of primitives $\boldsymbol{\Omega}_{kt}$ to surpluses is unique because of the belief structure posited. Banks choose merger partners by building counterfactual profits holding the other players fixed without additional M&A, so no expectations over opponent play could create lumps in the objective that would risk the disappearance of a unique maximum (e.g., a coordination game that requires the payoff of one player to affect another).

The chain resolution stage simply rewrites the allocation in a directed graph that is now defined by the adjacency matrix $P^{\mathcal{J}_{kt}-1}$. Since raising a matrix to a power is well-defined if the matrix is square, then a well-defined $P$ implies a well-defined $P^{\mathcal{J}_{kt}-1}$. □

## G.2 Microfoundations of the merger market allocation rule

For each bank $j \in \mathcal{J}_{kt}$, the eligible buyer pool is the set of weakly larger banks,

$$\mathcal{B}_{jt} := \{\, i \in \mathcal{J}_{kt} \setminus \{j\} : \text{book assets of } i \geq \text{book assets of } j \,\},$$

$V_{jt} + \Phi(Z_{ijt})$ denotes buyer $i$'s valuation of acquiring bank $j$ as in (24), where $V_{jt}$ is defined to add back in the standalone value of $j$, $V_{jt} \equiv \mathcal{C}^S + \pi_j(\boldsymbol{\Omega}_{kt})$. The allocation rule used to compute the likelihood of realized merger choices is:

$$d_{jt} = \begin{cases} \arg\max_{i \in \mathcal{B}_{jt}} \Phi(Z_{ijt}), & \text{if } \max_{i \in \mathcal{B}_{jt}} \Phi(Z_{ijt}) \geq 0, \\ 0, & \text{otherwise.} \end{cases} \tag{A}$$

We present three mechanisms that microfound this allocation rule (A): a centralized seller-proposing mechanism, simultaneous second-price auctions, and simultaneous ascending auctions.

The mechanisms imply different transfers, but because the likelihood depends only on acquirer

identities and never on transaction prices, they are observationally equivalent without transaction prices.

**Remark 4** (Role of the size restriction)**.** *Limiting $\mathcal{B}_{jt}$ to weakly larger banks makes simultaneous execution coherent by ruling out the possibility of cycles of buyers and sellers. For instance, if seller $A$ names buyer $B$ while $B$ itself names buyer $C$, the planner resolves the chain: $B$ is sold to $C$, and $A$ is absorbed by $C$ through $B$. Such chains ascend the size order and the resolution is well-defined. The size restriction thus substitutes for the sequencing device that renders allocation well-defined in a sequential model. Monotonicity is strict only under a strict size order; with ties, a tie-breaking rule is required to rule out two equal-size banks naming each other.*

We impose the following belief assumptions.

**Assumption 3** (Zero conjectural variation)**.** *Each agent's beliefs assign probability one to the event that no transaction other than its own occurs in period $t$.*

**Assumption 4** (Myopia)**.** *Each agent treats period-$t$ outcomes as once-and-for-all: the value of retaining bank $j$ is $V_{jt}$, the value to buyer $i$ of acquiring bank $j$ at price $p$ is $\Phi(Z_{ijt}) + V_{jt} - p$, and there is no option value of transacting in future periods.*

**Assumption 5** (Narrow bracketing)**.** *Buyer $i$'s valuation of target $j$ equals $\Phi(Z_{ijt})$ irrespective of the set of other banks $i$ owns or wins.*

### G.2.1 Centralized seller-proposing mechanism

**Game.**

1. Each seller $j \in \mathcal{J}_{kt}$ simultaneously reports an assignment $a_j \in \mathcal{B}_{jt} \cup \{j\}$, where $a_j = j$ denotes retention.
2. The planner executes every report: if $a_j = i \neq j$, bank $j$ is sold to bank $i$; buyers cannot decline.
3. Transfers: a seller assigned to buyer $i$ receives $p_{ij} = \Phi(Z_{ijt}) + V_{jt}$.

**Proposition 4.** *Under Assumptions 3–4, it is weakly dominant for every seller $j$ to report $a_j = \arg\max_{i \in \mathcal{B}_{jt}} \Phi(Z_{ijt})$ if $\max_{i \in \mathcal{B}_{jt}} \Phi(Z_{ijt}) \geq V_{jt}$ and $a_j = j$ otherwise. The induced allocation is* (A).

*Proof.* Under Assumption 3, seller $j$ believes its report affects only its own assignment and transfer. Under Assumption 4, staying independent yields 0 and reporting $a_j = i$ yields the transfer $p_{ij} = \Phi(Z_{ijt})$. The seller's payoff is hence $\Phi(Z_{ijt})$ or zero, independent of other reports, so the stated report maximizes $j$'s payoff against every profile of other reports and is weakly dominant. The mechanism executes reports directly, so the allocation is (A). □

**Remark 5** (Resolving chains)**.** *Under Assumption 3, sellers assign probability zero to such events when reporting. If the bank-level transaction records ($d_A = B$, $d_B = C$) could be incorporated into the likelihood, estimation would precisely reflect the model.*

**Remark 6** (Transfers)**.** *Full surplus extraction ($p_{ij} = \Phi(Z_{ijt})$) is sufficient but not necessary. Any transfer rule strictly increasing in the named buyer's valuation and satisfying $p_{ij} \geq V_{jt} \iff \Phi(Z_{ijt}) \geq 0$ leaves the seller's induced ranking over $\mathcal{B}_{jt} \cup \{j\}$, and hence Proposition 4, unchanged. One example of a transfer rule could be Nash Bargaining over the incremental surplus created, where price $p$ minus standalone value $V_j$ satisfies $p_{ij} - V_j \equiv \arg\max_q (\Phi_{it} - q)^{\lambda^B} q^{\lambda^S}$ and $\lambda^B/\lambda^S$ is constant for every potential merger pair. In this case, $p_{ij} = V_{jt} + \frac{\lambda^S}{\lambda^B + \lambda^S} \Phi_{ij}$.*

### G.2.2 Simultaneous second-price auctions

**Game.**

1. All banks in $\mathcal{J}_{kt}$ are put up for sealed-bid second-price auction simultaneously. The eligible bidder pool for bank $j$ is $\mathcal{B}_{jt} \cup \{j\}$; equivalently, each bank bids on every bank with weakly smaller book assets, itself included.

2. Bank $i$ submits a bid $b_{ij} \geq 0$ on each eligible target $j$; the self-bid $b_{jj}$ serves as $j$'s reservation price.

3. In each auction, the bank is awarded to the highest bidder at the second-highest bid; if the self-bid is highest, the bank is retained. All winning bids are binding and executed.

**Proposition 5.** *Under Assumptions 3–5, the strategy profile $b_{ij} = \Phi(Z_{ijt}) + V_{jt}$ for all $i \neq j$ and $b_{jj} = V_{jt}$ is weakly dominant. The induced allocation is* (A).

*Proof.* By Assumption 5, bidder $i$'s value of winning auction $j$ at price $p$ is $\Phi(Z_{ijt}) - p$, independent of the outcomes of all other auctions; by Assumptions 3–4, no other transaction enters $i$'s perceived objective. Each bidder's problem are therefore separable across auctions into isolated private-value second-price auctions, in which bidding one's valuation is weakly dominant (Vickrey, 1961). The analogous argument implies $b_{jj} = V_{jt}$ for the seller. Each auction thus awards $j$ to $\arg\max_{i \in \mathcal{B}_{jt}} \Phi(Z_{ijt})$ if and only if the maximand exceeds $V_{jt}$, which is (A): competitive bidding computes the maximum that the seller computes directly in the centralized mechanism. □

**Remark 7.** *Because winning bids are binding, simultaneity can produce chains: if $B$ wins $A$ while $C$ wins $B$, then $C$ acquires $B$ and, through $B$, executes $B$'s winning bid on $A$. Ex post, chain execution is in tension with Assumption 5, since the price paid for $A$ reflects $B$'s valuation rather than $C$'s. Under Assumption 3, however, bidders assign probability zero to such chains at the bidding stage, so equilibrium bids and hence auction outcomes are unaffected.*

### G.2.3 Simultaneous ascending auctions

**Game.** The game is identical to the simultaneous second-price game except for the auction format: each bank is put up for ascending-price auction, all auctions run concurrently, each seller $j$ sets reservation price $V_{jt}$, and in each auction the last remaining bidder wins at the price at which its final rival exits.

**Proposition 6.** *Under Assumptions 3–5, it is weakly dominant for each bidder $i$ to remain in the auction for bank $j$ until the price reaches $\Phi(Z_{ijt}) + V_{jt}$. The induced allocation is* (A).

*Proof.* With private values, the ascending auction is strategically equivalent to the sealed-bid second-price auction: exiting at one's valuation is weakly dominant and induces the same allocation and price (Vickrey, 1961; Milgrom and Weber, 1982). Assumptions 3–5 imply that each bidder's valuation in every auction equal to $\Phi(Z_{ijt})$, independent of concurrent auctions, exactly as in the proof of Proposition 5. The allocation in each auction is therefore (A). □

# H Estimation, Step 1: Merger market

Define the draw $r$ for one merged bank $j$ in market $k$ at time $t$ as $\tilde{\kappa}^{m,(r)}_{jkt}$, or $\tilde{\boldsymbol{\kappa}}^{m,(r)}_{kt}$ for the whole market vector. [29]

## H.1 Likelihood function for merger choices

We start by deriving $L^{sel}_{kt}(\boldsymbol{M}_{kt} \mid \theta)$.

**Deriving merger choice probabilities.** To derive the likelihood of the realized assignments of buyers to sellers, in a given market (U.S. state) and year, we construct the merger surpluses that drive selection (merger choices). Define "mean" surplus and "mean" profit that exclude the fixed cost pair-specific shock $\zeta_{jkt}$ with $\bar{\Phi}_{ij,kt}$ and $\bar{\pi}_{ij,kt}$. These quantities are still conditional on a particular vector of fixed costs for merged and unmerged firms, $\kappa^u$ and draw for merged firms $\tilde{\kappa}^m$. Using the merger-surplus expression (24), the mean surplus of pair $(i,j)$ is

$$\bar{\Phi}_{ij,kt} \equiv \frac{\Delta\bar{\pi}_{ij,kt}}{1-\beta} - \mathcal{C}^B_{ij,kt} - \tau A_{j,kt}, \tag{50}$$

In estimation, for tractability, in computing the deposit and loan prices charged a given pair's $\Delta\bar{\pi}_{ij,kt}$, we hold competitors' prices fixed.[30]

Applying the distributional assumption for $\zeta_{ijt}$ from Equation (18) and (24), the multinomial-logit probability that after Step 1 (before chain resolution), buyer $i$ is matched to any given seller $j$ is:

$$\Pr\big(P_{ij,kt}=1 \,\big|\, \tilde{\boldsymbol{\kappa}}^m_{kt}, \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, ; \theta\big) = \frac{\exp\big(\bar{\Phi}_{ij,kt}/s_\zeta\big)}{\sum_{\ell\in\mathcal{B}_{jkt}}\exp\big(\bar{\Phi}_{\ell j,kt}/s_\zeta\big)}, \qquad s_\zeta \equiv \frac{\sigma_\zeta A_j}{1-\beta}. \tag{51}$$

where acquiring oneself is included as $P_{jj}$ and the merger surplus is $\zeta_{jjt}$.

**Parameterizing RCW.** We parameterize merger frictions for estimation. We parameterize RCW to scale with seller assets and to apply differentially to privately held, rather than publicly traded buyer banks, consistent with Prediction 1 and empirical merger propensities. We define the indicator for privately held as $D_{ikt} \in \{0,1\}$.[31] We allow the bank's distance from the Tier 1 Leverage ratio to be

[29] The base shocks $\varepsilon^{(r)}_{kt} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ are drawn *once* at the start of estimation and held fixed across optimizer iterations; $\theta$-dependent affine transformations convert these standardized draws into conditional draws at any candidate $\theta$. This holds the simulated likelihood smooth as a function of $\theta$, which materially speeds optimization.

[30] Holding non-focal banks' prices fixed at observed values is the key simplification that makes the per-pair counterfactual tractable. The merged entity's logit denominator updates in closed form: $D^{(r)}_{ij,kt} = D^{(r)}_{kt,\text{baseline}} + \exp(V^{\mathcal{M},(r)}_{ij,kt}) - \exp(V^{(r)}_{ikt}) - \exp(V^{(r)}_{jkt})$, where $V$ denotes the indirect utility index. This avoids re-solving the equilibrium for every $(R,J,J)$ triple. However, the full solution is in principle feasible.

[31] Public banks still face one-time merger costs, but since banks are either private or public, they also must act as the reference category and are subsumed into the fixed cost estimated for $\tau$.

estimated by quantiles:

$$C^B_{ijkt} \equiv \sum_{q=1}^{6} c_q D_{ikt} \mathbf{1}\left(\frac{K_{ikt}}{A_{ikt} + A_{jkt} - K_{jkt}} \in \text{quantile q}\right) A_{jkt} \tag{52}$$

**Substituting final ownership matrix for auction outcomes.** Like estimation of matching in trading networks (Fox, 2017), the selection likelihood is the joint probability of all sellers' realized assignments, including the result of remaining standalone. Under our assumption of myopic surplus maximization, sellers' assignments are independent of one another conditional on the latent fundamentals.

Recall $\boldsymbol{P}_{kt}$ is the ownership matrix generated by sellers' initial assignments, before chains are resolved. Conditional on some realization of the market's shocks to fixed costs, $\widetilde{\boldsymbol{\kappa}}_{kt}$, the probability of realizing an initial ownership matrix $\boldsymbol{P}_{kt}$ is

$$f_{M|\kappa^u}\left(\boldsymbol{P}_{kt} \mid \boldsymbol{y}_{kt}, \widetilde{\boldsymbol{\kappa}}_{kt}; \theta\right) = \prod_{(i,j):P_{ij,kt}=1} \Pr(P_{ij,kt} = 1 \mid \boldsymbol{y}_{kt}, \widetilde{\boldsymbol{\kappa}}_{kt}; \theta) \tag{53}$$

Because the mapping from $\boldsymbol{P}_{kt}$ to $\boldsymbol{M}_{kt}$ is generally many-to-one, the exact conditional probability of the observed ownership matrix is

$$\Pr(\boldsymbol{M}_{kt} \mid \boldsymbol{y}_{kt}, \widetilde{\boldsymbol{\kappa}}_{kt}; \theta) = \sum_{\left\{\boldsymbol{P}:\boldsymbol{P}^{\mathcal{J}_{kt}-1}=\boldsymbol{M}_{kt}\right\}} \Pr(\boldsymbol{P} \mid \boldsymbol{y}_{kt}, \widetilde{\boldsymbol{\kappa}}_{kt}; \theta)\,, \tag{54}$$

is the set of initial ownership matrices consistent with the observed final ownership matrix. For estimation, we impose the restriction that no chains need to be resolved, so that

$$\boldsymbol{P}_{kt} = \boldsymbol{M}_{kt}. \tag{55}$$

Under this restriction, the observed matrix $\boldsymbol{M}_{kt}$ identifies each seller's initial assignment. So,

$$\Pr(\boldsymbol{M}_{kt} \mid \boldsymbol{y}_{kt}, \widetilde{\boldsymbol{\kappa}}_{kt}; \theta) = \prod_{(i,j):M_{ij,kt}=1} \Pr(M_{ij,kt} = 1 \mid \boldsymbol{y}_{kt}, \widetilde{\boldsymbol{\kappa}}_{kt}; \theta) \tag{56}$$

**Numerical integration.** We then use this expression to create a simulated likelihood of merger choices. Because $\widetilde{\boldsymbol{\kappa}}^m_{kt}$ is unobserved, it must be integrated out. As a result, the marginal selection likelihood is

$$L^{\text{sel}}_{kt}(\theta) = \int \Pr(\boldsymbol{M}_{kt} \mid \boldsymbol{y}_{kt}, \widetilde{\boldsymbol{\kappa}}^m_{kt}, \boldsymbol{\kappa}^u_{kt}, \theta) f_{\widetilde{\kappa}^m|\kappa^u}(\widetilde{\boldsymbol{\kappa}}^m_{kt} \mid \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta)\; d\widetilde{\boldsymbol{\kappa}}^m_{kt}. \tag{57}$$

We evaluate the integral by simulation. For draw $\widetilde{\boldsymbol{\kappa}}^{m,(r)}_{kt}$ indexed by draw number $r$ out of the

density $f_{\widetilde{\kappa}^m|\kappa^u}\left(\widetilde{\boldsymbol{\kappa}}^m_{kt} \mid \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta\right)$, define the per-draw conditional likelihood as

$$L^{\text{sel},(r)}_{kt}(\theta) \equiv \prod_{(i,j):M_{ij,kt}=1} \Pr\Big(M_{ij,kt} = 1 \mid \boldsymbol{y}_{kt}, \widetilde{\boldsymbol{\kappa}}^{m,(r)}_{kt}, \boldsymbol{\kappa}^u_{kt}, \theta\Big) \tag{58}$$

That density is constructed from the distribution specified in Equation (**??**) and conditioning on $\boldsymbol{\kappa}^u, \boldsymbol{y}_{kt}$ and a candidate value of $\theta$. For $R$ draws the simulated selection likelihood approximating Equation (57) is then:

$$\widehat{L}^{\text{sel}}_{kt}(\theta) = \frac{1}{R}\sum_{r=1}^{R} L^{\text{sel},(r)}_{kt}(\theta). \tag{59}$$

## H.2 Likelihood function for outcomes

In this section, we correct for selection on levels between merging and nonmerging banks using the outcome likelihood. The key difference in the outcome likelihood is that is based on the configuration of sellers matched to buyers that actually occurred. So, notably, if banks A and B were sold simultaneously to the same buyer C, the seller selection likelihood used the sellers A and B's selection of the highest surplus buyer C to rationalize realized merger configuration. But, once A and B's selections are made, the outcome of the new bank ABC combines $\zeta_{ACt}$, $\zeta_{BCt}$, $\psi_{At}, \psi_{Bt}, \psi_{Ct}$.

The outcome contribution to the likelihood is

$$L^{\text{out}}_{kt}(\theta) \equiv f_{\kappa^u}(\boldsymbol{\kappa}^u_{kt} \mid \boldsymbol{y}_{kt}, \theta)\, f_{\kappa^m|M,\kappa^u}(\boldsymbol{\kappa}^m_{kt} \mid \boldsymbol{M}_{kt}, \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta)\,. \tag{60}$$

The first factor is the unconditional density of the observed fixed costs of banks that remain standalone. The second is the conditional density of the observed fixed costs of the resulting merged banks. Let $\mathcal{G}_{kt}$ denote the set of resulting merged banks. For each $g \in \mathcal{G}_{kt}$, let $\mathcal{C}_g$ denote its constituent banks, $i_g$ its final acquirer, and $\mathcal{T}_g = \mathcal{C}_g \setminus \{i_g\}$ its targets. Let $\mathcal{N}^u_{kt}$ denote the banks that remain standalone after the merger market.

**Standalone outcome likelihood.** The recovered innovations of unmerged banks are

$$\frac{\widetilde{\kappa}^u_{jkt}}{A_{jkt}} = \frac{\kappa^u_{jkt}}{A_{jkt}} - \eta_\kappa \frac{\kappa_{j,k,t-1}}{A_{j,k,t-1}}. \tag{61}$$

Define the normal distribution density evaluated at $x$ as $\phi(x, .)$ with arguments mean and variance. Then the model implies,

$$f_{\kappa^u}(\boldsymbol{\kappa}^u_{kt} \mid \boldsymbol{y}_{kt}, \theta) = \prod_{j \in \mathcal{N}^u_{kt}} \phi\left(\frac{\kappa^u_{jkt}}{A_{jkt}}, \eta_\kappa \frac{\kappa_{j,k,t-1}}{A_{j,k,t-1}}, 0, \sigma^2_\kappa\right). \tag{62}$$

**Merged-outcome likelihood.** Naively, $f_{\tilde{\kappa}^m|\boldsymbol{M}_{kt},\kappa^u} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ unconditionally as well. However, the ability for banks to choose merger partners $(\boldsymbol{M}_{kt})$ based on shocks $\tilde{\kappa}$ induces a selection prob-

lem, so we will derive that selected density.

The merged-outcome density integrates over the posterior distribution of the latent standalone innovations of merger participants:

$$f_{\kappa^m|M,\kappa^u}(\boldsymbol{\kappa}^m_{kt} \mid \boldsymbol{M}_{kt}, \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta) = \int f_{\kappa^m|M,\kappa^u,\tilde{\kappa}^m}(\boldsymbol{\kappa}^m_{kt} \mid \widetilde{\boldsymbol{\kappa}}^m_{kt}, M_{kt}, \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta) f_{\tilde{\kappa}^m|M,\kappa^u}(\widetilde{\boldsymbol{\kappa}}^m_{kt} \mid \boldsymbol{M}_{kt}, \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta)\, d\widetilde{\boldsymbol{\kappa}}^m_{kt}. \tag{63}$$

By Bayes' rule, the second term in the integral can be written as a function of terms defined in Equations (56) and (59),

$$f(\widetilde{\boldsymbol{\kappa}}^m_{kt} \mid \boldsymbol{M}_{kt}, \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta) = \underbrace{\frac{\Pr(\boldsymbol{M}_{kt} \mid \widetilde{\boldsymbol{\kappa}}^m_{kt}, \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta)}{\Pr(\boldsymbol{M}_{kt} \mid \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta)}}_{\equiv w(\theta)} f_{\tilde{\kappa}^m|\kappa^u}(\widetilde{\boldsymbol{\kappa}}^m_{kt} \mid \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta). \tag{64}$$

**Conditional mean of the merged outcomes.** For a given realization of $\widetilde{\boldsymbol{\kappa}}^m_{kt}$, the T1EV assumption implies

$$E[\zeta_{ij,kt} \mid P_{ij,kt} = 1, \widetilde{\boldsymbol{\kappa}}^m_{kt}, \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta] = \sigma_\zeta \left[\gamma_{EM} - \log \Pr(P_{ij,kt} = 1 \mid \widetilde{\boldsymbol{\kappa}}^m_{kt}, \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta)\right], \tag{65}$$

$$\mathrm{Var}[\zeta_{ij,kt} \mid P_{ij,kt} = 1, \widetilde{\boldsymbol{\kappa}}^m_{kt}, \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta] = \frac{\pi^2 \sigma_\zeta^2}{6}. \tag{66}$$

where $\gamma_{EM}$ is the Euler-Mascheroni constant. Define

$$\lambda_{ij,kt}(\theta) \equiv \int \left[\gamma_{EM} - \log \Pr(P_{ij,kt} = 1 \mid \widetilde{\boldsymbol{\kappa}}^m_{kt}, \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta)\right] f_{\tilde{\kappa}^m|M,\kappa^u}(\widetilde{\boldsymbol{\kappa}}^m_{kt} \mid \boldsymbol{M}_{kt}, \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta)\; d\widetilde{\boldsymbol{\kappa}}^m_{kt}. \tag{67}$$

$$\mu_{hkt}(\theta) \equiv \int \widetilde{\kappa}^m_{hkt}\, f_{\tilde{\kappa}^m|M,\kappa^u}(\widetilde{\boldsymbol{\kappa}}^m_{kt} \mid \boldsymbol{M}_{kt}, \boldsymbol{\kappa}^u_{kt}, \boldsymbol{y}_{kt}, \theta)\; d\widetilde{\boldsymbol{\kappa}}^m_{kt} \tag{68}$$

From Equation (18), for each bank $j$ who sold to itself in the merger market or newly merged banks $\mathcal{M}(i,j)$ in $(k,t)$,the post-merger combination outcome $\kappa_{\mathcal{M}(i,j),kt}$ are equal to

$$\frac{\kappa_{\mathcal{M}(i,j),kt}}{A_{ikt} + A_{jkt}} = \frac{\widetilde{\kappa}_{ikt} + \widetilde{\kappa}_{jkt}}{A_{ikt} + A_{jkt}} + \eta_\kappa \left(\frac{\kappa_{i,k,t-1} + \kappa_{j,k,t-1}}{A_{ikt-1} + A_{jkt-1}}\right) + [\delta_\kappa + \delta_{\mathcal{O}} \mathcal{O}(B_i, B_j) - \zeta_{ij,kt}] \frac{A_{jkt}}{A_{ikt} + A_{jkt}} \tag{69}$$

As a result, the conditional mean of the merged bank's fixed cost as a fraction of assets is then

$$m_{g,kt}(\theta) \equiv \sum_{h \in \mathcal{C}_g} \mu_{hkt}(\theta) + \eta_\kappa \sum_{h \in \mathcal{C}_g} \frac{\kappa_{h,k,t-1}}{A_{h,k,t-1}} + \sum_{j \in \mathcal{T}_g} \frac{A_{jkt}}{A_{hkt}} \left[\delta_\kappa + \delta_{\mathcal{O}} \mathcal{O}(B_{i_g}, B_j) - \sigma_\zeta \lambda_{i_g j,kt}(\theta)\right]. \tag{70}$$

Not taking into account changes due to selection, the variance of fixed costs divided by assets is

$$\boldsymbol{C}_{\mathcal{M}}(\theta) = \boldsymbol{\Sigma}(\theta) + \frac{\pi^2\sigma_\zeta^2}{6}\sum_{j\in\mathcal{T}_g}(\frac{A_{jkt}}{A_{hkt}})^2. \tag{71}$$

**Gaussian approximation.** The exact conditional density in Equation (**??**) is generally a mixture distribution. For likelihood evaluation, we approximate it by a Gaussian density with the conditional mean in Equation (76). We retain the unconditional component variances and omit selection-induced changes in covariances. The approximation is therefore

$$f_{\kappa^m|M,\kappa^u}(\boldsymbol{\kappa}_{kt}^m \mid \boldsymbol{M}_{kt}, \boldsymbol{\kappa}_{kt}^u, \boldsymbol{y}_{kt}, \theta) \approx \prod_{g\in\mathcal{G}_{kt}} \phi(m_{g,kt}(\theta), \boldsymbol{\mathcal{C}}_{\mathcal{M},kt}(\theta))\,. \tag{72}$$

**Numerical integration.** We can then numerically evaluate Equation (72). For a draw $r$, we simulate the posterior moments in Equations (68) and (67). So, define simulation weights

$$w_{kt}^{(r)}(\theta) = \frac{L_{kt}^{\text{sel},(r)}(\theta)}{\sum_{s=1}^R L_{kt}^{\text{sel},(s)}(\theta)}. \tag{73}$$

The simulated posteriors over realizations of $\tilde{\kappa}^m$ are then

$$\widehat{\mu}_{hkt}(\theta) = \sum_{r=1}^R w_{kt}^{(r)}\widetilde{\kappa}_{hkt}^{m,(r)} \tag{74}$$

$$\widehat{\lambda}_{ij,kt}(\theta) = \sum_{r=1}^R w_{kt}^{(r)}\left[\gamma_{EM} - \log\Pr\Big(P_{ij,kt}=1 \mid \widetilde{\boldsymbol{\kappa}}_{kt}^{m,(r)}, \boldsymbol{\kappa}_{kt}^u, \boldsymbol{y}_{kt}, \theta\Big)\right]. \tag{75}$$

$$\widehat{m}_{g,kt}(\theta) \equiv \sum_{h\in\mathcal{C}_g}\mu_{hkt}(\theta) + \eta_\kappa\sum_{h\in\mathcal{C}_g}\frac{\kappa_{h,k,t-1}}{A_{h,k,t-1}} + \sum_{j\in\mathcal{T}_g}\frac{A_{jkt}}{A_{hkt}}\left[\delta_\kappa + \delta_{\mathcal{O}}\mathcal{O}(B_{i_g}, B_j) - \sigma_\zeta\widehat{\lambda}_{i_g j,kt}(\theta)\right]. \tag{76}$$

The implemented merged-outcome density is

$$\widehat{f}_{\kappa^m|M,\kappa^u}(\boldsymbol{\kappa}_{kt}^m \mid \boldsymbol{M}_{kt}, \boldsymbol{\kappa}_{kt}^u, \boldsymbol{y}_{kt}, \theta) \approx \prod_{g\in\mathcal{G}_{kt}} \phi(\widehat{m}_{g,kt}(\theta), \boldsymbol{C}_{\mathcal{M}}(\theta))\,. \tag{77}$$

In other words, we make two assumptions. First, we treat (77) as Gaussian. This is a distributional approximation: in reality, the centered Gumbel on the $\kappa$ row is right-skewed rather than symmetric, but its mean (zero) and variance ($\pi^2\sigma_\zeta^2/6$) are matched exactly. Second, we assume that the control function correction affects mean, but not variance. As a result of this assumption, we can represent the covariance of the full residual with $\boldsymbol{C}_{\mathcal{M}}(\theta)$.

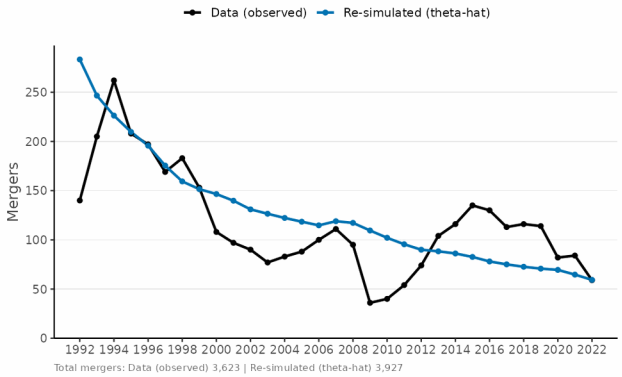
With the terms of the likelihood defined, we then use simulated maximum likelihood for draws $R = 100$ to recover the parameters. Initial values for $\eta_\kappa$ (Equation 21) and similar were set from estimating an AR(1) process in the data. Initial values for merger frictions come from estimating

logit demand on choices with $\Delta\pi$ replaced by a constant. Initial values for merger synergies $(\delta_\kappa, \delta_\mathcal{O})$ were estimated from realized pre and post merger differences in fixed costs.

# I Model fit

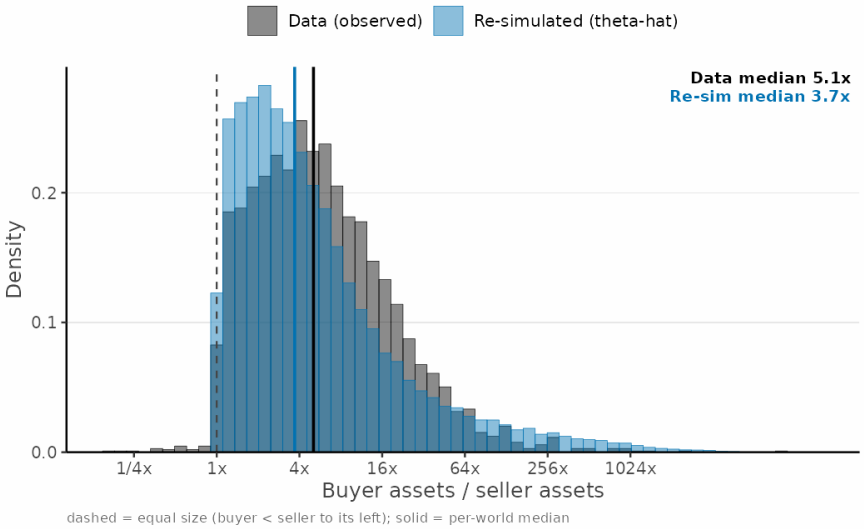
This section shows how data re-simulated from the model parameters compare to empirical data. The model is estimated using within-state merger choices and the outcomes that those mergers have on the profitability of banks in the market. As a result, the total number of mergers in the real data with the model's estimate, without additional assumptions, will not precisely match the number of mergers implied by simulations. Banks could have been purchased by out of state acquirers, failed, or exited, and the period has also been marked by modest bank entry, all of which are assumed not to exist in the re-simulation. We do not include the other phenomena in the computation of counterfactuals to create a transparent comparison between baseline and counterfactual policy. However, future sensitivity analysis could incorporate these channels.

Figure I.1: Count of mergers used for estimation versus implied by model parameters



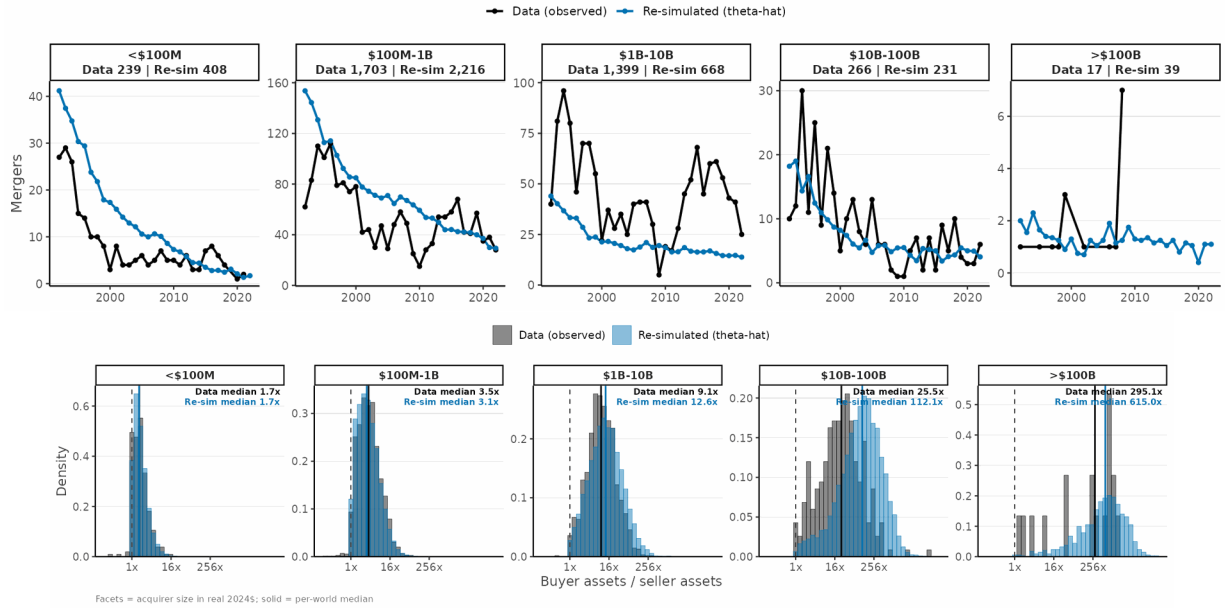

Note: This graph shows the number of in-state mergers used for estimation annually from 1992 to 2024 in black. The blue line plots the average number of mergers occurring in 500 re-simulations around the model's estimated parameters.

Figure I.2: Relative buyer and seller size




Note: This figure shows the mergers in the real data used for model estimation versus re-simulated mergers using the estimated parameter values. The re-simulated values produce mergers that are smaller (3.7x vs. 5.1x) than in the data.

Figure I.3: Model fit by bank size




Note: This figure shows the mergers in the real data used for model estimation versus re-simulated mergers using the estimated parameter values. The re-simulated values produce similar trends in mergers over time than in the data. The mergers in re-simulated data are roughly the same size as in the data. The difference between the simulation and data is in over-predicting the size of targets for regional banks (10-100B) and under-predicting the size of targets for the largest banks (100B+).

# J Evaluating counterfactual policy

This section describes how we evaluate the effect of counterfactual policy using our simulation based model. Changes in current market structure are created using the real empirical 1992 population of banks, then simulating counterfactual paths of industry structure forward given the new parameter values. We then compare those results to the effect of simulating forward with the estimated parameter values.

## J.1 Notation

**Timing.** For a merger in year $c$, $t-1$ merger values come from calendar year $c-1$. Post-merger $t$ values come from calendar year $c+1$. The simulation deals with this by transforming the time increment of two years from estimation to one year. The simulation runs $t = 1992, \ldots, 2024$ by state $k$, repeated for seeds $1, \ldots, S$ that each draw a matrix of $\boldsymbol{\zeta}$.

**Inputs.**

- Per bank $j$ and label $t$: primitives $\psi_{jt} = (\xi^D, \xi^L, c^D, c^L, \kappa)_{jt}$ with $\kappa$ a rate per dollar of assets; assets $A_{jt}$; Tier 1 capital $K_{jt}$; branch count $B_{jt}$; county footprint $F_{jt}$ and pairwise shared-county counts $N_{ijt}$; private indicator $D_j$.
- Per state: totals $M^D_{kt}, M^L_{kt}$, realized merger assignments $\boldsymbol{M}_{kt}$[32]
- Parameters: demand $(\alpha^d, \beta^d_{br})$; $\hat{\theta} = (\boldsymbol{\rho}_\psi, \sigma_\zeta, \tau, \boldsymbol{c}_{RCW}[\cdot]\delta_\mathcal{O}, \delta_\kappa)$; $\beta = 0.9$.

**State variables transitions.** The transition is an annual AR(1), $\psi_{jt} = \boldsymbol{\rho}_1 \odot \psi_{j,t-1} + \varepsilon_{jt}$, with $\boldsymbol{\rho}_1 = \boldsymbol{\rho}_\psi^{\circ 1/2}$ the elementwise square root of the estimated two-label persistence $\boldsymbol{\rho}_\psi$ because the model was estimated using one year prior to merger and one year after. Each bank carries $\psi_{jt}$, $A, K, B$, and its lineage of previous ownership.

**Initialization ($t_0 = 1992$).** Start from every bank present at values from 1992.

## J.2 Forward simulation

The following loop is repeated for each year.

[32] Note that state market sizes may not be exactly pulled from the data. Namely, entrants, exits, and changes of a bank's headquarter state would change these values. Entry and exit margins, since they are not in the model, are followed as they transpire in the data. We correct the market sizes of deposits and loans by state to correct for charter state changes. Explicitly, the counterfactual assumes no charter changes happen, so if banks were present in 1992 in a given state, they continue competing in that state throughout the panel.

1. **Next period values** Advance to $t+1$ values by holding values constant, except for $\kappa$ and subject to time fixed effects:

$$D_{it+1} = D_{jt} \tag{78}$$
$$K_{it+1} = (1+g_t^K)K_{jt} \tag{79}$$
$$A_{it+1} = (1+g_t^A)A_{jt} \tag{80}$$
$$B_{it+1} = B_{jt} \tag{81}$$
$$c_{it+1}^d = (c_{jt}^d - \bar{c}_t) + \bar{c}_{t+1} \tag{82}$$
$$\xi_{it+1}^d = \xi_{jt}^d \tag{83}$$
$$\kappa_{it+1} = \rho_\kappa \kappa_t + \tilde{\kappa}_{it+1} \tag{84}$$

   where $g_t$ is the median growth rate of capital and assets in the market, respectively.

2. **Equilibrium.** Solved among the banks present at $t$, i.e. before this label's mergers.[33] On each side $d \in \{D, L\}$, over banks with nonzero $\xi^d$ or $c^d$: utilities $V_{jt}^d = \alpha^d \rho_{jt+1}^d + \xi_{jt+1}^d + \beta_{br}^d \log B_{jt}$; shares $s_{jt}^d = e^{V_{jt+1}^d} / \sum_\ell e^{V_{\ell t}^d}$ (no outside option); spreads from the fixed point $\rho_{jt+1}^d = c_{jt+1}^d + 1/\Big(|\alpha^d|(1 - s_{jt+1}^d)\Big)$. Profits are then $\pi_{jt+1} = \sum_d M_{kt}^d s_{jt}^d (\rho_{jt+1}^d - c_{jt+1}^d) - \kappa_{jt+1} A_{jt}$.

3. **Surplus.** For every buyer–seller pair $(i, j)$, with overlap share $\mathrm{ov}_{ij,t} = \min(N_{ij,t}/F_{jt}, 1)$, and $\overline{\Delta\pi}_{ij,t} = -\theta_{ov,\kappa} A_{jt} \mathrm{ov}_{ij,t}$ (the fixed-cost change implied by step 4's $\kappa$ equation without $\zeta$):

$$\bar{\Phi}_{ij,t} = \frac{\overline{\Delta\pi}_{ij,t}}{1-\beta} + D_j c_{RCW}[b(i,j)]\, A_{jt} - \tau A_{jt}$$

   where $b(i,j)$ are quantile bins $K_{ijt}/(A_{ijt} + A_{ijt} - K_{ijt})$.

4. **Selection.** For each seller $j$: draw $\zeta_{ij,t} \sim \text{Gumbel}(0, \sigma_\zeta)$ i.i.d. over buyers and an outside draw $\zeta_{0j,t}$. Feasible buyers $\mathcal{B}_{jt} = \{i \neq j : A_{it} \geq A_{jt}\}$; $j$ sells to $i^* = \arg\max_{i \in \mathcal{B}_{jt}} \{\bar{\Phi}_{ij,t} + \zeta_{ij,t} A_{jt}/(1-\beta)\}$ iff the max exceeds $\zeta_{0j,t} A_{jt}/(1-\beta)$, else stays standalone. Sellers decide independently; a buyer may purchase several sellers; a buyer who also sold resolves to the top of its chain.

5. **Combination.** For each surviving buyer $j$ and its sellers $s$, with weights $w^{\mathcal{M}}$ on $w_{jt}^{\mathcal{M}} = \frac{s_j}{s_{jt} + \sum_s s_{st}}$,

$$V_{\mathcal{M}t^d} = \log(V_{jt} + \sum_s V_{st}) \quad c_t^M = w_{jt}^{\mathcal{M}} c_{jt} + \sum_s w_{st}^{\mathcal{M}} c_{st},$$
$$\kappa_t^M = w_{jt}\kappa_{jt} + \sum_s w_{st}\kappa_{st} + \sum_s w_{st}\left(\theta_{ov,\kappa}\, \mathrm{ov}_{is,t}\right)$$

   plus, when there are $\geq 2$ sellers, a mean-zero Gumbel$(0, \sigma_\zeta)$ on $\kappa_t^M$ per seller–seller pair, at weight $\min(w_{st}, w_{s't})$. Balance sheet: $A_t^M$ as above, $K_t^M = K_{jt} + \sum_s K_{st} - 0.03 \sum_s A_{st}$,

[33] An entity formed in $t$ does not compete until $t+1$, its first full post-merger year

$B_t^M = B_{jt} + \sum_s B_{st}$, $P_M = P_i$; the entity's lineage is the union of the parents' lineages. [34] In this way, mergers cost capital and limit the ability for banks to do another merger in future periods.

6. **Transition to $t+1$.** A non-merging bank advances its standalone primitives advance. Merging banks follow the merger combination equations above.

7. **Population.** Entrants enter the state market as observed in the data, banks not yet absorbed by merger exit by their real exit date.

### J.3 Policy experiments

All worlds run the identical simulation, using the same data, same $\hat{\theta}$, same seeds, same $t_0$, except: (1) for the baseline, $\hat{\theta}$ is unchanged; (2) when removing the RCW, every $c_{RCW}$ loading set to zero ($c_{RCW}$ enters only $\bar{\Phi}$, so this changes which mergers happen, not the merger technology); (3) when imposing the RCW on public buyers, the $D_i = 1$ restriction on $b_t(i,j)$ is switched off with $\hat{\theta}$ unchanged (4) deposit caps $\boldsymbol{\mathcal{D}}$ compute the sum of pre-merger market shares and exert infinite cost (remove the merger from the seller's choice set) when $s_{ikt} + s_{jkt} > \boldsymbol{\mathcal{D}}$.

To report welfare, we compute welfare for each seed and each market year $k, t$ from the simulated panel. Namely,

$$CS_{kt}^d = \frac{M_{kt}^d}{|\alpha^d|} \log \sum_j e^{V_{jt}^d}, \qquad PS_{kt} = \sum_d M_{kt}^d \sum_j s_{jt}^d(\rho_{jt}^d - c_{jt}^d) - \sum_j \kappa_{jt} A_{jt}, \qquad W_{kt} = CS_{kt}^D + CS_{kt}^L + PS_{kt}.$$

In comparing welfare to the baseline result, we compare separately for each seed, market and year. $CS$ levels carry an arbitrary constant absent an outside option.

[34] When unobservable pair-specific selection on gains $\zeta_{ist}$ is omitted from the outcome and just used for merger selection in estimation, it is omitted as well.